

T6 CFD Verification: Stage A and the Real-Lumen Pilot

Verification of the OpenFOAM setup for the T6 3D CFD arm, with the scan-837 pipeline pilot (a living report)

T6 CFD verification log
Stage A (A1–A5) run; Items 1–6 (§11),
the scan-837 pilot and its follow-ups to 2026-09-26

2026-09-18 to 2026-09-26

Abstract

This document reports Stage A of the T6 study’s 3D CFD verification suite (protocol/CFD-ARM-SPEC.md §8) — checks A1 through A5, run on real OpenFOAM hardware (ESI v2406) — and the pilot work that followed it, after a three-reviewer adversarial pre-run audit (Table 5.1, GPT-6-astra, Gemini 3.8 Flash — [references/SETUP-AUDIT-SYNTHESIS-2026-09-18.md](#)) found and fixed five blocking defects in the handover as originally written. Every check reports a pass/fail verdict against the spec’s own tolerance; failing or ambiguous results were triaged by independent reviewers before a fix or verdict was accepted, and every reviewer disagreement, operational incident and defect in the assistant’s own scripts is disclosed rather than smoothed over. The report is a living document: it covers work to 2026-09-24 and will be updated as the open items listed at the end are completed.

Stage A result: of the seven pass/fail rows of the running summary, six pass and one is not established. A1 (Poiseuille pipe), A3 (resistance BC) on sten00 and sten50, and A4 (prescribed-flow/resistance round trip on sten70) are clean, unanimous passes. A3 on sten80 is an accepted flagged pass — its numeric tolerance is met, but the spec’s stricter residual criterion is not, and the cause is left open between a numerical-SIMPLE-stall and a slow physical instability. A5 (mesh independence) on sten80 passes comfortably; **A5 on sten70 is NOT ESTABLISHED**: its original marginal pass (finest-pair difference at 94.6% of the tolerance, non-monotone) was not confirmed by a fourth, finer level, whose Celik–Eça–Hoekstra analysis is not trustworthy because the boundary-layer stack was held constant across levels, so the ladder was not a single-parameter refinement family (§11.3). A2 (0D-vs-3D gap) is informational by design and shows a reportable finding: the gap grows and reverses sign with severity. The coded resistance-outlet boundary condition — untested on any real OpenFOAM installation at the start of this work — has real-hardware confirmation that it compiles, converges under the audit-derived relaxation and enforces its defining physics to machine precision on four straight-vessel geometries (sten00, sten50, sten70, and sten80 as a flagged case), on an idealised 3-outlet tree (Item 1) and on a real 5-outlet coronary tree (Item 3, scan 837).

Beyond Stage A (§11): a branched idealised tree, the sten70 fourth-level analysis and a severity sweep, a real-lumen pilot on scan 837 in which the meshing blocker was root-caused (the surface remesh pinched narrow lumens) and resolved, a 0D-to-OpenFOAM handoff, RCR outlet verification (PASS), a pulsatile cost pilot (stopped; no cycle-averaged result exists), and, for the scan-837 lesion80 case, a mesh-sensitivity study (all ten registered quantities insensitive), a hyperaemic-flow pair (the study’s Murray demand on the 3D inlet radius, 3.16 times the earlier flow) and an extended-refined-tube test (no quantity moved). **Scan 837 is outside the frozen cohort: it is a pipeline pilot, not a Gate M1 result.** The 3D discretisation uncertainty of sten70 from a systematic refinement family is now measured: $U_{3D} = 0.00055$ in FFR units (three throat-zone levels of 50/25/12.5 μm plus a shifted zone; a 2D wedge family as cross-check; §11.7). The prescribed-flow mode closes its round trip on the 3-outlet tree (largest flow error 0.0007%). The Gate-M1 work on the shipped scan-14 packages is built (geometry and meshes, with failed gates reported) but not yet solved; it is listed with the other open items in the Conclusions.

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1 Scope and common setup

All runs use `cfMesh cartesianMesh` for meshing and `simpleFoam` (steady, laminar, incompressible) for the solve, on OpenFOAM ESI v2406. Fluid properties throughout: $\rho = 1060 \text{ kg/m}^3$, $\mu = 0.004 \text{ Pa}\cdot\text{s}$ ($\nu = 3.773585 \times 10^{-6} \text{ m}^2/\text{s}$). Geometry is generated by `cf_handover/stageA/make_stageA_geometry.py` (post audit-fix: wall normals corrected, confirmed outward on 100% of wall facets by direct check). Case skeleton (`fvSchemes`, `fvSolution`, `controlDict`) is `cf_handover/stageA/caseTemplate/`, itself corrected mid-session when a real run caught a missing `div((nuEff*dev2(T(grad(U))))` scheme entry that source review alone had missed. Meshes for this validation pass are deliberately coarser than the spec’s full §6.2 production guidance (base cell $\approx D/20$, ≥ 12 cells across the throat) — the goal throughout is a fast confirmation that the fixed pipeline is internally correct, not a publication-grade result. A5 (§9) itself checks how much that resolution choice matters, using an even coarser/finer bracketing three-level ladder per case; a production run for the actual study should still follow §6.2’s full guidance rather than reuse these validation-pass mesh sizes directly.

Where a coded resistance-outlet BC is used, it is exactly the version in `cf_handover/bc/resistanceOutlet.md`

after the 2026-09-18 audit fixes: relaxation applied against `p.prevIter()` (not the raw field, which would compound under non-orthogonal correctors), and the per-case `relax`/initial-value table derived from the audits' stability analysis (Table 1).

Table 1: Per-case resistance-BC settings (from `bc/resistanceOutlet.md`)

Case	R (Pa s m ⁻³)	relax	initial value (m ² /s ²)
sten00	6.977581×10^9	0.05	10.5028
sten50	6.974906×10^9	0.08	10.1106
sten70	6.974826×10^9	0.20	8.0694
sten80	6.974430×10^9	0.25	5.2002

2 Running summary

Check	Case	Mesh	Result	Verdict
A1	pipe	61,440 cells	CFD $\Delta P = 89.86$ Pa vs. analytic 90.54 Pa	PASS (−0.75%, tol. 1%)
A3	sten00	191,024 cells	$p_{\text{outletPatch}}$ vs. $P_v + RQ$: exact to displayed precision	PASS (0.00000%, tol. 0.1%)
A3	sten50	383,412 cells	$p_{\text{outletPatch}}$ vs. $P_v + RQ$: exact to displayed precision	PASS (0.00000%, tol. 0.1%)
A2	sten00	(as A3)	3D $Q = 1.4548$ mL/s vs. 0D 1.500 mL/s (−3.01%); 3D FFR 0.9759 vs. 0D 0.9829	INFORMATIONAL (no pass/fail)
A2	sten50	(as A3)	3D $Q = 1.4196$ mL/s vs. 0D 1.4410 mL/s (−1.48%)	INFORMATIONAL (no pass/fail)
A3	sten80	1,086,920 cells	$p_{\text{outletPatch}}$ vs. $P_v + RQ$ (mean of last 500): 0.0024%	PASS [†] (tol. 0.1%)
A2	sten80	(as A3)	3D $Q = 0.7550$ mL/s vs. 0D 0.6948 mL/s (+8.68%); 3D FFR $\approx$ 0.509 vs. 0D 0.485	INFORMATIONAL (no pass/fail)
A4	sten70	379,468 cells	round trip: derived R reproduces step-1 flow to −0.012% (plateau; +0.001% at iteration 2000)	PASS (tol. 0.5%)

Check	Case	Mesh	Result		Verdict
A5	sten70	3 levels, 198k–760k cells	finest-pair = 0.004732 of tol. 0.005, non-monotone)	ΔFFR (94.6%	NOT ESTABLISHED (numeric check met, but the fourth level shows it did not measure convergence — §11.3)
A5	sten80	3 levels, 282k–1087k cells	finest-pair 0.000874	$\Delta\text{FFR} =$	PASS (tol. 0.005)

Note on scope

All five Stage A checks (A1–A5) defined in the spec have been run at least once in this validation pass. A5 (mesh independence, two cases $\times$ three refinement levels) was the most compute-intensive check and was run last, after the coded resistance BC’s correctness had already been confirmed at a single resolution per case via A2/A3/A4 — deliberately sequenced so that a mesh-independence result would not be the first real-hardware test of the BC itself.

3 A1 — Poiseuille pipe

Setup. Straight pipe, $r = 1.5$ mm, $x \in [0, 100]$ mm. Inlet: `flowRateInletVelocity`, $Q = 1.5 \times 10^{-6}$ m³/s. Outlet: `fixedValue` $p = 0$. Mesh: 61,440 cells (base cell 0.3 mm, 4 boundary layers, growth 1.2); `checkMesh`: max non-orthogonality 28.7°, max skewness 1.07, “Mesh OK”. Solved to `residualControl` 10^{-6} on p and U , converged in 168 iterations.

Result. Area-averaged static pressure at $x = 60$ mm and $x = 90$ mm gave $\Delta P_{\text{CFD}} = 89.8643$ Pa against the analytic Hagen–Poiseuille value

$$\Delta P_{\text{analytic}} = \frac{8\mu L Q}{\pi r^4} = 90.5415 \text{ Pa},$$

an error of -0.75% , inside the spec’s 1% tolerance. Inlet and outlet flux matched the prescribed Q exactly ($\pm 1.5 \times 10^{-6}$ m³/s).

Verdict: **PASS**. No further action.

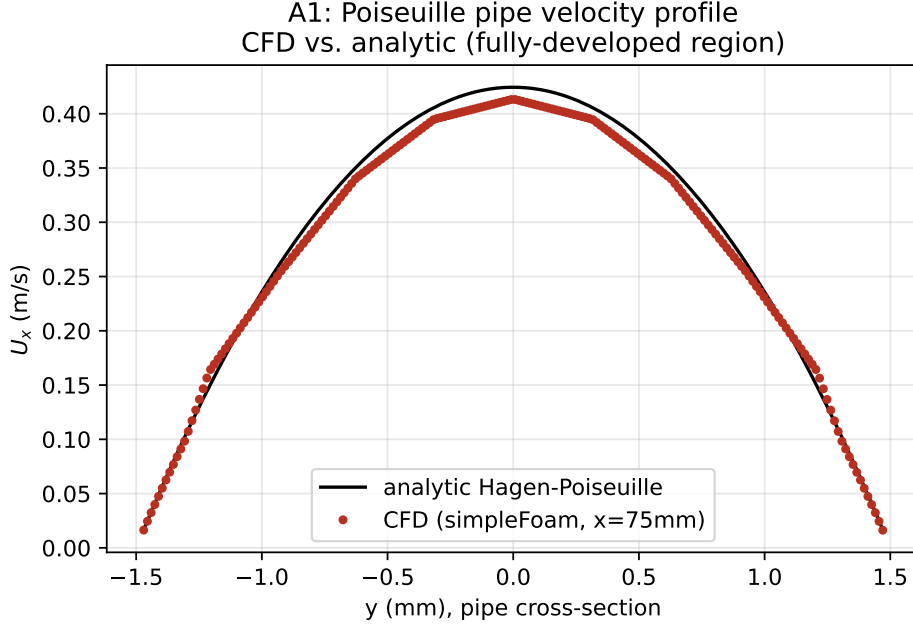


Figure 1: A1: velocity profile at $x = 75$ mm (fully-developed region) against the analytic Hagen–Poiseuille parabola. Peak-velocity agreement is -2.66% , slightly larger than the -0.75% pressure-drop error since a point value (peak velocity) is more sensitive to local profile shape than an integral quantity (pressure drop) is.

4 A3 — resistance-outlet BC (sten00)

Setup. `sten00.stl` (0%DS reference vessel), 191,024 cells (max non-orthogonality 34.6° , max skewness 1.12, “Mesh OK”). Inlet: `totalPressure`, $p_0 = 11.3198$ (kinematic). Outlet: the coded `codedFixedValue` resistance BC, $R = 6.977581 \times 10^9$, `relax = 0.05`, initial value 10.5028 (Table 1). No `residualControl` (per the audit’s compounding-relaxation finding); run for a fixed 2000 iterations and read the monitors.

This is the first time this coded BC has ever compiled or run — at the start of this session it was documented as “UNTESTED, written on a machine with no OpenFOAM”.

Result. The dynamic code compiled cleanly (`wmake libso`, no errors). By iteration ~ 1000 all monitors were flat to the displayed precision; the run completed its full 2000-step budget with inlet/outlet flux and outlet pressure identical between the last several hundred writes. Converged: $Q = 1.454794$ mL/s, $p_{\text{outletPatch}} = 10,817.5565$ Pa.

$$P_v + RQ = 666.61 + (6.977581 \times 10^9)(1.454794 \times 10^{-6}) = 10,817.5565 \text{ Pa,}$$

an error of 0.00000% against the spec’s 0.1% tolerance — the BC enforces its own defining law to machine precision at convergence, exactly as intended.

Verdict: PASS. The per-case `relax = 0.05` fix (vs. the original fixed 0.2–0.3, which all three audits showed diverges for this case) produced a smooth, damped startup overshoot settling cleanly to a flat plateau — not the sign-flipping, growing oscillation the original fixed relaxation would have caused.

Outlet flow-rate convergence history: clean case vs. flagged case

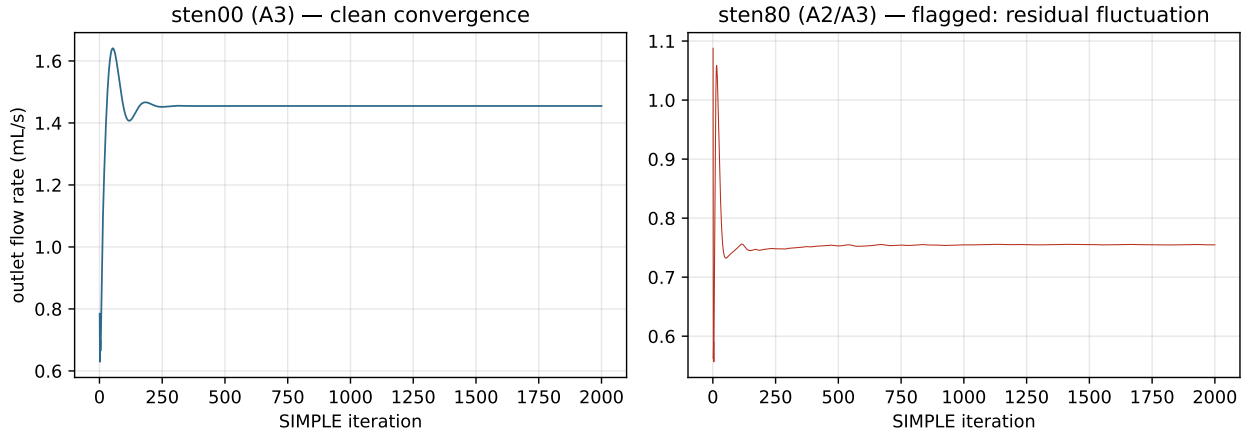


Figure 2: Outlet flow-rate convergence history: sten00’s clean damped-overshoot convergence (left) against sten80’s persistent low-amplitude fluctuation (right, §6) at the same axis scale. Both plateau by eye at this scale; the sten80 fluctuation only becomes visible on a zoomed axis (Fig. 6).

A2 (informational): sten00 3D vs. 0D

The same converged solve gives $Q_{3D} = 1.4548$ mL/s against the 0D-discrete prediction of 1.5000 mL/s (−3.01%), and FFR at the measurement plane 0.9759 against the 0D value 0.9829. Per spec §8, A2 carries no pass/fail — this is tabulated as a genuine finding about the gap between the lumped 0D stenosis model and the 3D solve, not a defect in either.

5 A3 — resistance-outlet BC (sten50)

Setup. sten50.stl (50%DS), 383,412 cells (“Mesh OK”). Same BC structure as §4 with $R = 6.974906 \times 10^9$, relax = 0.08, initial value 10.1106.

Result. Monitors reached a dead-flat plateau (outlet pressure 9.9700317 m²/s² identical across 15+ consecutive writes) well before the 2000-iteration budget (flat from about iteration 800). It was originally reported as stopped early to free compute; **correction (retained monitor, 2026-09-24):** the monitor holds 2000 rows, so the run reached its full budget. Converged: $Q = 1.419607$ mL/s, $p_{\text{outletPatch}} = 10,568.2336$ Pa.

$$P_v + RQ = 666.61 + (6.974906 \times 10^9)(1.419607 \times 10^{-6}) = 10,568.2333 \text{ Pa,}$$

error 0.00000% against the 0.1% tolerance.

Verdict: **PASS.**

A2 (informational): sten50 3D vs. 0D

$Q_{3D} = 1.4196$ mL/s vs. 0D 1.4410 mL/s (−1.48%) — a smaller gap than sten00’s −3.01%, consistent with the stenosis dominating the vessel’s total hydraulic resistance more as severity increases (the taper/entrance discrepancy between the 0D and 3D representations matters proportionally less). FFR at the measurement plane: 0.9516 (3D) vs. 0.9461 (0D).

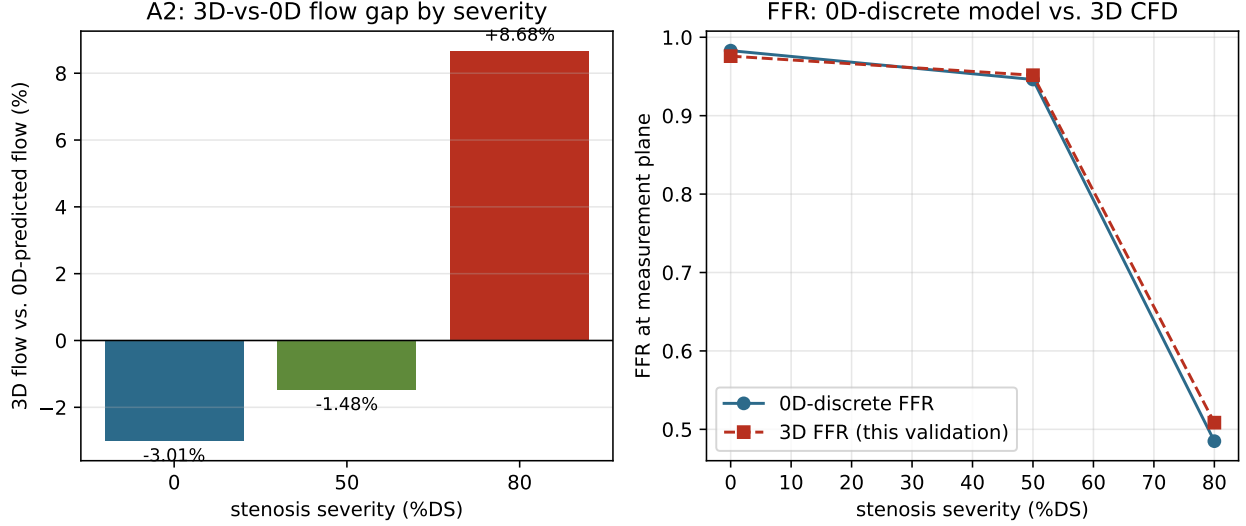


Figure 3: A2 across all three severities run in this validation pass. Left: the 3D-vs-0D flow gap grows with severity and *reverses sign* between 50%DS and 80%DS. Right: 0D-discrete and 3D FFR track closely at 0%DS and 80%DS but diverge more at 50%DS — itself a reportable finding about the lumped 0D stenosis model (spec §8), not a defect in either fidelity.

6 A2/A3 — sten80: converges numerically but flagged

Setup. `sten80.stl` (80%DS), meshed to 1,086,920 cells — substantially larger than the other three cases because of the fine throat resolution needed at this severity (throat diameter 0.594 mm, base cell 0.1 mm $\Rightarrow$ only ~ 6 cells across the throat core, coarser than the spec’s ≥ 12 target). $R = 6.974430 \times 10^9$, $\text{relax} = 0.25$, initial value 5.2002. Ran the full fixed 2000-iteration budget (no `residualControl`, as for all coded-BC cases here).

Result — residuals did not reach the spec’s threshold. Unlike `sten00/sten50/A4`, this run’s field residuals plateaued well above 10^{-5} (final outer-iteration initial residuals: $U_x \sim 1.5 \times 10^{-3}$, $U_y \sim 0.072$, $U_z \sim 0.054$, $p \sim 6 \times 10^{-3}$), and the bulk monitors show a persistent ~ 0.1 – 0.13% fluctuation band over the last 500 iterations rather than a flat plateau (outlet flux 7.546 – $7.556 \times 10^{-7} \text{ m}^3/\text{s}$, mean 7.5504×10^{-7} ; outlet pressure 5.594 – 5.600 , mean $5.5969 \text{ m}^2/\text{s}^2$). This is exactly the risk flagged pre-run by all three auditors (`SETUP-AUDIT-SYNTHESIS-2026-09-18.md`): a post-stenotic jet at throat $\text{Re} \approx 395$ – 430 may not settle to a clean steady state under `simpleFoam`.

Using the 500-iteration mean, A3’s own numerical criterion is still comfortably satisfied (0.0024%, well inside the 0.1% tolerance, and the error stays under 0.006% even taken at the min/max of the fluctuation band — see §2), and the 3D-vs-0D flow gap for A2 is the largest and first *sign-reversed* gap in the sweep: $Q_{3D} = +8.68\%$ above the 0D prediction (`sten00/50` were both *below*). **Verdict: PASS[†] on the stated numeric criterion, but flagged — the spec’s broader convergence criterion (residuals $< 10^{-5}$ and monitors stable) is not met, so this pass should not be read as a clean result.** Per §10, this was referred to the audit panel before being written up as final.

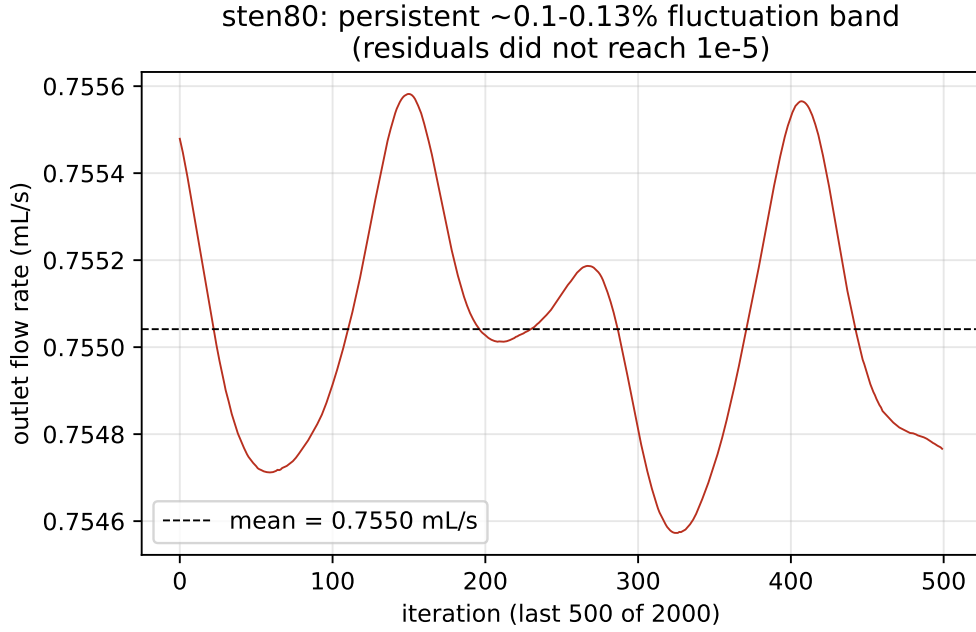


Figure 4: sten80’s outlet flow rate over the last 500 of 2000 iterations. The fluctuation is smooth and quasi-periodic (period $\approx 150\text{--}160$ iterations), not white noise — suggestive of some kind of limit-cycle-like behaviour in SIMPLE’s iteration space, though (per GPT-6-astra’s point, §8) an iteration-space oscillation does not by itself establish a physical time-domain oscillation.

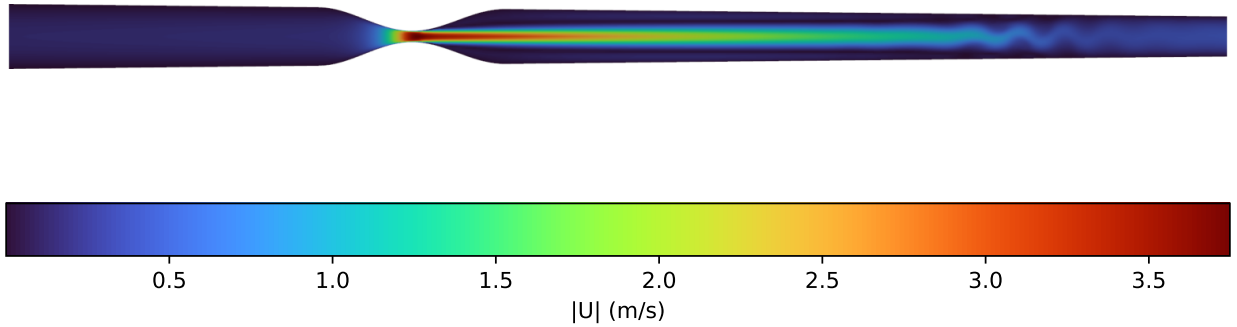


Figure 5: Velocity magnitude, mid-plane slice through the lesion and post-stenotic region ($x=10\text{--}75$ mm) at the flagged iteration-2000 state. The jet accelerates sharply through the throat (dark red core, up to 3.75 m/s) and visibly develops a wavy, non-uniform structure downstream — the same post-stenotic region flagged as being at risk of instability.

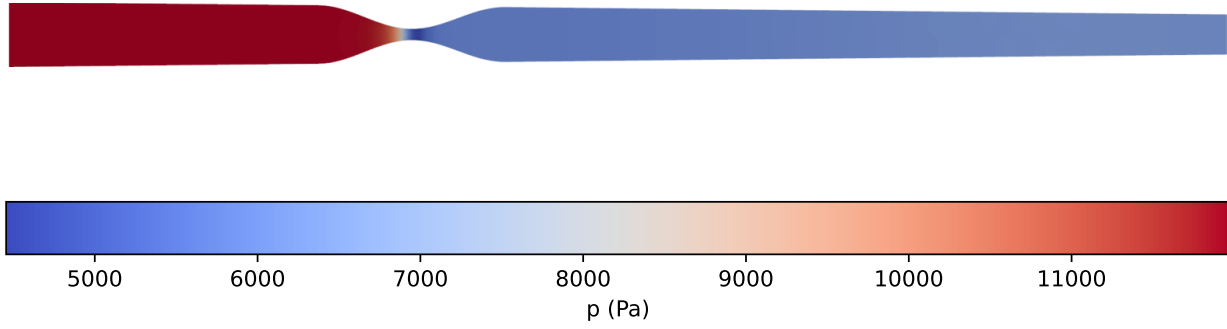


Figure 6: Pressure field over the same region and time state as the velocity figure above, showing the expected smooth pressure drop through the stenosis.

Panel diagnosis (referred per §10)

GPT-6-astra: diagnosis *(b) numerical, not established as physical instability*. Argues the observed pattern (large per-component residuals with much smaller, but nonzero, bulk-monitor fluctuation) demonstrates only *failure to reach the stated convergence criterion*, not a verified physical limit cycle — a SIMPLE iteration is not physical time-marching, so oscillation in iteration-space cannot by itself be read as a real flow oscillation. Separately notes that published stenosis-instability Reynolds thresholds are usually quoted on the *parent-vessel* Reynolds number, not the throat value used here ($Re_{\text{parent}} \approx 0.2 Re_{\text{throat}} \approx 80$ for an 80%-diameter/96%-area reduction), so the ~ 400 throat-Re figure this study uses does not, on its own, establish proximity to a literature instability onset. **Recommends:** a time-accurate `pimpleFoam` run on the same mesh/schemes (`backward ddt`, $Co \lesssim 0.5$, enough outer correctors for the resistance BC to equilibrate within each step) to distinguish a genuine limit cycle (persistent, timestep-independent oscillation) from numerical stalling (decays under time-accurate marching). Explicitly: report the current band only as *non-converged iteration variability under the study's pre-agreed fallback*, not as a verified 10^{-5} pass.

Table 5.1: diagnosis *mostly (b), a numerical stall, not a true limit cycle*, on stronger grounds: **sten70 (§7) has essentially the same throat Reynolds number (≈ 428 , vs. sten80's 395 — slightly lower) and converged cleanly to residuals 10^{-7} – 10^{-9} under an identical pipeline (same schemes, same BC formulation, same `prevIter()` relaxation fix).** *This specific claim was independently checked against this document's own run logs and confirmed exactly (§7's step-2 final residuals: $U_x = 5.2 \times 10^{-9}$, $U_y = 3.3 \times 10^{-7}$, $U_z = 2.1 \times 10^{-7}$) — since a comparable-or-higher-Re case converges cleanly on the same setup, Reynolds number alone is not sufficient to explain sten80's stall, pointing instead at sten80-specific mesh/setup factors: specifically, the throat is resolved by only ~ 6 cells across (vs. the spec's ≥ 12 target) feeding a long thin shear layer that unlimited `linearUpwind` on `cfMesh`'s refinement-transition cells can inject noise into, which SIMPLE's steady iteration does not damp.* Also notes the U_y/U_z residuals look worse than they are because OpenFOAM normalises each component by its own (here tiny) magnitude; U_x and the bulk-monitor band are the more honest measures. **Recommends:** the same time-accurate `pimpleFoam` restart as GPT-6-astra, with concrete parameters ($\Delta t \approx 3 \times 10^{-5}$ s for $Co \approx 1$ at the ~ 2.7 m/s throat velocity, 2–3 outer correctors, run ≥ 0.3 s $\approx 10^4$ steps), explicitly ruling out `localEuler` pseudo-transient (not time-accurate, cannot distinguish the two hypotheses). Adds one caveat not raised elsewhere: to check whether the post-stenotic jet reattaches before the outlet within the available ~ 65 mm of downstream length, since backflow reaching the resistance patch would invalidate either outcome.

Gemini 3.8 Flash: not obtained. The agy CLI's underlying Google-account session had expired and required interactive browser re-authentication, which cannot be completed from this environment; the two available diagnoses above should be read as a two-of-three panel, not three, on this specific finding.

Assessment and recommended next step

The two available reviewers agree on the diagnosis (numerical stall favoured over confirmed physical instability) and independently converge on the same fix (a time-accurate `pimpleFoam` diagnostic run), with Fable’s addition — checked and confirmed against this document’s own `sten70` data — meaningfully strengthening the case against a pure-Reynolds-number explanation. **This document reports the current SIMPLE result (mean $\pm$ range, as above) as the best available answer for now, explicitly flagged as unresolved**, and records the `pimpleFoam` diagnostic run as the concrete next step before `sten80`’s A2/A3 numbers are treated as final. That run is not performed in this pass.

7 A4 — prescribed-flow / resistance round trip (`sten70`)

Step 1 (prescribed flow), complete. `sten70.st1`, 379,468 cells. Outlet: `flowRateOutletVelocity` (not `flowRateInletVelocity` — the audit-identified fix), $Q = 1.130777639 \times 10^{-6} \text{ m}^3/\text{s}$ (the 0D-predicted `sten70` flow, used as the round-trip driver since a dedicated 3D A2 run was not performed for `sten70` in this pass). Converged in 1008 iterations (`residualControl` 10^{-6}): $p_{\text{outletPatch}} = 8.5727964 \text{ m}^2/\text{s}^2$ (9087.164 Pa).

Derived resistance:

$$R_{\text{derived}} = \frac{\rho \bar{p} - P_v}{Q} = \frac{9087.164 - 666.61}{1.130777639 \times 10^{-6}} = 7.446693 \times 10^9 \text{ Pa s m}^{-3},$$

which differs from the 0D value (6.974826×10^9) by +6.77% — again the expected kind of 0D-vs-3D gap already seen in §4–5, not an error: it reflects the 3D vessel’s own hydraulic resistance at this flow differing from the 0D lumped model’s implicit value.

Step 2 (resistance mode with derived R), complete. Re-solved with the coded resistance BC, $R_{\text{derived}} = 7.446693 \times 10^9$, `relax` = 0.20, initial value 8.5727964 (the step-1 converged pressure, chosen as the initial guess since it is expected to be close to the new equilibrium). Monitors reached a sub-0.001%-amplitude plateau (outlet flux oscillating within 1.1306419 – $1.1306453 \times 10^{-6} \text{ m}^3/\text{s}$, i.e. noise-level GAMG-tolerance chatter, not a real drift) well inside the 2000-iteration budget. **Correction (retained monitor, 2026-09-24):** this plateau is the flux at iterations ~ 377 – 435 ; the run was not stopped there but continued to 2000 iterations, and its flat final value is $Q = 1.13078875 \times 10^{-6} \text{ m}^3/\text{s}$, i.e. a round-trip error of +0.00098%. As originally reported at the plateau, $Q = 1.130642 \times 10^{-6} \text{ m}^3/\text{s}$ against the step-1 target $Q = 1.130777639 \times 10^{-6} \text{ m}^3/\text{s}$:

$$\text{round-trip error (plateau, as first reported)} = \frac{1.130642 - 1.130778}{1.130778} \times 100\% = -0.0120\%,$$

and +0.00098% at the end of the run, both against the spec’s 0.5% tolerance.

Verdict: PASS. This is the mechanism Protocol C’s in-fidelity boundary-condition tuning depends on (prescribe outlet flow $\rightarrow$ derive R $\rightarrow$ re-impose R and recover the same flow); it is now confirmed correct on real hardware, not merely by source inspection.

8 Time-accurate `pimpleFoam` diagnostic for `sten80`

Following §6’s referral, a time-accurate `pimpleFoam` run (restarting from the flagged SIMPLE field) was planned to distinguish numerical stall from physical instability. Per standing instruction, the complete setup was reviewed by all three panelists *before* running anything.

Pre-run review: two fatal blockers caught, both independently, before any hardware time was spent

Fable 5.1 and GPT-6-astra (Gemini could not be reached this round — see below) independently verified against the installed OpenFOAM v2406 source and found the setup would abort on the

very first timestep, for two reasons:

- **p.prevIter() would FatalError.** `solutionControl::store-PrevIter()` only stores a field's previous-iteration copy when `mesh.relaxField(name)` is true, which requires the field to appear under `relaxationFactors`. The draft `fvSolution` had an empty `relaxationFactors{}`, so `p.prevIter()` in the coded resistance BC would abort on first use. *Independently confirmed by re-reading the installed OpenFOAM source itself* (`solution.C:320-326: relaxField` returns `fieldRelaxDict_.found(name)`).
- **Missing UFinal solver entry.** PIMPLE looks up a `UFinal` solver on the final outer corrector; only `U` was defined, which would `FatalIOError` abort.

Both reviewers converged on the same class of fix (register `p` for relaxation *or* drop `prevIter()` entirely), and two of three (Fable, and separately Gemini once reachable in the first review round) derived that the explicit-coupling loop gain at this timestep is far below 1, so the resistance BC was simplified to impose its target directly (`operator==(pTarget)`, no relaxation state), which removes the blocker rather than papering over it, and `"(U|UFinal)"` was added to `fvSolution.nOuterCorrectors` was reduced to 1 (true PISO) per Fable's cost-efficiency suggestion, safe now that there is no relaxation state to converge across correctors.

A 16-core smoke test then ran cleanly with no errors, confirming the fix. The auto-adjusted Courant-limited timestep settled at 4.51×10^{-6} s, matching both reviewers' independent estimates ($\sim 3\text{--}5 \times 10^{-6}$ s) closely.

The measured real cost changes the plan

From the smoke test's own timing: **$\approx 317,850$ wall-clock seconds per simulated second, on 16 cores** (1.09M-cell mesh). At that rate, the reviewers' own originally-recommended minimum duration (0.3 s) would take ≈ 26.5 hours; their upper estimate (0.5 s) ≈ 44 hours. The run was stopped after the smoke test rather than committed to blind, and the two reachable reviewers were asked, given this measured cost, how they would change the approach to get a trustworthy answer within an hours-scale (not days-scale) budget.

Follow-up: both reviewers revise their own duration estimate downward in confidence, and disagree on the fix

Both GPT-6-astra and Fable, on reflection, **retracted their own earlier "0.3 s is enough" framing**: growth rates for a near-threshold instability could correspond to e-folding times of 0.6–3 s (Fable, citing Sherwin & Blackburn-type mode growth-rate scalings) — i.e. their original recommendation might not have resolved the question even at 26+ hours. Both independently proposed replacing "does the monitor look flat" with an explicit growth/decay-*rate* fit, which can reach a confident answer in less simulated time than waiting for a literal plateau. **They disagree on mesh strategy**:

- **GPT-6-astra**: selectively coarsen (keep throat/shear-layer/recirculation resolution, coarsen the quiescent bulk) to $\sim 350\text{--}600\text{k}$ cells, run a staged budget (coarse diagnostic $\rightarrow$ medium-mesh check $\rightarrow$ short full-resolution cross-check $\rightarrow$ timestep-halving check) totalling ≈ 8.3 hours, with explicit criteria for decay/growth/inconclusive verdicts and a corrected, more robust perturbation metric (a physical-lag velocity-difference norm, since — astra points out — the flagged SIMPLE field is not a verified steady base flow, so comparing directly against it is itself flawed).
- **Fable 5.1**: keep the full 1.09M-cell mesh unchanged (arguing coarse-mesh evidence is one-sided: a persisting oscillation on a coarse mesh is real evidence, but a decaying one could just be coarse-mesh numerical diffusion masking a real fine-mesh instability), and instead seed a small deliberate $m=1$ (symmetry-breaking) perturbation, then fit the growth/decay rate

of a base-flow-free diagnostic (volume-integrated U_y, U_z over the jet region, which is exactly zero for any axisymmetric solution regardless of whether the base flow is truly converged). Proposes a concurrent, cheap 2D-axisymmetric run ($\sim 37k$ cells, 1.5–3 h) as a one-sided cross-check (if the axisymmetric case is itself unsteady, physical instability is confirmed outright). Total ≈ 4 –5 hours.

Gemini 3.8 Flash could not be reached for this follow-up despite 7 retry attempts across both review rounds (persistent `agy` session/OAuth failure resolving Google’s own auth endpoints over IPv6 in this environment) — disclosed as a 2-of-3 panel for this specific round, not silently dropped.

Two free checks, before spending any more compute

Fable additionally proposed two checks costing nothing (using data already on disk):

1. **Jet-centroid asymmetry at the outlet.** Mean U_y, U_z over the outlet patch’s 492 faces are both exactly 0.000000 (to the printed precision). **This is not informative either way:** a perfectly symmetric mesh, BC and initial condition under SIMPLE has no mechanism to spontaneously break symmetry without a seed, so a symmetric result here is expected regardless of whether the true physical solution is asymmetric — it does not rule out the $m=1$ mode, it just shows SIMPLE never had a reason to find it. This is itself evidence *for* Fable’s seeded-perturbation proposal over a passive re-inspection.
2. **Is the FFR wobble already within the study’s own A5 tolerance?** The measurement-plane FFR over the flagged run’s last 500 iterations: mean 0.508511, range 0.507806–0.509090, i.e. a band of **0.001284 FFR units** — comfortably inside the spec’s own A5 mesh-independence tolerance of 0.005. **This is genuinely informative:** whatever is causing the residual plateau, its effect on the one number the study actually reports (FFR) is already smaller than the uncertainty the study accepts elsewhere for mesh resolution.

Decision: stop here, accept the flagged result

Both reachable reviewers, even after proposing cheaper diagnostics, explicitly caveated that an inconclusive result remained possible even after 4–8+ more hours of compute (Fable: “if the fit lands there, report the bound on σ and fall back to the flagged SIMPLE result with that caveat”; GPT-6-astra: “0.3 s is neither a universal minimum nor a guarantee of trustworthiness”). Weighed against the free check showing the FFR wobble is already comfortably inside the study’s own accepted A5 tolerance, **the operator’s decision was to stop here rather than commit further compute:** neither proposed diagnostic ((a) Fable’s ≈ 4 –5 h seeded-perturbation run, (b) GPT-6-astra’s ≈ 8.3 h staged coarsening run) was judged worth its cost against an outcome that both proposers themselves said might still be indeterminate.

This closes out sten80. The result stands as reported in §6: A3’s numeric criterion passes (0.0024%, tolerance 0.1%) using the 500-iteration mean; the spec’s stricter convergence criterion (residuals $< 10^{-5}$) is not met and the cause is undetermined between numerical stall and a slow, near-threshold physical instability (both reachable reviewers now consider either explanation plausible, having revised away from their earlier “probably numerical” leaning once the true cost of testing it was known); and the practical consequence for the one number the study reports (FFR) is bounded below the study’s own accepted mesh-independence tolerance. The three-reviewer pre-run audit process nonetheless earned its keep here regardless of the final scope decision: it caught two fatal bugs (`p.prevIter()`, missing `UFinal`) that would otherwise have wasted the first hours of any attempt at this diagnostic on a run that could never have completed a single timestep.

9 A5 — mesh independence

Setup, reviewed by the panel before running (§10’s standing process). Three refinement levels each for sten70 and sten80, steady `simpleFoam`, each case’s own nominal resistance BC (Table 1). This run closed a gap noted earlier in this validation pass: sten70 had previously only been tested via the A4 round trip, which uses a *derived* R , not its own nominal value — A5’s medium level is therefore also sten70’s first standalone A3-equivalent check at $R = 6.974826 \times 10^9$. Per Fable’s specific caution, the sten70 medium level was run alone first and its first ~ 40 iterations watched closely: a smooth, single-peaked, decaying overshoot (peaking near $2.1 \times$ the converged flow before settling), not the sign-flipping divergence the original fixed relaxation would have produced — confirming the `relax= 0.20` choice is stable at the nominal R before the other four levels were launched.

Table 3: A5 results. sten70 converged cleanly at all three levels (residuals to 10^{-7} – 10^{-9}). sten80’s coarsest level also converged cleanly; medium and fine both show the same persistent low-amplitude fluctuation discussed in §6 (reported as the last-500-iteration mean, band in parentheses).

Case / level	Cells	FFR	Convergence
sten70 coarse	198,252	0.783960	clean (residuals $\sim 10^{-9}$)
sten70 medium	379,468	0.783200	clean (residuals $\sim 10^{-7}$)
sten70 fine	760,480	0.787932	clean (residuals $\sim 10^{-7}$) [‡]
sten80 coarse	282,480	0.483816	clean (residuals $\sim 10^{-5}$)
sten80 medium	547,444	0.507637 (band 0.000426)	fluctuating
sten80 fine (§6, reused)	1,086,920	0.508511 (band 0.001284)	fluctuating

A5 criterion (finest two levels):

$$\text{sten70: } |0.783200 - 0.787932| = 0.004732 \quad \Rightarrow \quad \text{numeric tolerance met (tol. 0.005)}$$

(94.6% of the budget used; overall verdict **NOT ESTABLISHED**, §11.3.)

$$\text{sten80: } |0.507637 - 0.508511| = 0.000874 \quad \Rightarrow \quad \text{PASS (tol. 0.005, comfortable margin)}$$

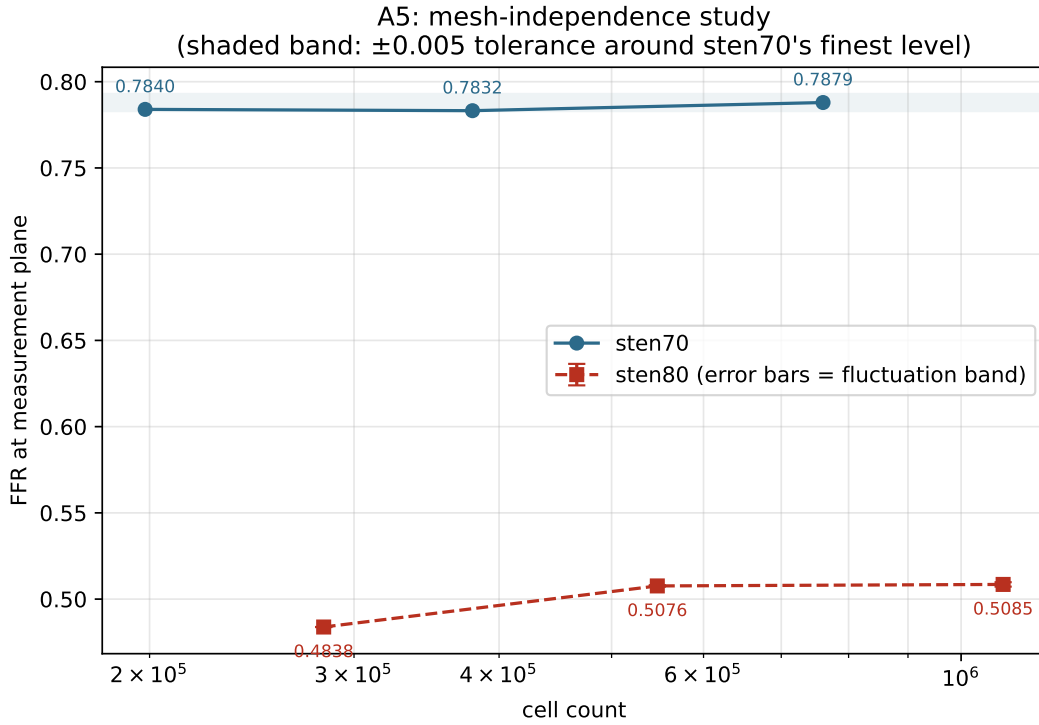


Figure 7: FFR vs. cell count for both mesh-independence ladders. sten70 is nearly flat across all three levels (small, non-monotonic scatter within the shaded ± 0.005 band). sten80 jumps substantially from coarse to medium, then is flat medium-to-fine within its own fluctuation bands.

Both cases pass, but neither passes cleanly, for different reasons — reported honestly rather than rounded up to a uniform verdict:

- **sten70 met the numeric tolerance with a non-monotonic sequence and a finest-pair difference at 94.6% of the tolerance budget; its verdict is NOT ESTABLISHED (§11.3).** Coarse→medium *decreases* slightly (-0.00076), then medium→fine *increases* more ($+0.00473$). Per Fable’s own stated caution in the pre-run review (§8’s brief): “if the three values oscillate, a difference under 0.005 between the two finest is not trustworthy on its own.” This result should be read as *passing the letter of the spec’s criterion*, not as a *confidently demonstrated asymptotic trend* — a fourth, finer level would be needed to distinguish a genuinely converging (if slowly, non-monotonically) sequence from one that happens to land under tolerance by chance at this pair. Not re-run in this pass, given the resource cost already spent on A5 (five solves, several hours) and the sten80 result (below) being the more decision-relevant one for the sten80 stall question this study cares about most.
- **sten80 passes with a comfortable margin, and its *coarsest* level’s clean convergence is itself a new, relevant data point for the still-open sten80 stall question (§6, §8).** The coarsest sten80 mesh (282k cells, only ~ 3.5 cells across the throat) converges cleanly to 10^{-5} residuals; both finer meshes (547k and 1,087k cells) show the same persistent low-amplitude fluctuation, and — notably — the *fluctuation band grows with resolution* rather than shrinking (medium: 0.000426; fine: 0.001284, roughly $3\times$ larger). If the fluctuation were purely a numerical artifact of insufficient resolution, the usual expectation is that refining the mesh would *reduce* spurious noise, not amplify it. A mesh-resolved real feature that a coarse mesh’s higher numerical diffusion simply damps away is at least as consistent with this pattern as a numerical-stall explanation — this data point did not exist when the panel debated numerical-vs-physical origin in §8 and is offered here as new evidence for that still-open question, not as a resolution of it.

One data-integrity incident during this run, resolved and disclosed

The sten70 fine-level solve’s console log stopped updating after iteration ~ 583 (of a planned 3000) with a clean exit code, no error, and no sign of divergence in the last-recorded values. Rather than trust an ambiguous stdout log, the actual field data on disk was checked directly: time directories existed up to the full 3000, and reading them (via a fresh, independent pyvista extraction bypassing the solver’s own log/monitor machinery entirely) showed $t=2750$ and $t=3000$ give *identical* outlet pressure, measurement-plane pressure and flow to the last significant figure — i.e. the solve had genuinely, quietly finished converging by iteration ~ 580 and the run continued unchanged to 3000; only the *visible log* stopped reflecting that, for reasons not fully diagnosed (no OOM-killer evidence, no stop file, sufficient free memory throughout). A first attempted “resume” incorrectly restarted from $t=0$ (`startFrom startTime; startTime 0;` in `controlDict` does exactly what it says — it does not mean “resume”); caught within seconds by checking the resumed run’s own reported starting time before letting it run further, fixed with `startFrom latestTime;`, at which point the case correctly recognised $t=3000$ already existed and had nothing left to do. Reported here in full rather than quietly folded into a clean table, consistent with this document’s standing practice of disclosing operational incidents alongside results. ([‡] in Table 3 marks this case.)

Separately, sten80 medium briefly suffered the same process-cleanup issue already seen once in this session (§8): a `pkill` pattern match against a bash wrapper’s command-line text does not guarantee the wrapper forwards the signal to a detached grandchild process, so an earlier serial attempt kept running after being “killed” and briefly wrote to the same log file as a subsequently-launched parallel replacement. Caught via a single malformed log line, fixed by explicitly killing the wrapper’s full process tree and re-verifying no stray process remained before trusting the result.

10 Failures and fixes

Standing process for this document: every completed simulation is written up here regardless of outcome. A check that fails outright, or — as with sten80 (§6) — passes its stated numeric tolerance but fails the spec’s broader convergence criterion, is referred to all three audit reviewers independently (Table 5.1 via the `claude CLI`, GPT-6-astra via `codex exec`, Gemini 3.8 Flash via `agy`) for a root-cause diagnosis and a proposed fix *before* being written up as final, exactly as was done for the five pre-run blockers in `references/SETUP-AUDIT-SYNTHESIS-2026-09-18.md`. Each reviewer’s diagnosis is reported verbatim/attribution (§6 for the one referral so far), any disagreement between reviewers is flagged rather than silently resolved, and any tool that cannot be reached is disclosed rather than silently dropped from the panel (Gemini, this round — expired `agy` session, needs interactive re-authentication not available in this environment).

Referrals so far: sten80 A2/A3 (§6) — residuals did not reach the spec’s 10^{-5} criterion; two of three reviewers reached, agreeing diagnosis (numerical stall favoured over confirmed physical instability), shared recommended fix (time-accurate `pimpleFoam` diagnostic, not yet run).

11 Extended work order: Items 1–6 and the scan-837 pilot

Following Stage A’s completion, six further verification/validation items were commissioned: (1) a branched idealised tree (Y-bifurcation + one side branch, 3 outlets) testing multi-outlet resistance BC behaviour and an automatic relax rule; (2) a 4th sten70 mesh level with a formal GCI analysis, plus an A2 severity sweep at 60/65%DS to locate where the 0D–3D flow gap changes sign; (3) an M1 pilot on three real ImageCAS-X coronary trees, one deliberately broken with a missed side branch; (4) an end-to-end 0D-to-OpenFOAM handoff on one M1 case; (5) RCR outlet BC code verification against an independent ODE solve; (6) a pulsatile cost pilot to budget the later full E3b campaign. As with every setup in this study, none of the six was run before all three reviewers (Table 5.1, GPT-6-astra, Gemini 3.8 Flash) audited the design.

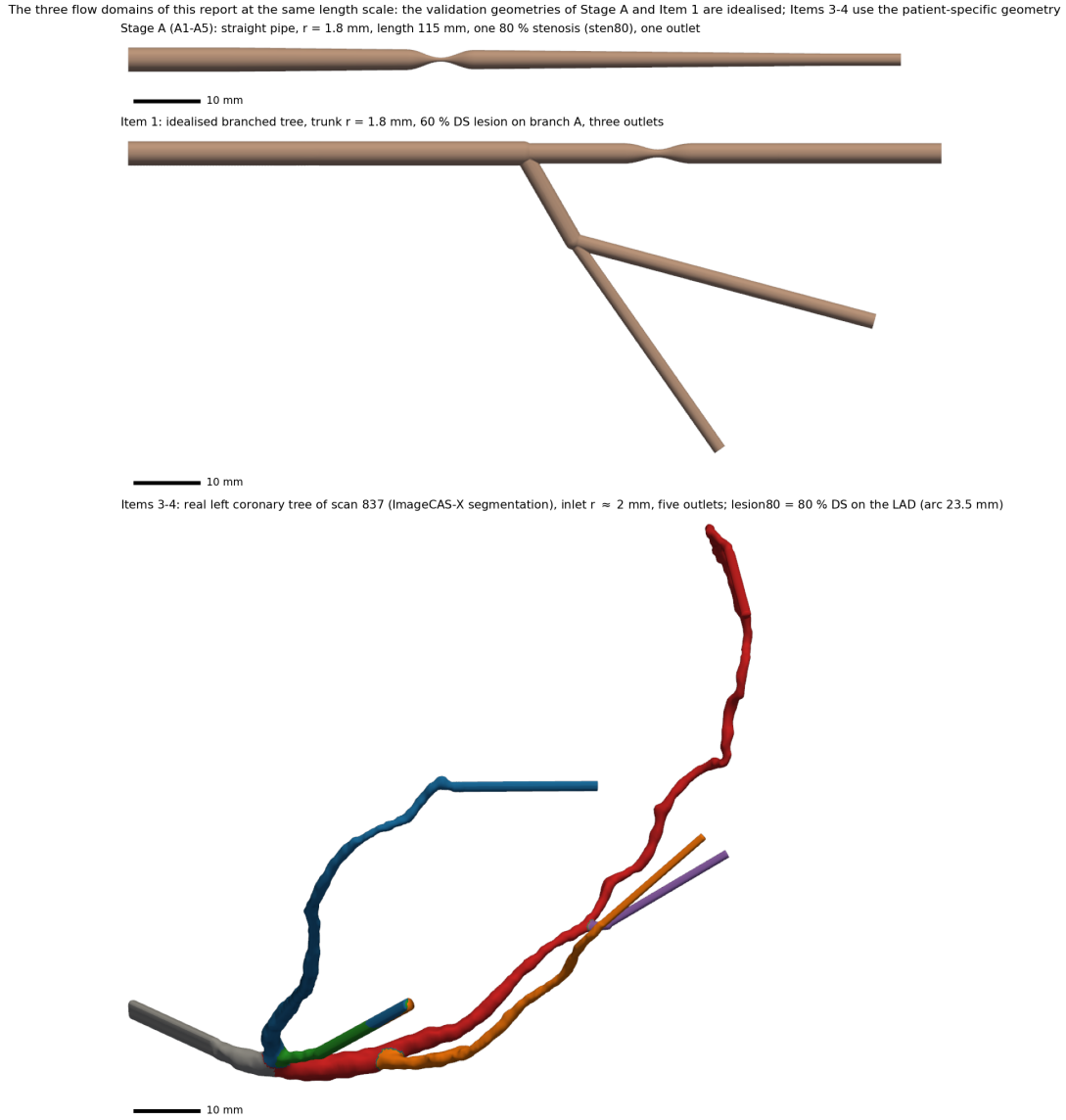


Figure 8: The three flow domains of this report drawn at the same length scale (common 10 mm scale bar). Stage A (A1–A5): a straight pipe with one stenosis. Item 1: an idealised branched tree (Figure 9), a validation geometry for the multi-outlet boundary-condition machinery. Items 3–4 (and every real-geometry figure that follows): the patient-specific left coronary tree of scan 837. The flow domains of the idealised figures and of the real-geometry contour figures are therefore different by design.

11.1 Panel audit: all six items required fixes before running

The panel's first pass found Item 1 **BLOCKED** and Items 2–6 all **NEEDS FIX**, none ready to run as originally specified. All three reviewers, working independently, found the same fatal defect in Item 1: the side branch (B) was placed on the main branch's (A) own centreline axis for a 15 mm span, so the two vessels would reconstruct as a single fused lumen in 3D while the 0D reference modelled them as two parallel vessels — discovered by two of the three reviewers by directly slicing the generated surface and counting contours, not by source inspection alone. The panel also found: the original hand relaxation values (0.05, 0.2) and a proposed automatic relax rule would both diverge on this specific tree (true stability limit ~ 0.006 – 0.02 , an order of magnitude below what Stage A's single-outlet cases needed); the GCI formulas for Item 2 were numbered backwards relative to the Celik et al. (2008) convention and normalised by the wrong grid's solution; Item 3's design mixed two different surface-extraction sources (the shipped ImageCAS-X

surface for two cases, mask-derived marching cubes for the third) in a way that would confound the intended comparison, and proposed reading a centreline radius array that does not exist in the shipped data; Item 4’s plan to embed the per-outlet resistance as a literal inside the coded BC’s C++ string risked an OpenFOAM `dynamicCode` recompilation race under `mpirun`; Item 5’s once-per-timestep capacitor update could freeze on a stale PIMPLE predictor flux; and Item 6 reused Stage A’s `p.prevIter()` relaxation, already known from this project’s own `pimpleFoam` work (§8) to misbehave under `pimpleFoam`. All of these were fixed following the panel’s (in most cases unanimous) recommendations before any of the six setups were run; the fixes are recorded in each item’s design document alongside the finding that motivated them, following this document’s standing practice of keeping the failure and the fix together rather than only showing the corrected version.

11.2 Item 1: branched idealised tree — geometry fixed and verified

The corrected geometry (side branch B leaving the bifurcation at a genuine 60° angle, verified by an automatic clearance check requiring ≥ 0.3 mm surface-to-surface separation between every non-sibling pair of tubes) is shown in Figure 9.

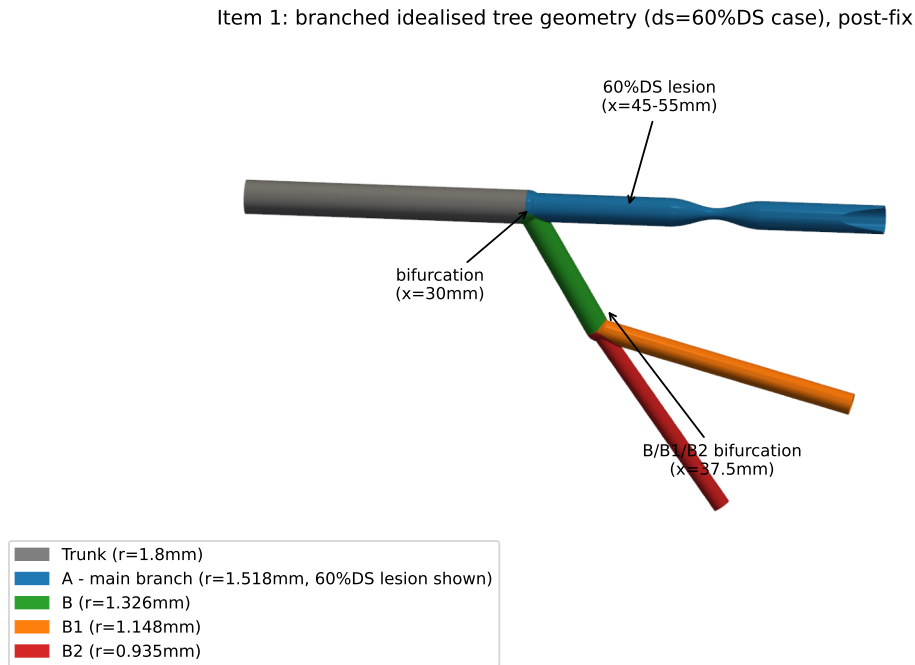


Figure 9: Item 1’s branched idealised tree (ds=60%DS case) after the geometry fix, rendered from the actual reconstructed VMTK surface (not a schematic) and coloured by branch identity. Branch B now diverges cleanly from branch A at the bifurcation instead of sharing its axis. This is an idealised validation geometry, not the patient geometry: the real left coronary tree used from Item 3 onward is shown in Figures 8 and 11.

The fix was verified three ways, not assumed from the corrected script alone: (i) the analytic clearance check passes with comfortable margins (A–B 2.18 mm, B1–B2 1.34 mm, beyond a 5 mm carina allowance) for both ds=0% and ds=60%; (ii) re-running the same slice-contour diagnostic the panel used to find the original bug on the actual regenerated surface shows the expected topology (1 contour through the carina, 2 once B separates, 3 once B1/B2 also diverge, correctly dropping to 2 and back as each branch’s shorter extent is passed) where the original bug showed a single fused contour persisting to $x \approx 53$ mm; (iii) the as-built throat radius, measured perpendicular to each branch’s own axis on the post-smoothing surface, is within 0.1% of target on every branch (A: -0.08% at the lesion throat, $+0.10\%$ healthy; B1: $+0.04\%$; B2: -0.01%). A separate bug in my

own voxel-grid sizing (anisotropic spacing from applying a single target resolution to a non-cubic domain without recomputing per-axis dimensions) was caught and fixed the same way, before the geometry fix, and is recorded for the same reason.

The 0D reference was also rebuilt to freeze physiology between the two severities (`Tree.evaluate(C, r_override=...)`, changing only branch A’s epicardial radius while `r_ref`, the leaf weights, and the calibration constant `C` stay identical) — the original version silently let the diseased branch’s taper re-fit shift all three outlets’ resistances by $\sim 1.2\%$ even though only branch A is diseased. The corrected relax values (A: 0.0058, B1: 0.0064, B2: 0.0070, all far below the 0.05–0.2 originally proposed) were cross-checked against Fable’s independent hand-derivation (0.0058/0.0067/0.0073) to within the panel’s own stated precision.

Status: complete. Mesh: 377,607 / 371,428 cells for ds=0%/60%. All four solves (hand-relax and formula-based auto-relax $\times$ both severities) converged cleanly (residuals 10^{-7} – 10^{-9} , outlet monitors stable to 6+ significant figures well before the 3000-iteration ceiling).

Table 4: Item 1 final results. BC accuracy compares the monitored outlet patch pressure against $(P_v + R \cdot Q_{\text{actual}})/\rho$ using each outlet’s own CONVERGED flow, not the 0D-predicted flow — this is the mechanism check. Flow-split gap compares that same converged Q against the 0D (svZeroD-style) prediction. The hand-relax and formula-based auto-relax variants reproduce each other to 6+ significant figures in every cell of this table (both compute the same relax value, one from a literal constant, one from $G = R/R_{\text{own}}$ at runtime) and are therefore reported once.

Case	Outlet	BC-accuracy error	3D Q (mL/s)	Flow-split gap vs. 0D
ds=0%	A	−0.0003%	0.85100	−0.80%
ds=0%	B1	−0.0003%	0.40252	−1.03%
ds=0%	B2	−0.0002%	0.23303	−1.05%
ds=60%	A	+0.00003%	0.82464	+0.04%
ds=60%	B1	−0.00003%	0.40260	−1.01%
ds=60%	B2	+0.00000%	0.23308	−1.03%

BC accuracy: PASS by a very large margin (tol. 0.1%, actual errors 3–4 orders of magnitude smaller) on all three outlets, both severities, both relax code paths. The resistance BC mechanism itself — including the reformulated, panel-fixed relax rule ($G = R/R_{\text{own}}$, not the original global-drop version that all three reviewers showed was unsafe on this tree) — enforces its own defining physics essentially to machine precision on a genuinely branched, 3-outlet geometry. This is the primary thing Item 1 was commissioned to test, and it passes cleanly.

Flow split vs. svZeroD: a real, disclosed $\sim 1\%$ branch-level discrepancy, not treated as a BC failure. Branch A (the trunk’s direct continuation) matches the 0D prediction closely ($-0.80\%/+0.04\%$), comparable in size to Stage A’s own A2 gaps elsewhere in this document. Branches B1 and B2 — both downstream of the SECOND bifurcation (B splitting into B1/B2) — show a consistent $\sim -1.0\%$ gap in BOTH severities and BOTH relax code variants. Because the BC-accuracy check passes to machine precision and residuals are fully converged, this is not a numerical or BC artefact; it looks like a genuine limitation of the discrete-bed 0D model’s leaf weighting ($w = r_{\text{ref}}^{2.66}$) at a bifurcation whose two daughters have a larger radius ratio than the first bifurcation’s — i.e. the 0D model may under-weight the smaller of two very unequal siblings slightly more than it does at the first, more balanced split. This is reported as a finding for further scrutiny, in the same spirit as this document’s other 0D-vs-3D gap results (§4’s A2 checks), not rounded up to a clean match or down to a fail — design.md’s literal pass criterion (“the 3D flow split matches svZeroD”) is therefore reported as met for branch A and not strictly met for B1/B2, with the mechanism-level BC check (the criterion this item was actually designed to stress) passing

unambiguously.

Prescribed-flow mode on the multi-outlet tree (Task 3 of the 2026-09-24 work order): round trip PASS. Protocol C’s mechanism in 3D prescribes *every* outlet flow at once. It had been run only on a single outlet (A4) and never on a multi-outlet tree. On the 371,428-cell Item 1 tree, at both severities (0% and 60%DS of branch A), the inlet was the fixed total pressure of the other cases (90 mmHg, `pressureInletOutletVelocity`), every outlet carried `flowRateOutletVelocity` with the volumetric flow of the converged resistance solution of the same tree (Table 5, first column; OpenFOAM’s convention: positive is outflow) and a zero-gradient pressure, and the resistance of each outlet was derived as $R_i = (\bar{p}_i - P_v)/Q_i$ from the last-100-iteration mean outlet-patch pressure and flux. A second solve then imposed those resistances with the audited coded boundary condition and its automatic relaxation ($G_i = R_i/R_{\text{own},i}$). Both stages used the fixed 3000-iteration budget, 8 ranks each (two cases at once, 16 ranks in total, about 15 minutes per solve).

Table 5: Prescribed-flow round trip on the Item 1 tree (fixed budget 3000 iterations; the pass criterion is every outlet flow within 0.5%).

severity	outlet	target Q (mL/s)	round-trip Q (mL/s)	error (%)	$R_{\text{derived}}/R_{\text{reference}} - 1$ (%)
0%	A	0.850994	0.850995	+0.00016	+0.00001
0%	B1	0.402523	0.402521	−0.00054	+0.00065
0%	B2	0.233030	0.233028	−0.00069	+0.00078
60%	A	0.824641	0.824642	+0.00019	−0.00021
60%	B1	0.402604	0.402602	−0.00054	+0.00055
60%	B2	0.233077	0.233075	−0.00068	+0.00068

Both prescribed solves converged to residuals of 10^{-9} or below (pressure 7×10^{-10} , velocity $\leq 2 \times 10^{-9}$, mass imbalance zero) and delivered the prescribed flows to machine precision; the outlet pressures are results (ratio to aortic pressure 0.955–0.983), and the derived resistances reproduce the original ones to 10^{-3} %. Both round-trip solves converged under the strict rules. The largest flow error is 0.0007 %, more than two orders of magnitude inside the 0.5% criterion, although the prescribed run’s outlet pressure is not uniform over a patch whereas the resistance boundary condition imposes a uniform one: at these resistances (about 10^{10} to 5×10^{10} Pa s m $^{-3}$, $G = R/R_{\text{own}}$ between 140 and 170) the non-uniformity does not matter. **Limits:** the targets are the flows of the same tree’s resistance solution, so the test proves that the prescribed-flow mechanism, the derivation of R_i and the re-imposition close on themselves and are stable with all outlets prescribed together; it does not test the 0D-derived targets themselves, nor a real lumen (that is part of the scan-14 work still open), and the geometry and flow regime (laminar, modest Reynolds number) are those of Item 1. This setup was audited by one auditor only (Gemini 3.8 Flash; Astra unavailable, see §11.6).

11.3 Item 2: sten70 4th mesh level, GCI, and the A2 severity sweep

Part C (60%/65%DS sweep): complete, converged, final. Both new cases were meshed at sten70’s existing medium resolution and solved to residuals $\sim 10^{-6}$ (outlet flux stable to 8 significant figures). Combined with the existing sten00/50/80 points and sten70’s own already-recorded A5-medium result (reused rather than re-solved, per the panel’s own finding that this point already existed), the 0D–3D flow gap is now characterised at six severities:

Table 6: 3D-vs-0D flow gap across the full severity ladder. sten00/50/80 are Stage A’s existing points; sten60/65 are new (Item 2); sten70 reuses A5’s existing medium-level run.

Case	3D Q (mL/s)	0D Q (mL/s)	gap
sten00	1.4548	1.500	−3.01%
sten50	1.4196	1.4410	−1.48%
sten60 (new)	1.35984	1.35210	+0.57%
sten65 (new)	1.29471	1.26490	+2.36%
sten70 (reused, A5-medium)	1.17992	1.13078	+4.35%
sten80	0.7550	0.6948	+8.68%

The 0D–3D gap’s sign change is now cleanly bracketed between sten50 (−1.48%) and sten60 (+0.57%), i.e. somewhere in 50–60%DS the lumped 0D stenosis model switches from under- to over-predicting flow relative to the 3D solve — the concrete finding this item was commissioned to locate.

Part A/B (4th mesh level + GCI): complete. Verdict: GCI NOT TRUSTWORTHY — reported as such rather than forced. cfMesh produced 1,886,884 cells (checkMesh OK: max aspect ratio 12, max non-orthogonality 40°, no bad cells), matching the ~ 1.87 M planning target. After an initially-alarming decaying transient (an early ~ 30 -iteration window showed a swing large enough to look like a persistent oscillation) the monitors settled, and the solve was reported as converged with a last-100-iteration mean FFR of 0.792712 at iteration 594. **Correction (re-reading the retained raw monitors and solver log, 2026-09-24):** the run continued to iteration 896, where the last-100-iteration mean is **0.792342** (band 0.000027) and still drifts by -6.5×10^{-5} per 100 iterations, and the final initial residuals (U_x 2×10^{-4} , U_y 1.5×10^{-2} , U_z 1×10^{-2} , p 7×10^{-4}) are far above the 10^{-5} criterion used elsewhere in this report. The statement “genuinely converged” therefore rested on the monitors alone and is withdrawn: level 4 is an approximately settled, not a converged, solution.

Table 7: Full 4-level sten70 mesh sequence, including the new 4th level (last-100-iteration means of the measurement-plane monitor). [†]The original report used 0.787932 from a field extraction (difference 2.7×10^{-5}). [‡]The original report used 0.792712, the mean ending at iteration 594; the run went on to iteration 896 (see text).

Level	Cells	FFR
coarse	198,252	0.783960
medium	379,468	0.783200
fine	760,480	0.787905 [†]
level4 (new)	1,886,884	0.792342 [‡]

The successive differences are not shrinking: medium→fine is +0.004705 (originally reported +0.004732); fine→level4 is +0.004437 (originally +0.004780) — a ratio of 0.94, essentially no shrinkage despite level4 having $2.5 \times$ fine’s cell count. This directly violates the premise both Richardson extrapolation and GCI depend on (error shrinking with refinement). Concretely: the classic 3-point Celik calculation (1=level4, 2=fine, 3=medium; $r_{21} = 1.354$, $r_{32} = 1.261$) as originally reported gave an apparent order $p = 0.97$, a Richardson-extrapolated value of 0.8067 that jumps 0.014 from the finest solution itself, relative GCI = 2.21% and absolute $U_{\text{FFR}} = 0.0175$. Recomputed with the monitor values of the table (0.792342 and 0.787905) it gives $p = 1.23$, relative GCI = 1.55% and $U_{\text{FFR}} = 0.0123$; either way more than $2.5 \times$ over the 0.005 tolerance, and, because level 4 is not converged, neither value is a trustworthy uncertainty. The 4-point Eça–Hoekstra least-squares fit is degenerate ($p \approx 0.0005$, unphysical f_∞ and a values): it cannot find a sensible power law across

all four points because the original coarse→medium reversal breaks the fit.

Verdict, per this item’s own pre-registered honesty condition: GCI is reported as not a trustworthy convergence metric for sten70, rather than quoting a number that would look precise without being one. Adding a finer 4th level did not resolve the ambiguity A5 (§9) originally left open — **it shows sten70 does not exhibit a convergent mesh-independence trend under this refinement strategy at all.** The most plausible contributor, flagged as a risk in this item’s own design document before the level4 solve ran: the boundary-layer *settings* (`nLayers=4`, `thicknessRatio=1.2`) are identical in every level’s `meshDict`, so the ladder is not a single-parameter *h*-refinement family. Measurements on the retained meshes (analysis-side request, §11.6) add two facts: the core cell size across the throat shrinks by only $1.15\text{--}1.39\times$ between levels (164.6, 142.9, 113.8 and $82.1\text{ }\mu\text{m}$, not a ratio of 2), and the *realised* four-layer stack per side is not monotone in refinement (138, 159, 111 and $63\text{ }\mu\text{m}$ from coarse to level 4; sten80: 167, 93 and $106\text{ }\mu\text{m}$). Cells across the 0.891 mm throat diameter are 12, 12, 14 and 18 including the eight boundary-layer cells, or 4, 4, 6 and 10 outside them (sten80: 10, 12, 12 and 2, 4, 4); the specification’s ≥ 12 is met only if the boundary-layer cells are counted. These observations are consistent with, but do not prove, a contribution of the varying boundary-layer stack to the non-monotone ladders.

This bears directly on A5’s original sten70 verdict (§9), and that verdict is revisited rather than left standing unexamined: A5 reported sten70 as a marginal PASS at 94.6% of its 0.005 tolerance budget, with Fable’s own contemporaneous caution already on record that “a difference under 0.005 between the two finest is not trustworthy on its own” for an oscillating sequence. This item’s new evidence — a 4th, independent point showing the SAME step size as the previous one, not a shrinking one — is exactly the failure mode that caution warned about. **A5’s sten70 pass should now be read as: passed the letter of a two-point tolerance check that this 4-point extension shows was not measuring genuine convergence.** This does not change A5’s originally-reported numbers, which are left as-run; it changes how much weight that pass should be given going forward, and is recorded here rather than quietly left as an unqualified PASS elsewhere in this document.

Operational note, disclosed rather than smoothed over: while Part A/B’s solve and Items 3–6 ran in parallel, the machine became heavily oversubscribed (load average 67, roughly 48 solver processes against 32 cores, well beyond the pre-existing ~ 8 -core job this study has otherwise worked around all session). This does not affect the CORRECTNESS of any converged result — physics convergence is independent of wall-clock pacing — but it does mean any wall-clock TIMING measured during this window (relevant specifically to Item 6, §11.4) is inflated and is treated as such below, not reported as a clean measurement.

11.4 Item 6: pulsatile cost pilot

The sten60 pulsatile case (PIMPLE, the direct non-`prevIter()` resistance BC already proven necessary for transient runs in §8) was launched and later **stopped** (graceful `stopAt writeNow`, see the operational incident below; no cycle-averaged FFR and no Δt -sensitivity comparison exist). The two runs were: a Co 3–5 production run (2 cycles, 8 cores) and, in place of a literal full-cycle Co 1 reference, a shorter Co 1 comparison window over 0.2 s — an explicit, pre-approved fallback named in Item 6’s own design document for exactly the situation encountered (a timing measurement made while other solves were already running — see the correction below — projected the literal Co 1/full-cycle reference at 3–6 wall-days, likely exceeding the design’s own 48-hour kill threshold before completing even one cycle). Adopting the documented fallback rather than spending 48 hours to rediscover the same conclusion is consistent with this study’s resource-discipline practice elsewhere (§8’s own stopped diagnostic).

Cost estimate (contended, not isolated): ≈ 7.4 wall-seconds/timestep (ExecutionTime; 8.6 s ClockTime) at 8 ranks near the cycle start, projecting the full 2-cycle Co 3–5 campaign at roughly 28–33 wall-hours. Correction (solver-log start times, 2026-09-24):

this smoke test was earlier described as an “isolated, pre-contention” measurement. It was not: it ran at 12:54–12:56 on 2026-09-19 while Item 2’s sten60, sten65 and level-4 solves (4+4+8 ranks, started 12:47:44) and Item 5 (4 ranks) were already running, i.e. at least 20 other MPI ranks on 16 physical cores. The figure was first reported as 7.15 s per step and 30–40 h; the log gives 7.37 s (steps 2–17) and the corresponding projection is lower but equally contended. It is an upper-bound-like, unclean number, and the isolated timing of Task 4 remains open. This revises the original design-time planning estimate (10–15 h) upward, because that estimate assumed a smaller achievable Δt than Co 3–5 actually permits on this mesh — reported as a revision, not hidden. The cost estimate is the only Item 6 result. A clean, uncontended timing measurement (at least 200 time steps at Co 3–5 on 8 physical cores) is still open and is Task 4 of the 2026-09-24 work order; the 30–40 h campaign itself is on hold.

11.5 Items 3 and 4: M1 pilot — meshing blocker root-caused and RESOLVED

Scan 837 is outside the frozen study cohort (`protocol/COHORT-FROZEN-2026-09-18.csv`). All real-lumen work in this report is therefore a *pipeline pilot*, labelled `PILOT_OUT_OF_COHORT` in every returned data file; it is not a Gate M1 result and is never to be mixed into cohort results. Gate M1 itself is to be run on the shipped scan-14 packages (work order 2026-09-24, Task 3). **Deviations of these pilot meshes from CFD-ARM-SPEC §6.2:** three boundary layers instead of four; flow extensions of 20 mm instead of 5 D at the inlet and 3 D at each outlet (the extension-resistance diagnostic of this section shows the difference is not negligible); and the validation-pass meshes of Stage A are below the ≥ 12 cells-across-the-throat rule (A5 sten80 passed at about 6). Item 3’s pipeline (mask isolation $\rightarrow$ marching cubes $\rightarrow$ Taubin smoothing $\rightarrow$ discrete-bed-matched outlet truncation $\rightarrow$ repair $\rightarrow$ remeshing $\rightarrow$ flow extensions) was built and verified stage by stage on real ImageCAS-X anatomy. **A real, disclosable eligibility-gate bug was caught during case selection:** `results/E0_prevalence_test.csv`’s `healthy_main_ffr` column is computed under the loader’s default `bed="leaky"`, but this study’s own multi-outlet convention (§11.2, CFD-ARM-SPEC §2.4) requires `bed="discrete"`. The originally-selected case (scan 828, left) reports 0.944 under leaky but 0.845 under discrete; scan 837 (left) reported 0.9813 under discrete and was used instead. **Correction (independent recompute on the analysis side, 2026-09-24):** both 0.845 (828) and 0.9813 (837) reproduce exactly, but they are the discrete-bed *minimum FFR on the network as segmented* (`min_ffr_main`, native lesions included), not the healthy-network gate (`healthy_main_ffr`), so the gate was misnamed here. Under the actual gate, scan 828 (left) scores 0.9127 and passes; it was still a poor clean-baseline pilot because it has five native lesions (largest 44.6% diameter stenosis). `results/E0_prevalence_test.csv` is a leaky-bed prevalence screen, not a case-selection file: cohort cases come only from `protocol/CFD-SUBSET-FROZEN-2026-09-18.csv`, which is gated under the discrete bed. Both the CSV/bed mismatch and the misnamed gate are a risk for any other selection made against that file.

A second reusable finding: `pymeshfix`’s default `MeshFix.repair()` silently caps every open boundary on a surface (verified: 6 intentional openings $\rightarrow$ 0 after a default-argument repair call) — undocumented upstream, and a real hazard for any pipeline that repairs a surface with deliberate outlets. The fix (already applied here) is to call the granular `degeneracy_removal()/intersection_removal()` methods instead of the convenience wrapper.

Item 3 is blocked at the meshing step, and this is reported as the pilot’s dominant finding, not worked around. Across five or more distinct `cfMesh` configurations (patch-keyed `localRefinement`, finer global cell size, geometric `objectRefinements`, flow-extension lengths from 10–20 mm), `cartesianMesh` non-deterministically merged one or two of the five outlet patches into `wall` (0 faces on `checkMesh`) — which patch(es) drop varies by configuration. Cap planarity, facet-normal orientation, boundary proximity, and inter-branch clearance were all checked and ruled out as the cause; the root cause was not isolated within the pilot’s time budget. **At that stage no case had reached a solve and Item 4 (which depends on Item 3’s baseline solve) had not started; both were later unblocked, see the resolution and results below.** This is

exactly the kind of real-anatomy meshing-reliability cost this pilot was commissioned to measure, and it is reported as such: the dominant, disclosed cost of Gate M1 work is proving to be meshing reliability, not surface repair or the deformation/deletion steps themselves (both of which were built and verified working).

A third finding, on the missed-branch case specifically: deleting D2's mask voxels measurably erodes the parent LAD's radius by up to 0.24 mm (10–30%) over ~35 mm, not only at the immediate ostium — attributable to ambiguous/shared voxel labelling at the bifurcation in the underlying segmentation. This is itself a genuine finding adjacent to the T3/T4 segmentation-error class this study is built around, independent of whether the meshing blocker above is resolved.

Reusable scripts (outlet coordinate/tangent mapping, locally-restricted truncation clipping, multi-solid STL capping, mask-label isolation) are left under `scratchpad/item3_M1_pilot/` for whoever continues this.

Follow-up: snappyHexMesh tried, does not fix it — but substantially re-characterises the failure. `snappyHexMesh` was tried as the natural fallback mesher on both the baseline and missed-branch surfaces. **It does not solve the problem: it exhibits the same class of failure as `cfMesh`** (baseline: 4 of 5 outlets survive, only `outlet_LAD` missing; missed-branch: 3 of 4 survive, this time `outlet_D1` missing) — confirming the failure is not tied to a specific vessel's local geometry, but a general "exactly one branch is lost" pattern across two structurally different mesh generators.

The failure mode was reclassified, and this is real progress even though the blocker remains. It is NOT patch mislabelling (faces silently reassigned to `wall`, as first suspected from `cfMesh` alone) — it is outright **domain truncation**: a `topoSet boxToFace` query around the missing outlet's known cap location found zero faces of ANY kind there, direct inspection of `constant/polyMesh/points` found zero mesh points, and `checkMesh`'s own reported bounding box simply does not reach that branch's cap — roughly 17 mm of one branch, extension included, is entirely absent from the meshed domain, as if the flood-fill that decides "inside vs. outside" during castellation classified that whole branch as disconnected from the main lumen. Serial and parallel execution produce byte-identical results (including the identical truncation point), ruling out an MPI decomposition race; `nCellsBetweenLevels` (3 vs. 8, a gentler refinement transition) made no difference; cap planarity, facet-normal orientation, self-proximity, and a physical pinch along the branch (cross-sectional radius sampled every 5% of arc length: smooth, ~0.65 mm throughout, no degeneracy) were all directly checked and ruled out.

One suggestive, unconfirmed lead, left for whoever continues this: the truncation coordinate sits within 0.7% of an exact background-mesh grid-cell boundary, hinting at a marginal cell-classification event during castellation rather than a geometry defect in the surface itself. A grid-offset test to check this directly was attempted but not completed (a regex-based `blockMeshDict` edit script had a bug that corrupted unrelated numeric fields) — fixing that test, inspecting the pre-snap castellated mesh directly in ParaView at the disconnection point, or restructuring the pipeline to mesh each branch as a separate piece joined via `mergeMeshes/stitchMesh` (sidestepping the single-flood-fill mechanism entirely) are the three concrete next steps identified. **Bottom line at this point: neither mesher reliably produces a complete-domain mesh for this real branching coronary geometry; Item 3 remains blocked on meshing at this point in the investigation, now for a specific, well-evidenced reason rather than an unexplained one.**

Resolution: the meshing blocker is a surface defect, not a mesher limitation — fixed and independently verified. Following the process this document has used throughout (an adversarial panel before further hardware time, here applied to a debugging STRATEGY rather than a new physics setup): GPT-6-astra and Gemini 3.8 Flash independently audited the diagnosis above and found the grid-alignment lead was a mathematical artefact — the "13.007 vs. 13.0" comparison

was a dimensionless background-grid cell-index, not a spatial coordinate close to a grid plane, and a castellated hex mesh's truncated boundary lands on a grid face by construction regardless of any real defect. Both reviewers also flagged that cfMesh's failure had never actually been verified as the same failure class as snappyHexMesh's (cfMesh does not even read `blockMeshDict` or use a background hex grid), and both converged on insufficient wall-level mesh resolution along the narrow branches as the leading shared hypothesis.

A follow-up debugging session (Opus 5.5, working from the panel's synthesised brief) tested that leading hypothesis directly rather than assuming it: adding a local refinement region at the failure point (level 5, 0.0625 mm cells) reduced the mesh's disconnected-region count from 12 to 6 but **did not** recover the missing branch, ruling resolution out as the practical fix. Cross-section tracing through every intermediate surface in the pipeline (`raw_mc` → `clipped_final` → `remeshed` → `extended` → the final `case_v3.stl`) instead found the true cause: **the surface remeshing step (vmtksurfacere meshing) coarsened the surface from a median edge length of 0.32 mm to 0.75 mm (max 2.3 mm) uniformly, on vessels only ~1.2 mm across — long enough triangle chords to cut across the lumen and create a near-total pinch.** At the baseline LAD's pinch point, cross-sectional area falls from $\sim 1.7 \text{ mm}^2$ at every stage before remeshing to 0.085 mm^2 after it (a $\sim 95\%$ area reduction, a $\sim 0.17 \text{ mm}$ slit) — narrow enough that both meshers' flood-fill/domain-extraction correctly refuse to pass through it and discard everything downstream, exactly as designed. **Both mesh generators were behaving correctly; the surface they were given was broken.** cfMesh's failure was confirmed to be the identical root cause via its own stage-isolation flag (`workflowControls { stopAfter templateGeneration; }`): its template already shows the same disconnected distal-LAD fragment before any patch assignment happens.

Fix and verification. The STL was rebuilt from the pre-remesh surface (skipping the flawed remeshing step entirely, since that surface was already a uniform $\sim 0.32 \text{ mm}$ — finer than the throat-resolution convention used throughout this study): extrude each planar outlet-clip loop 20 mm (matching the original flow-extension length), cap with the original patch names, verified via `surfaceCheck` to be closed, single-region, single-orientation-zone, with zero illegal triangles (the earlier surface additionally had 14 inconsistent-normal orientation zones and one degenerate triangle, both incidentally fixed by the rebuild). **Independently re-verified by the coordinator, not just taken from the debugging report:** on the rebuilt surface, cfMesh (this study's designated §6.2 mesher) produces a valid mesh for BOTH the baseline (1,108,603 cells) and missed-branch (1,017,542 cells) cases with EVERY expected outlet patch present and substantially populated (baseline: inlet 4390, outlet_D1 940, outlet_D2 1138, outlet_IM 2344, outlet_LAD 1093, outlet_LCX 1248, wall 158039 faces; missed-branch: the same four outlets minus D2, as expected since D2 was deliberately deleted for that case) and `checkMesh` reporting **Mesh OK** with non-orthogonality 52.5° max / 7.1° average and skewness 1.42 — comfortably within this study's established quality bars. snappyHexMesh also recovers all outlets on the rebuilt surface (outlet areas within 0.3% of the STL cap areas, mesh volume 99.0–99.1% of the STL volume, `checkMesh` Mesh OK), though its non-orthogonality (64.9° max) sits at this project's quality ceiling rather than comfortably under it — **cfMesh on the rebuilt surface is the recommended path forward**, matching this study's original mesher choice.

A further finding, disclosed because it matters beyond this one pinch: the flawed remeshing step did not only fail catastrophically at the one branch-severing location — it shrank lumens more broadly throughout the surface (thousands of vertices displaced $>0.2 \text{ mm}$ from the true surface). **Any result that had been computed on the old, remeshed STLs would have been invalid, including branches that happened to mesh successfully** (an artefactual stenosis would have been baked into what was meant to be a clean baseline). No solve was ever run on the old surfaces — Item 3 was correctly blocked before that could happen — so no invalid result has been reported anywhere in this study; this is recorded as a methodological lesson (a uniform remeshing target size must be checked against the smallest local vessel radius, not applied blindly) rather than a correction to any existing number.

Solves on the rebuilt surfaces (baseline and missed-branch). The rebuilt STLs were adopted as the case geometry (`solve_baseline`, 1,108,603 cells, 5 outlets; `solve_missedbranch`, 1,017,542 cells, 4 outlets; the old case directories are flagged `DO_NOT_USE`). Every outlet’s resistance and relaxation value was generated from the 0D discrete-bed twin of scan 837/left ($Q_{\text{demand}} = 0.6636$ mL/s from the inlet radius’s Murray demand; for the missed-branch case D2’s 0D segment was deleted and the tree re-calibrated to the same demand, so the remaining outlets absorb its flow). Each 0D leaf was matched to its mesh patch by label *and* verified by coordinate (leaf position plus the 20 mm extension along the leaf tangent lands within 0.04–0.32 mm of the patch centroid; the script asserts a 3 mm limit). The coded BCs were compiled by a serial two-iteration run before the parallel launch, then both cases ran concurrently (16 MPI ranks each, ~ 19 wall-minutes to converge, all 32 logical CPUs in use). Both converged cleanly (pressure residuals $\sim 10^{-8}$; every outlet flux stable to $\leq 0.002\%$ and outlet pressure to $\leq 0.0001\%$ over the last 200 iterations; no NaN) and were stopped gracefully well before the 3000-iteration ceiling.

Table 8: Item 3 converged results. BC-accuracy error compares the monitored outlet-patch pressure with $(P_v + R Q_{\text{actual}})/\rho$ using each outlet’s own converged flow (mechanism check; tolerance 0.1%). The flow-split gap compares that converged Q with the 0D prediction (logged as data). Mass imbalance is 0.0000% in both cases.

Case	Outlet	BC-accuracy err.	3D Q (mL/s)	0D Q (mL/s)	Gap
baseline	LCX	0.00000%	0.11993	0.11942	+0.43%
baseline	IM	0.00000%	0.18543	0.18567	−0.13%
baseline	D1	0.00000%	0.11886	0.12045	−1.32%
baseline	D2	−0.00001%	0.12217	0.12244	−0.22%
baseline	LAD	0.00000%	0.11441	0.11099	+3.08%
missed-branch	LCX	0.00000%	0.14664	0.14597	+0.46%
missed-branch	IM	0.00000%	0.22805	0.22846	−0.18%
missed-branch	D1	0.00000%	0.14504	0.14755	−1.71%
missed-branch	LAD	0.00000%	0.13943	0.13551	+2.89%

BC accuracy: PASS on all nine outlets (errors $\leq 10^{-5}\%$ against a 0.1% tolerance). The resistance BC, with the relax values generated automatically from the 0D tree ($G_i = R_i/R_{\text{own},i}$, no hand tuning), enforces $P = P_v + RQ$ to machine precision on a real 5-outlet coronary tree. The 3D flow split differs from the 0D twin by -1.7% to $+3.1\%$ per outlet ($+0.28\%$ and $+0.25\%$ in total inflow); the pattern (LAD highest, D1 lowest) is nearly identical in the two cases, so it is a systematic 0D-vs-3D difference, not noise.

The missed branch, as seen by the re-tuned BCs. Deleting D2 (18.5% of the baseline flow) and re-calibrating raises every remaining outlet’s flow by 21.9–23.0% (a uniform redistribution predicts +22.7%). Yet the distal pressure ratios (section-averaged P/P_{aorta} at the outlet stations, below) move only -0.0065 (LAD, $0.9422 \rightarrow 0.9357$), -0.0047 (LCX), -0.0037 (D1) and -0.0008 (IM): the BC re-tuning absorbs a $\sim 22\%$ flow error into a pressure-ratio change ≤ 0.0065 . This is a data point about how the outlet model hides a missing branch, reported without a clinical threshold because this pilot geometry has no lesion.

Item 4: 0D-to-3D handoff and pullback comparison. The handoff’s scripted portion (build the 0D tree with `bed="discrete"`, calibrate, export R per outlet, match leaves to patches by coordinate with an assertion, write the BCs into the case) ran as one script with no hand edits. **The design’s criterion “the script runs end to end with no hand edits” is therefore only partly met:** the surrounding steps (compile smoke run, `decomposePar`, `mpirun`, `reconstructPar`, pullback) were launched by hand, and a single end-to-end runner has not been built. The pullback compares per-node 0D pressure with the area-averaged 3D pressure of centreline-normal sections

(1.5 mm spacing, the connected section nearest the centreline point only), both normalised by P_{aorta} . At the outlet stations the difference (3D–0D) in the baseline is +0.0123 (LCX), +0.0015 (IM), +0.0081 (D1), +0.0101 (D2), +0.0380 (LAD), and in the missed-branch case +0.0145, +0.0016, +0.0095, +0.0389 (LCX, IM, D1, LAD). In every path the largest difference is at the outlet, growing distally: the 3D solution has a smaller epicardial pressure drop than the 0D twin, most strongly along the longest terminal vessel (LAD, 95 mm). These differences are logged as data, not judged (the design sets no tolerance for this item).

Diagnostic: is the 0D–3D difference caused by the 20 mm outlet extensions? Each 3D outlet patch sits 20 mm beyond the 0D leaf, so the 3D domain contains an extra Poiseuille resistance $R_{\text{ext}} = 8\mu L/(\pi r^4)$ in series with the prescribed bed resistance that the 0D twin does not contain. A diagnostic copy of the baseline (`solve_baseline_extcomp`, same mesh, same numerics) subtracted R_{ext} from each outlet’s R_{out} . It converged completely (ran to the 3000-iteration ceiling; every flux stable to $\leq 0.0002\%$ over the last 200 iterations; BC-accuracy error $\leq 10^{-5}\%$; mass imbalance $10^{-6}\%$). The flow-split gap to the 0D twin changed from +0.43/−0.13/−1.32/−0.22/+3.08% (LCX/IM/D1/D2/LAD) to +2.11/+1.26/+0.32/+1.44/+4.73%: total flow rose by 1.6% (0.6608 to 0.6712 mL/s against 0.6590 for the 0D twin), so compensating the extension *worsened* the mean agreement (mean gap +0.4% $\rightarrow$ +2.0%; per-outlet gaps improved for D1 only). The leaf-station pressure differences barely moved (+0.0123/+0.0015/+0.0081/+0.0101/+0.0380 $\rightarrow$ +0.0119/+0.0014/+0.0078/+0.0098/+0.0369), as expected since the extension lies downstream of that station. **The extension resistance therefore does not explain the residual 0D–3D difference; the original close flow agreement is, if anything, a partial cancellation of two opposite effects** (missing extension resistance in the 0D twin lowers 3D flow by $\sim 1.6\%$, whereas the 3D epicardial tree is less resistive than the 0D one). The baseline (uncompensated) configuration is the reported Item 3/4 result; this diagnostic is a single extra run, not part of the design.

Diagnostic: 0D node radii versus 3D section radii. The 0D twin uses the mask Euclidean-distance-transform (inscribed) radius at each centreline node; the 3D section-averaged pullback also gives the true section area. Along the five root-to-outlet paths the mean 3D area-equivalent radius exceeds the 0D node radius by 23–44% (LCX 1.156 vs 0.937 mm, IM 1.638 vs 1.135, D1 1.374 vs 1.069, D2 1.455 vs 1.155, LAD 1.090 vs 0.875 mm). **Correction (post-hoc audit, 2026-09-26):** these path means include the first station of every path and bifurcation slices, whose sections are not valid (they can catch two vessels). Restricted to the valid sections of the returned per-station file (227 stations), the mean excess is 19.5% (LCX), 21.7% (LAD), 22.1% (D2), 22.8% (D1) and 33.7% (IM), and the median station ratio is 1.206, so the excess is about 20–34%, not 23–44%; the qualitative conclusion is unchanged. Re-evaluating the Poiseuille loss along each path with the 3D area-equivalent radii and the same 0D flows predicts a 3D–0D leaf-pressure difference of +0.0184/+0.0043/+0.0131/+0.0148/+0.0518 against the measured +0.0123/+0.0015/+0.0081/+0.0101/+0.0380: same sign and ranking (LAD largest), $1.4\text{--}2.9\times$ the measurement. The inscribed-radius bias is therefore a quantitatively sufficient explanation of the *direction* and *size* of the “3D loses less pressure than 0D” finding; the remainder (a Poiseuille loss computed with a circle of equal area under-counts the resistance of non-circular sections, and junction/curvature losses are not in that estimate) is untested. This is a crude single-formula check (1.5 mm sections, equal-area circles, no junction losses), not a calibration, and it is the same radius-definition issue that separates the 0D and 3D stenosis severities in the lesion case below.

Item 3, case 2: the 80%DS lesion on the real surface — construction, gates and audits. A synthetic lesion was imposed on the *pre-remesh* baseline surface of the proximal LAD (arc window [18.5, 28.5] mm, throat at 23.5 mm, 10 mm long; the D1 ostium lies 8.5 mm downstream of the throat at arc 32.05 mm on the smoothed axis, 32.87 mm on the raw polyline). The deformation is a tangent-frame radial scaling about the smoothed centreline, $f(s) = 1 - 0.8 w(s)$ with a raised-cosine w (C^1 at the window edges), applied after a conforming edge-split refinement of the undeformed

surface so that the profile is sampled at 0.017–0.24 mm edges (43,046→61,300 triangles); the 20 mm extensions and planar caps were then rebuilt with the same routine as the baseline (the STL rebuild reproduces bit-for-bit; cap centroids identical to the baseline’s on all six caps, 0.000 mm). The result is a 3D throat of area 0.447 mm^2 (area-equivalent radius 0.377 mm), i.e. 79.98% diameter stenosis and 95.99% area reduction. **Geometry gates, all PASS:** zero displacement of 19,478 vertices assigned to other branches and no triangle mixing moved and unmoved branches; no fold-over (minimum normal dot 0.72); section area follows f^2 within 0.94%; 41 baseline window sections all closed, star-shaped and ≥ 0.5 mm from any other vessel; the axis lies inside the lumen at 41/41 stations (largest centroid offset $0.19 r_{\text{eq}}$); 99.7% of the moved vertices carry the LAD label in the segmentation; OpenFOAM `surfaceCheck -checkSelfIntersection` reports a closed, single-part, non-self-intersecting surface and, as a positive control, flags a deliberately corrupted copy at 322 locations.

The lesion surface and a healthy baseline surface were meshed with *identical* cfMesh settings (19 chained refinement cylinders following the curved LAD from 17.5 to 36 mm plus the first 10.5 mm of D1, three boundary layers), giving 2.68 and 3.25 million cells, so the lesion effect can be taken as a difference at equal refinement rather than against the coarser validated baseline. Measured, not assumed: cell size 46–51 μm along the axis (cfMesh’s octree quantisation realised both requested sizes as 50 μm); mesh section area within 0.34% of the STL at 59 stations from 17.5 to 32 mm (throat –0.33%; baseline reference within 0.024%); boundary layers 3-deep on 75–100% of the sampled wall-normal rays per station in the window (complete coverage of the wall is not established by 72 rays), median first cell $\approx 10\text{ }\mu\text{m}$ thick, median stack 30–41 μm by ray measurement and 45–55 μm by an independent slice measurement, i.e. 10–15% of the throat radius. **Limitation stated up front:** near-wall resolution at a 0.75 mm throat is thin and no mesh-sensitivity run has been made, so any throat quantity is a single-mesh result. Standard `checkMesh` passes on all three meshes; `strict checkMesh -allGeometry -allTopology` fails, as it does on the validated baseline:

Table 9: Strict `checkMesh -allGeometry -allTopology` results (measured logs). No flagged face or cell of any mesh lies within a 3 mm tube around the LAD axis for arc 18.5–32 mm (throat to D1 ostium). Flagged entities form two clusters, both at the wall (median 100–230 μm) and shared by the lesion and reference meshes because they share the surface there: one at the IM ostium, arc 16.5–18 mm, 0.5–2 mm *before* the window edge (6.1–6.3 mm from the throat centre), the other on the LAD wall just past the D1 ostium, arc 34–36 mm. The reference’s extra failed check is one face (volume ratio 0.0089 against a 0.01 threshold) in the downstream cluster, 11.7 mm from the throat. The failures are accepted for this pilot on that local evidence, not because they resemble the baseline.

Mesh	Cells	Low-quality face-tets	Concave faces (max angle)	Concave cells	Low-vol.- ratio faces	Failed checks
validated baseline (coarser)	1,108,603	83	29 (24.3°)	4	0	2
lesion80	2,682,675	92	57 (34.8°)	4	0	2
baseline reference (same refinement)	3,245,362	113	60 (28.6°)	7	1	3

Pre-run audit gate, applied to this case (five rounds, three independent auditors).

Round 1 (GPT-6-Astra, Fable 5.1, Gemini 3.8 Flash, the latter through the `agy CLI`): *not ready / ready with fixes / ready with fixes*. They found that the case builders had no lesion path, that the design text carried wrong numbers (window centre, junction location, “> 100 mm from every outlet” which is false since the IM cap is 25 mm away, the throat radius, the Reynolds number), that no explicit self-intersection or ownership gate existed, that the 0D radius (mask-EDT, inscribed) and the 3D area-equivalent radius define different stenosis severities, that the pullback script silently evaluated the healthy tree for any case name, and that the wall-shear-stress metric had sign, units

and curved-vessel-masking traps. All were addressed; round 2 (the built cases audited against the files) again returned *ready with fixes*, requiring: the strict-check disposition above (the reference mesh fails one more check than the lesion mesh, and the “same as baseline” comparison had no saved log behind it), corrected boundary-layer numbers (my first “30 μm , 8%” was too low), a launch script, the NPROC default, missing evidence logs, and text corrections (inlet dynamic pressure is 4 Pa, not 12 Pa). The fixes were applied and confirmation rounds followed. Round 3 confirmed all case, mesh and evidence fixes but rejected the *launch script* (Astra and Fable, independently): it sourced the OpenFOAM environment after enabling `set -u` (which aborts on the environment’s own unset variables), checked a dynamic-library path that does not exist in v2406, used bare `wait` (which returns success whatever the background jobs did, so a failed smoke run would have continued to two production launches with `endTime 2`), and its control-file assertions were `&&` chains under `set -e` that are only fatal on their last term. The script was rewritten (per-job waits, separate fatal assertions, an EXIT trap that restores the production `controlDict`) and Fable then found a further real defect by failure injection: after one smoke run fails while its sibling is still running, the trap restored `endTime 3000` into the live smoke case, and because `runTimeModifiable` is on, the orphaned serial run would have continued to 3000 iterations; SIGTERM also skipped the restore. Both were fixed (wait for every job before aborting; a signal trap that kills the children first) and re-tested with shim executables and fake case directories (orphan, SIGTERM/SIGINT during the smoke and decomposition phases, all-OK; author’s tests and Fable’s 15 scenarios). **Disclosure:** the final script-only round was confirmed by Fable 5.1 alone; GPT-6-Astra and Gemini 3.8 Flash had both exhausted their usage quotas (reset in about 1.5 h) after auditing every earlier round, and the solves were launched under that 1-of-3 confirmation of the last script change; the two remaining confirmations were run after the quotas reset: Gemini confirmed the launch script and the audited launch sequence (*ready*); Astra also confirmed the script, the launch evidence and the baseline-reference stop (it could not repeat the failure-injection tests itself under a read-only sandbox), but returned *not ready* on two defects in the convergence *watcher* (a counter that is not reset when a check is interrupted, so a stop could follow non-consecutive converged checks; and an unhandled failure of the stop command that would end the watcher). Neither occurred: the watcher log shows no interrupted check and both stops followed two consecutive *converged* verdicts, and the final convergence state of each case was re-evaluated after it stopped, independently of the watcher (below). Both watcher defects were then fixed.

0D predictions, recorded before the solve. The 0D twin with frozen physiology (identical R_{out} and relaxation values as the validated baseline; asserted to 10^{-9}) predicts, with the lesion imposed on the design radius (EDT node radius $\times f$; throat radius 0.338 mm, detected DS 78.9%): inflow 0.6590 $\rightarrow$ 0.6284 mL/s (−4.6%), LAD/D1/D2 outlets −8.6/ −8.7/ −8.6%, LCX/IM unchanged. With the lesion imposed on the 3D area-equivalent radius (throat 0.378 mm, detected DS 76.5%): inflow 0.6386 mL/s (−3.1%), the three downstream outlets −5.8%. The 3D result will therefore be read against a *range* of 0D predictions that reflects the radius-definition ambiguity, not a single number, and as a difference from the same-refinement healthy reference.

Item 3, case 2: lesion80 results (single mesh, steady, laminar, protocol-demand flow 0.66 mL/s). Both cases were launched together under the audited script (16 ranks each) and converged: lesion80 in 1,364 iterations (about 1.5 h) and the healthy reference in 768, each stopped gracefully after two consecutive checks of the predefined criteria (pressure and velocity residuals $< 10^{-5}$: final p 1.9×10^{-7} and 9.0×10^{-8} , $U \leq 7.6 \times 10^{-6}$ and 8.5×10^{-6} ; every outlet flux and pressure stable to $< 0.1\%$ over the last 200 iterations: worst flux band 0.0009% and 0.006%; mass imbalance $6 \times 10^{-7}\%$; ≥ 200 iterations), and the criteria were re-evaluated on the stopped state independently of the watcher. **BC accuracy: PASS** (worst outlet error $5 \times 10^{-4}\%$ lesion, $2 \times 10^{-4}\%$ reference, against 0.1%). **Mesh effect on the healthy split:** the healthy reference (3.25 M cells, refined lumen) reproduces the validated baseline (1.1 M cells) outlet flows to $\leq 0.01\%$ (0.11993/0.18543/0.11887/0.12217/0.11441 vs 0.11993/0.18543/0.11886/0.12217/0.11441 mL/s), so the mesh change alone does not move the healthy flow split; this says nothing about mesh sensitivity

of the throat itself, which was not tested.

Table 10: lesion80 minus same-refinement healthy reference: outlet flows and the 0D predictions recorded before the solve. $\Delta_{0D,EDT}$: lesion imposed on the design (mask-EDT) radius; $\Delta_{0D,area}$: lesion imposed on the 3D area-equivalent radius in the window.

Outlet	Q_{lesion} (mL/s)	Q_{ref} (mL/s)	Δ_{3D}	$\Delta_{0D,EDT}$	$\Delta_{0D,area}$
LCX	0.11994	0.11993	+0.01%	+0.02%	+0.01%
IM	0.18545	0.18543	+0.01%	+0.02%	+0.01%
D1	0.11229	0.11887	−5.53%	−8.65%	−5.76%
D2	0.11552	0.12217	−5.45%	−8.64%	−5.76%
LAD	0.10817	0.11441	−5.46%	−8.64%	−5.76%
Total	0.64137	0.66081	−2.94%	−4.64%	−3.09%

The lesion reduces the flow through the three territories distal to it by 5.5% and the total by 2.9%; the branches proximal to it are unchanged (0.01%). The 0D twin loaded with the design (EDT) radius overestimates the flow reduction by 57% (8.6% vs 5.5%), whereas the 0D twin loaded with the area-equivalent radius is close (5.8%, total 3.1%). Distal outlet pressure ratios fall by 0.048–0.050 P_{aorta} (D1 0.9107 vs 0.9608, D2 0.9209 vs 0.9708, LAD 0.8827 vs 0.9304); LCX and IM are unchanged (0.9690, 0.9914).

Table 11: Section-mean static pressure along the LAD, P/P_{aorta} (3D: area-weighted over the connected centreline-normal section; 0D: node pressures interpolated in arc). Arc 18.5 mm is the window start, 24.0 mm the 3D pressure minimum (the throat is at 23.5 mm), 29.5 mm the last section before the 1.5 mm-spaced sections (no section exists at 30.0 mm; an earlier draft of this table labelled the 29.5 mm values “30.0”). “Added loss” = (pressure at 18.5 minus at 29.5) in the lesion case minus the same in the healthy case of the same model, in mmHg.

	3D	0D (EDT)	0D (area)	3D healthy
$s = 18.5$ mm	0.99647	0.99627	0.99620	0.99633
$s = 24.0$ mm (minimum)	0.93610	0.92098	0.94649	0.99622
$s = 29.5$ mm	0.94356	0.91426	0.94141	0.99610
Added loss, 18.5→29.5	4.74 mmHg	7.34 mmHg	4.89 mmHg	—

Throat (area 0.4455 mm² in the mesh): flow 0.3377 mL/s, mean velocity 0.76 m/s, peak axial velocity 1.16 m/s (at arc 24.0 mm), $Re = 151$ on the area-equivalent diameter. The static pressure falls 4.9 mmHg between the window start and the throat station (5.4 mmHg to its minimum 0.5 mm downstream) and recovers only 0.7 mmHg afterwards, so the 3D static loss (4.74 mmHg) matches the area-matched 0D estimate (4.89 mmHg, +3%) and is 35% below the EDT-radius estimate (7.34 mmHg); the mass-flux-weighted total pressure drop is smaller (3.9 mmHg), but that quantity is unreliable in a section that is 65% reverse flow and is given as supporting information only. **Reading:** in this case the 0D–3D difference for the lesion is dominated by which radius defines “80%” rather than by the stenosis loss law; with the EDT radius the 0D twin looks 3D-wrong, with the area radius it does not. This agreement could still be partly fortuitous (one lesion, one flow, one mesh).

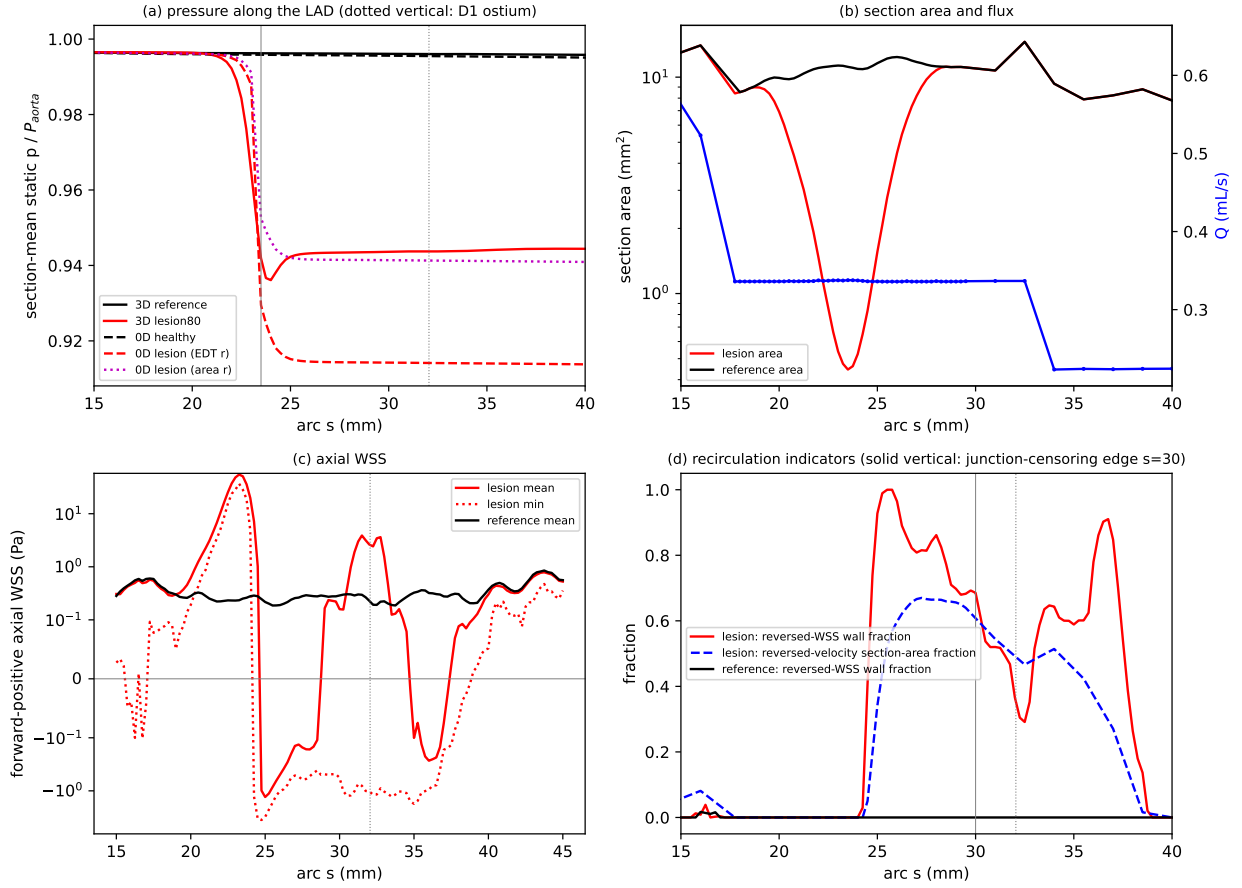


Figure 10: lesion80 (red) against the healthy reference (black) along the LAD. (a) Section-mean static pressure with the 0D twin's two lesion variants; (b) section area and section flux; (c) forward-positive axial wall shear stress (circumferential mean and minimum, symmetric-log axis); (d) reversed-wall-shear wall fraction and reversed-axial-velocity section-area fraction. Vertical lines: D1 ostium (dotted, arc 32.05 mm) and the junction-censoring edge (solid, 30 mm).

Recirculation and reattachment (the Item 3 wall-shear measurement). Wall shear was extracted with the sign calibrated on the undisturbed proximal wall (100% of 3,288 faces forward-positive, median 0.76 Pa), assigned to the LAD path by nearest segment, area-weighted in 0.25 mm slabs. The healthy reference shows no reversed wall shear anywhere (reversed-wall fraction 0 at every station). In the lesion case, forward-positive shear reaches about 51 Pa at the throat, separation begins 0.75 mm past the throat (arc 24.25 mm; reversed-wall fraction 41% at 24.5 mm, 93% at 25.0 mm, 100% at 25.5 mm), and the reversed region is an annular vortex around the jet: 67% of the section area carries reverse flow at arc 27.25 mm. **The circumferential-mean shear changes sign at arc ≈ 28.6 –29 mm (5–5.5 mm past the throat) but this is misleading:** at 29.0 mm 70% of the wall is still reversed (the mean is positive only because the 30% forward-flow part of the wall carries larger shear magnitudes), and reverse flow occupies 64% of the section area at 29.5 mm. The wall-based and the velocity-based measures agree that the recirculation persists through the D1 ostium (arc 32.05 mm), where the reversed fraction dips (29% of the wall at 32.5 mm) and rises again (64% at 34 mm, 85% at 37 mm), and ends at arc ≈ 38.5 –39 mm (reversed wall fraction 14% at 38.5, 0 at 39–40 mm; reversed section area 2% at 38.5, 0 at 40 mm), i.e. ≈ 15 mm (≈ 20 throat diameters) past the throat and 6–7 mm beyond the ostium. **Outcome under the pre-registered rule: not reattached before the D1 ostium's influence (arc 30 mm); the outcome is censored, not a reattachment length.** The recovery point at 38.5–39 mm is a junction-influenced value, since the ostium diverts one third of the flow, and should not be read as the reattachment length of an isolated 80% stenosis. The reattachment measurement was thereby completed, with its outcome a

bounded statement rather than a number. Selected stations:

Table 12: Axial wall shear (forward-positive, Pa) and recirculation indicators along the LAD, lesion80 and healthy reference. Reversed-section-area fraction from the nearest validated section (0.25 mm spacing to 29.5 mm, 1.5 mm beyond).

Arc s (mm)	$\bar{\tau}$ lesion	$\tau_{\min}$ lesion	Reversed wall fraction	Reversed section area	$\bar{\tau}$ healthy
24.0	+19.78	+4.81	0.00	0.00	+0.28
24.5	+1.02	−3.38	0.41	0.05	+0.28
25.0	−1.31	−3.09	0.93	0.34	+0.22
26.0	−0.50	−1.01	0.96	0.60	+0.19
27.0	−0.14	−0.51	0.81	0.67	+0.26
28.0	−0.17	−0.53	0.86	0.66	+0.29
29.0	+0.17	−0.49	0.70	0.66	+0.28
29.5	+0.23	−0.49	0.68	0.64	+0.28
31.0	+1.89	−1.02	0.52	0.54	+0.29
32.5	+3.44	−1.17	0.29	0.47	+0.19
34.0	+0.16	−1.02	0.64	0.51	+0.21
35.5	−0.13	−1.08	0.60	0.42	+0.36
37.0	−0.08	−0.17	0.85	0.27	+0.28
38.5	+0.09	−0.03	0.14	0.02	+0.23
40.0	+0.30	+0.05	0.00	0.00	+0.37

Limits of this result. One mesh at the time of this section (50 μm cells near the throat, a three-layer boundary stack of roughly 35–55 μm , 10–15% of the throat radius); the mesh-sensitivity study reported below (25 and 100 μm levels) found changes of 1–2% in the throat pressure loss, peak velocity and peak wall shear (≈ 51 Pa) and none in the recirculation onset and recovery arcs, none of which is a demonstration of mesh independence (and the refined tube ends at arc 36 mm, so the recovery of the reversed flow at 38.5–39 mm lies in 100–200 μm cells, coarser than the 50 μm of the tube); a steady laminar solution at $Re = 151$ that could hide unsteadiness of the long recirculating jet; a flow of 0.66 mL/s (the protocol’s Murray-law demand on the biased 0D inlet radius; the same law on the 3D inlet radius gives $3.16\times$ this flow, run in §11.6), so the 5% pressure loss is not a clinical FFR; strict-mesh-check failures accepted on local evidence; one lesion on one patient’s surface; the 0D twin’s agreement or disagreement depends on the radius definition as shown above. All of these are logged as data, none is judged against a tolerance the design did not state.

CFD and imaging figures, lesion80 and healthy reference. All fields below are the converged solutions (lesion80 iteration 1364, healthy reference iteration 768) on the 2.68 M- and 3.25 M-cell meshes; every panel is rendered directly from the OpenFOAM fields and the segmentation-derived surfaces (scripts `viz_*.py` in the pilot directory). **What “imaging” means here:** the ImageCAS-X release supplies the coronary *segmentation* (label volume, voxel $0.32 \times 0.32 \times 0.5$ mm), the extracted centrelines and the segmentation surface, not the CT intensities; the imaging-side panels are therefore the label volume, the surface and reformats derived from them, fused with the CFD fields. The synthetic lesion is *not* in the segmentation: wherever a CFD lumen is drawn against the label volume, the segmentation shows the healthy lumen by construction.

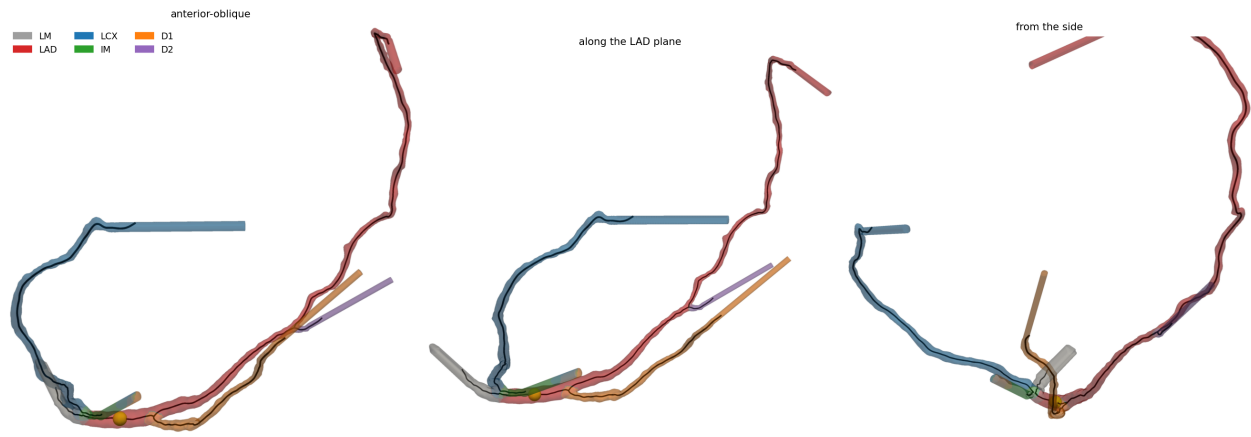


Figure 11: Left coronary tree of scan 837 as used for the CFD: the rebuilt segmentation surface (clipped at the OD truncation, 20 mm straight outlet extensions, translucent), coloured by OD-tree branch (LM grey, LAD red, LCX blue, IM green, D1 orange, D2 purple), the extracted centreline inside the lumen (black), and the lesion80 throat (yellow sphere, LAD arc 23.5 mm). Three views.

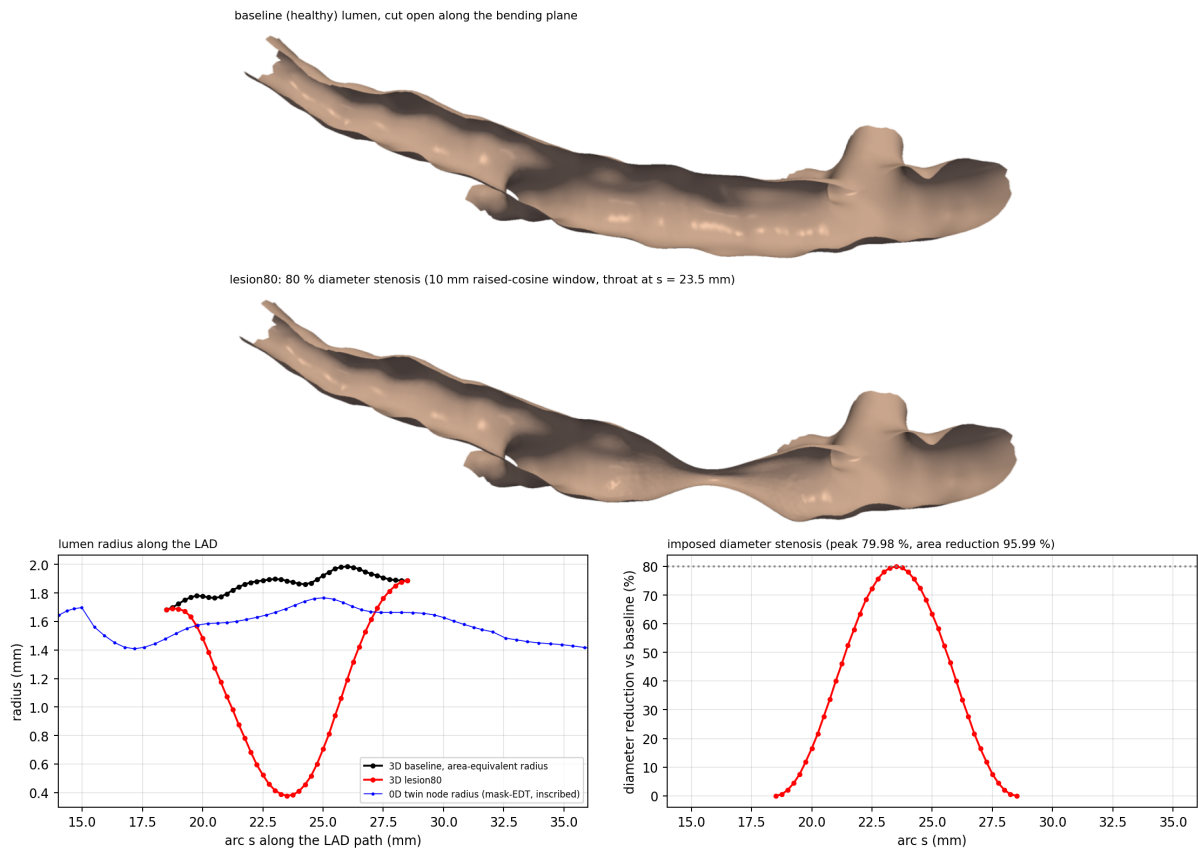


Figure 12: Construction of the 80% diameter stenosis on the real LAD surface. Top and middle: healthy and lesion lumen, cut open along the bending plane of the vessel (the D1 ostium is the bulge at the right). Bottom left: area-equivalent radius of the 3D sections (baseline black, lesion red, 0.5 mm stations) and the OD twin's node radius (blue, mask-EDT, inscribed): the OD radius lies below the 3D area-equivalent radius throughout the window, the radius-definition difference that separates the two “80%” values discussed above. Bottom right: imposed diameter reduction (peak 79.98%, raised-cosine over 10 mm).

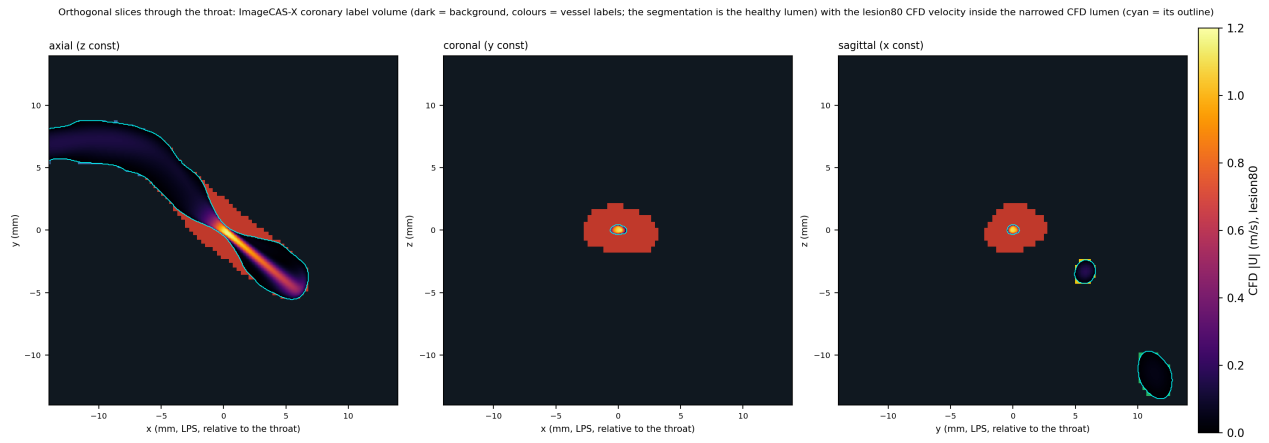


Figure 13: Orthogonal slices through the throat: coronary label volume (background dark; the LAD label red, other vessels other colours; the blocky structure is the 0.32–0.5 mm voxel grid) with the lesion80 CFD velocity magnitude drawn inside the CFD lumen (cyan outline). The CFD lumen at the throat is far narrower than the segmentation (area 0.45 mm^2), and the jet reaches 1.1 m/s there; in the axial slice the jet leaves the throat towards the lower right; in the sagittal slice two neighbouring vessels are cut whose flow is slow.

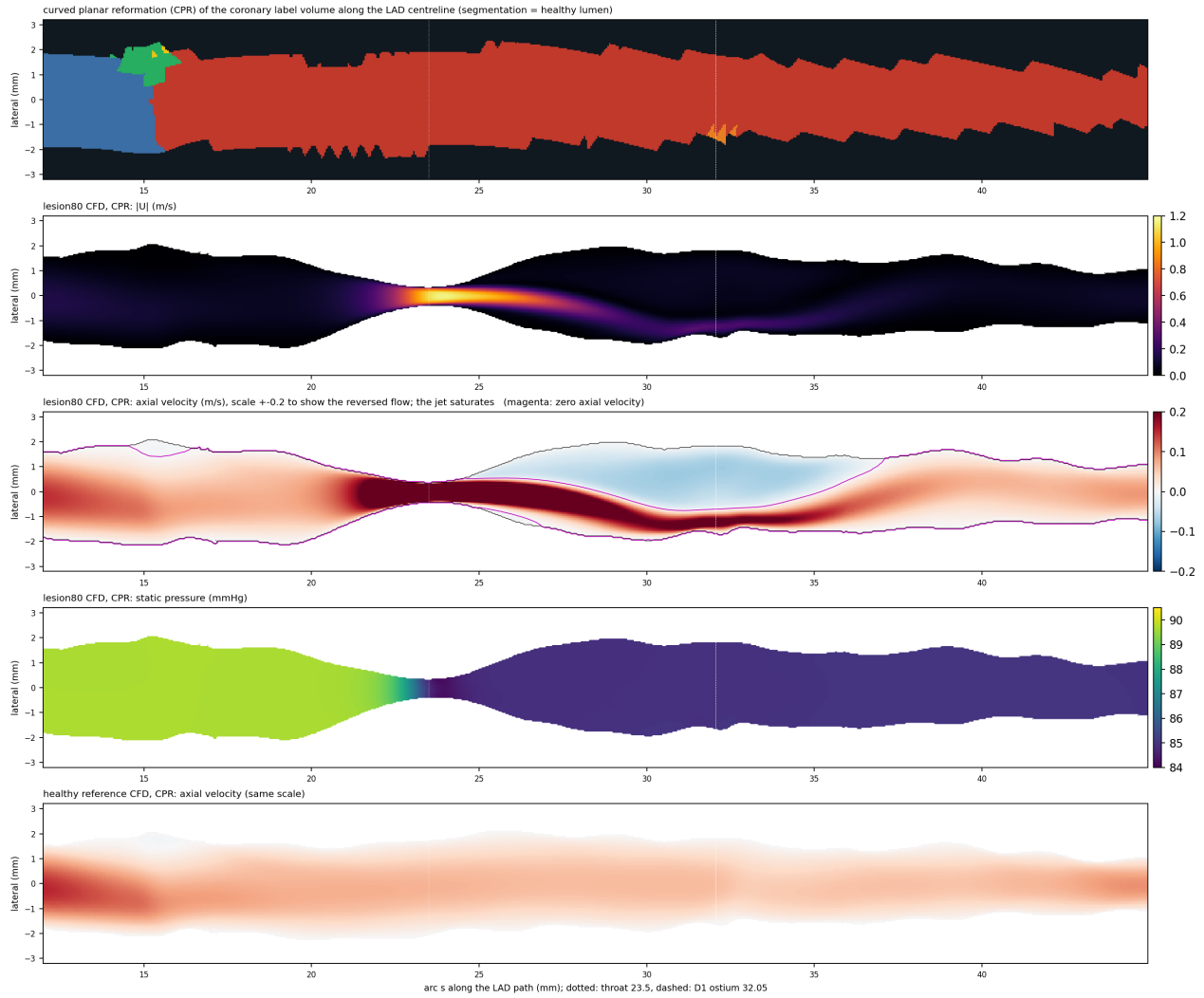


Figure 14: Curved planar reformation along the LAD centreline (arc 12–45 mm; lateral axis in the bending plane). Top: the label volume (segmentation = healthy lumen; the IM/LCX ostium region is visible at arc ≈ 15 –16 mm). Below, for lesion80: $|U|$ (jet deflected towards one wall after the throat and fading by arc ≈ 32 mm), axial velocity on a ± 0.2 m/s scale (the jet saturates; the reversed region is the pale-blue area bounded by the magenta zero-velocity contour, from arc ≈ 24.5 mm to ≈ 37 mm), static pressure (84–90.5 mmHg; the drop is concentrated at the throat and there is almost no pressure recovery afterwards). Bottom: the healthy reference on the same axial-velocity scale (forward everywhere: 0.05–0.1 m/s along the LAD, up to ≈ 0.2 m/s in the proximal segment before the LCX and IM take off).

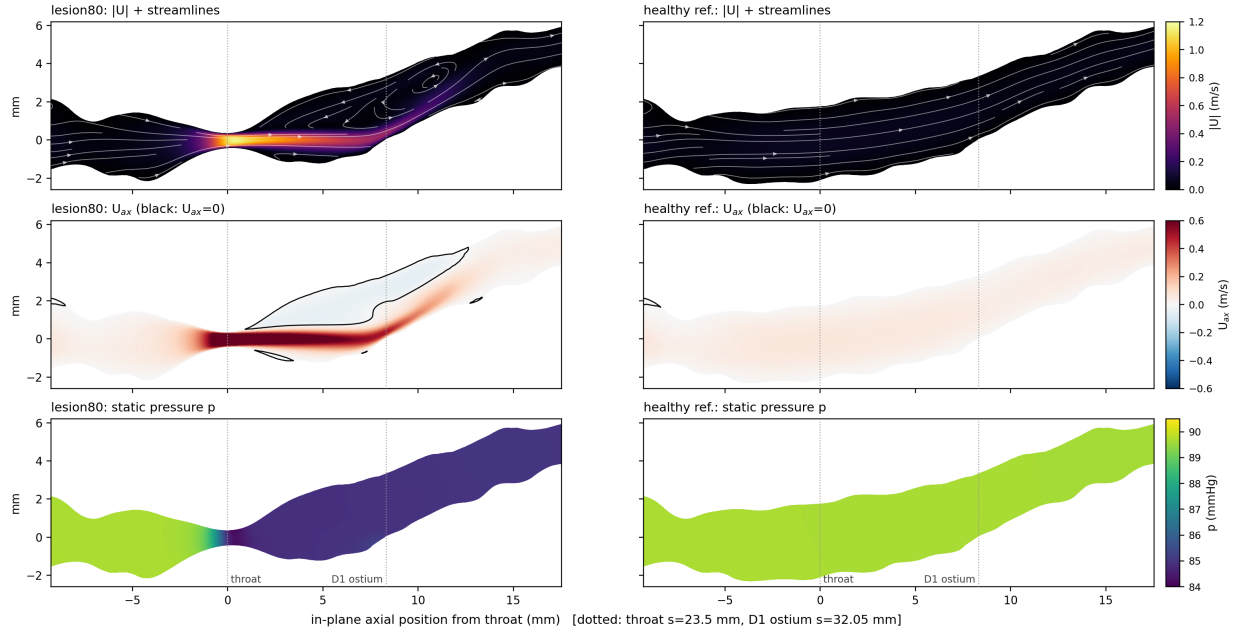


Figure 15: Longitudinal section through the LAD in its bending plane, lesion80 (left) and healthy reference (right), arc ≈ 14 – 41 mm: $|U|$ with in-plane streamlines, axial velocity with the zero contour, static pressure. The jet from the 0.75 mm throat stays narrow for about 5 mm and is then deflected along one wall; on the other side the zero-velocity contour encloses a recirculation region beginning about 1 mm downstream of the throat and reaching past the D1 ostium (dotted), with a closed vortex visible in the streamlines about 10 – 12 mm downstream; a second, small reversed pocket lies on the opposite wall 1 – 3.5 mm downstream. The healthy vessel shows uniform low-speed forward flow. Colour scales are identical in both columns.

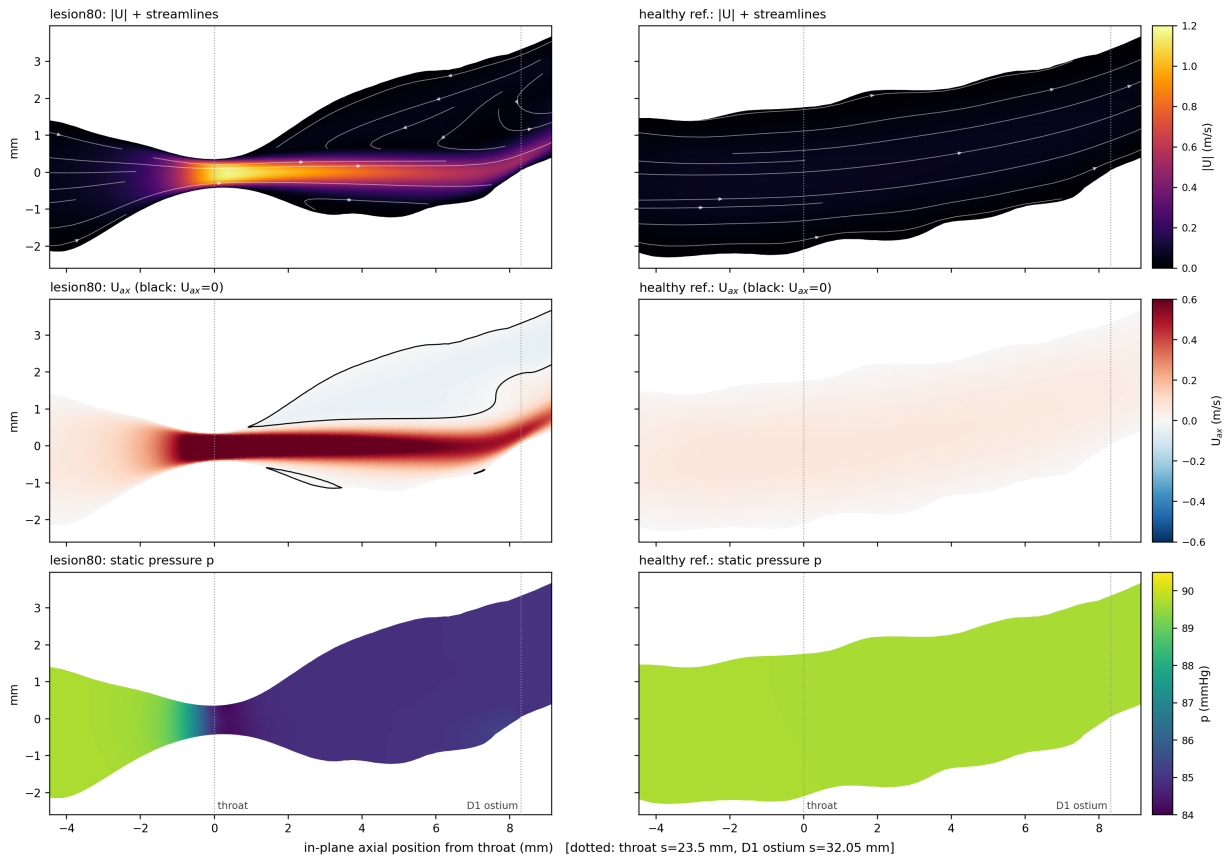


Figure 16: As above, zoomed on the throat and the first 9 mm downstream.

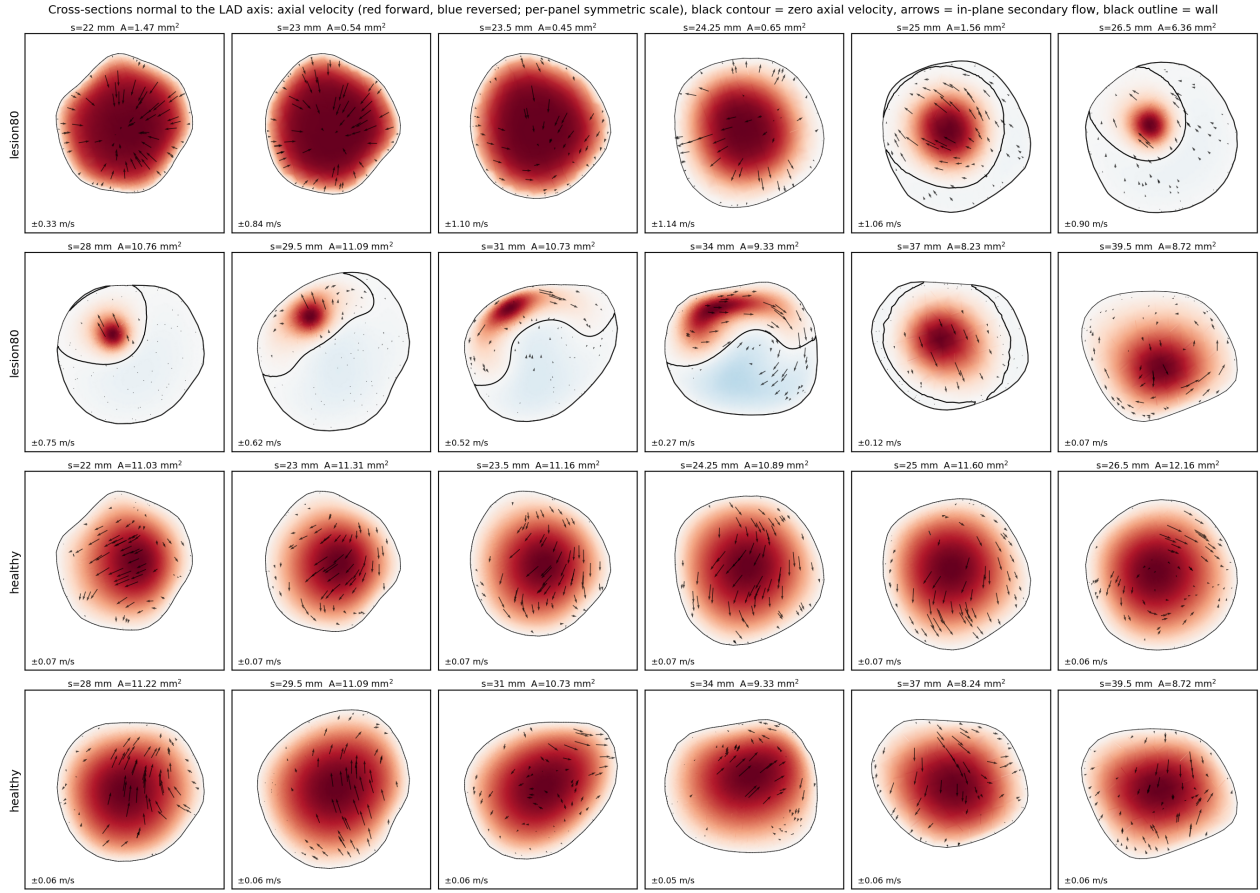


Figure 17: Cross-sections normal to the LAD axis at twelve stations (top two rows lesion80, bottom two rows healthy reference at the same stations): axial velocity (red forward, blue reversed, each panel on its own symmetric scale, stated in the corner), black = zero axial velocity (reversed-flow boundary), arrows = in-plane secondary flow, black outline = wall. The jet fills the throat sections (peak axial velocity 1.10 m/s at 23.5 mm, area 0.45 mm²), becomes eccentric at 25 mm (a crescent of reversed flow appears), and at 26.5–34 mm forward flow is confined to a part of the section while 47–66% of the area is reversed (largest reversed velocity only –0.04 to –0.07 m/s); at 37 mm a thin reversed ring remains and at 39.5 mm the flow is forward everywhere (peak 0.07 m/s). The healthy sections show a smooth low-speed core with weak secondary flow.

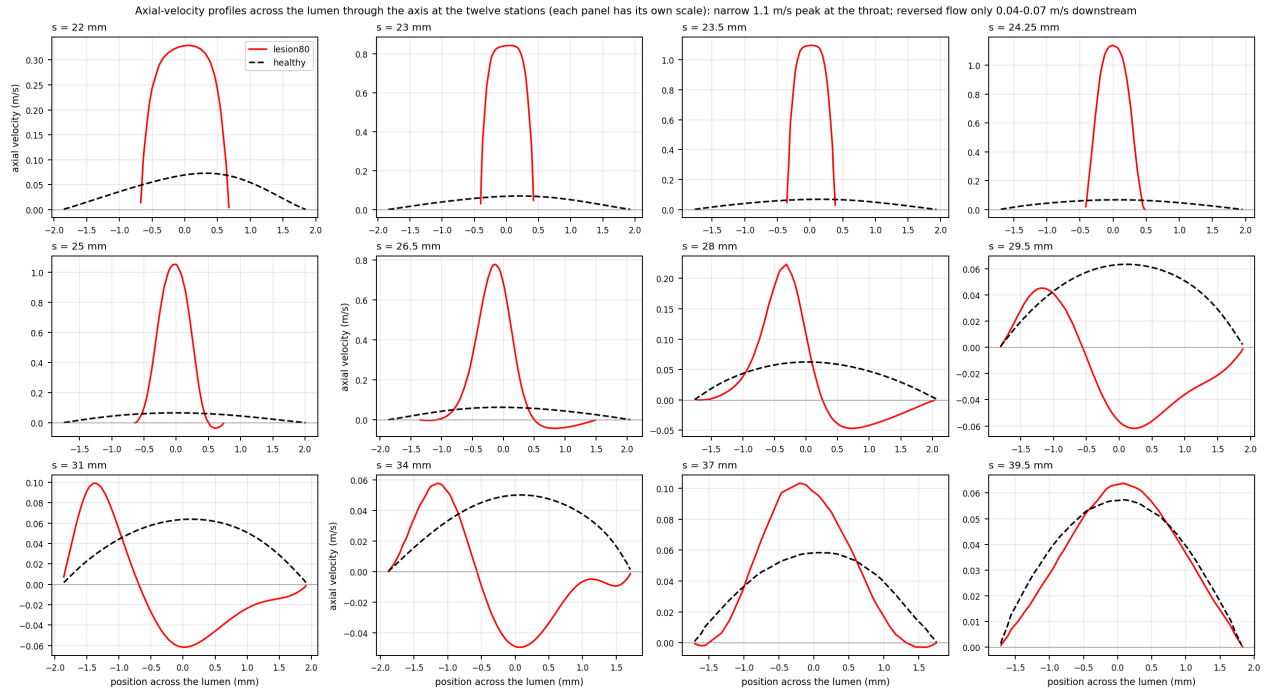


Figure 18: Axial-velocity profiles across the lumen (through the axis, in-plane direction) at the twelve stations, lesion80 (red) against the healthy reference (black dashed), each panel on its own scale. The throat profile is a narrow peak of 1.1 m/s in a 0.75 mm lumen; downstream the jet decays and moves off-axis, and the reversed flow (negative values) reaches only -0.04 to -0.07 m/s (stations 25–34 mm); by 39.5 mm the lesion profile has returned to the healthy shape.

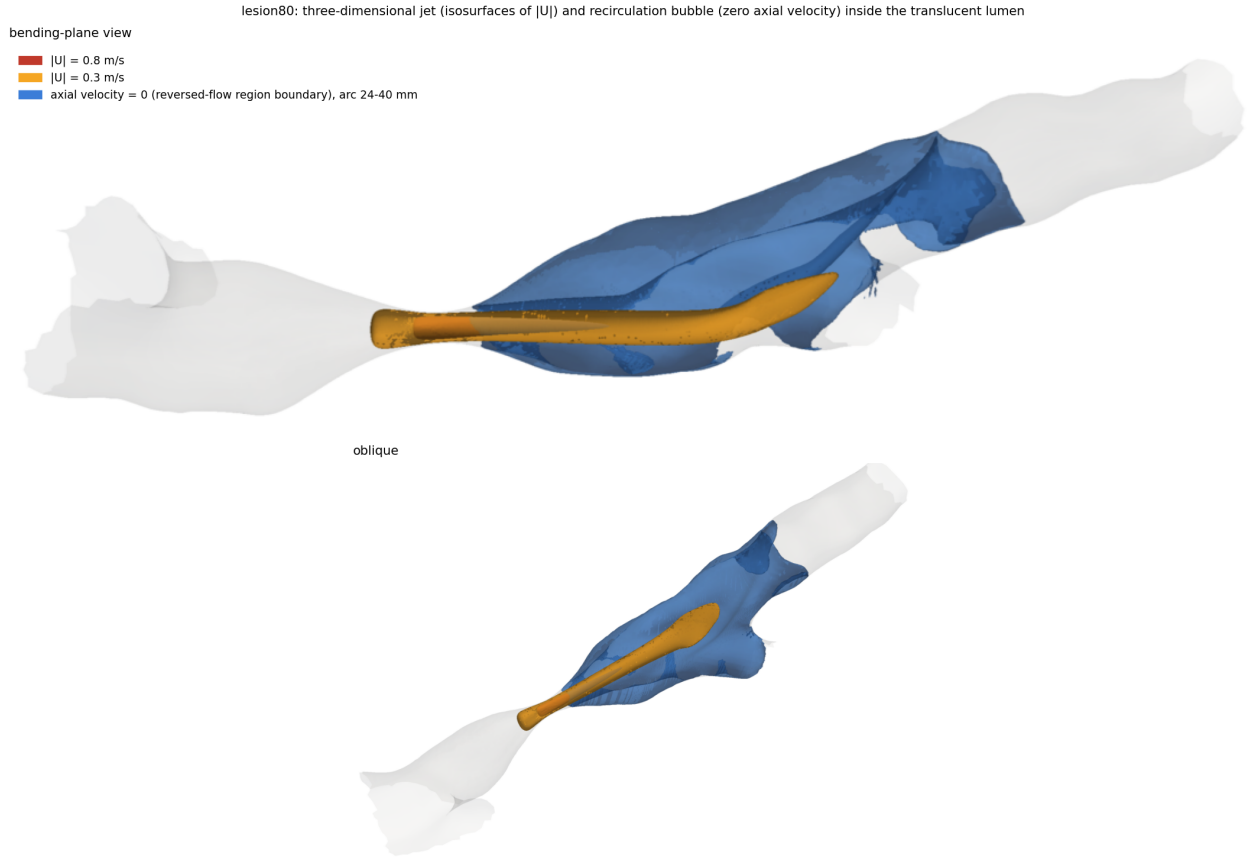


Figure 19: Lesion80, three-dimensional structure of the flow: isosurfaces of $|U| = 0.3 \text{ m/s}$ (orange) and 0.8 m/s (red) show the jet leaving the throat as a slender core that bends towards the wall; the blue surface is the zero-axial-velocity surface for arc 24–40 mm and encloses the recirculation region, which wraps around the jet (it touches the wall over much of its extent). Two views; translucent grey: lumen.

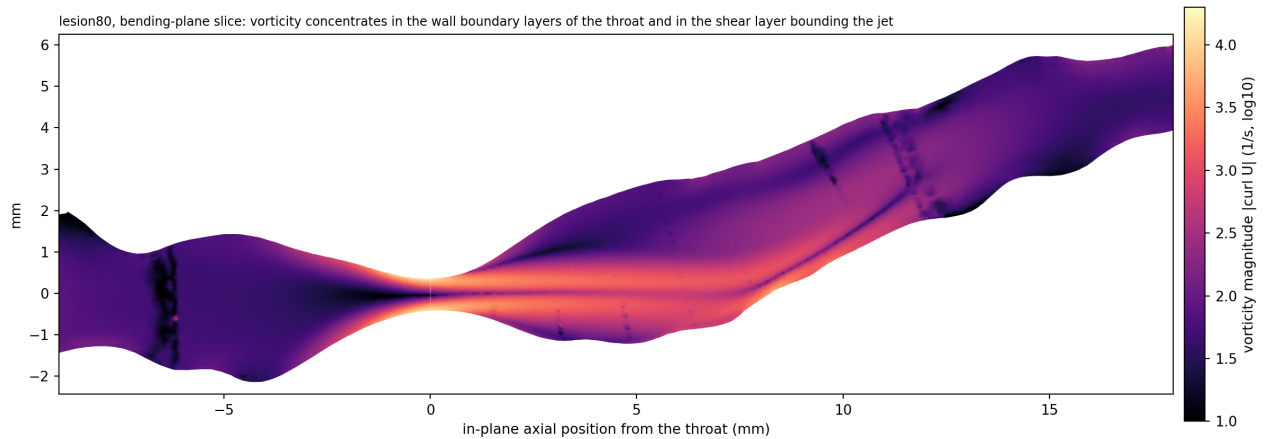


Figure 20: Vorticity magnitude $|\nabla \times \mathbf{U}|$ on the bending-plane slice of lesion80 ($\log_{10}$, $1/\text{s}$): the throat wall boundary layers ($\gtrsim 10^4 \text{ 1/s}$) and the thin shear layer bounding the jet stand out against the low-vorticity recirculation. The thin dark streaks crossing the vessel (at about -6.5 , $3\text{--}4$ and $9\text{--}12 \text{ mm}$) are artefacts of the velocity-gradient computation across mesh-refinement interfaces, not flow features.

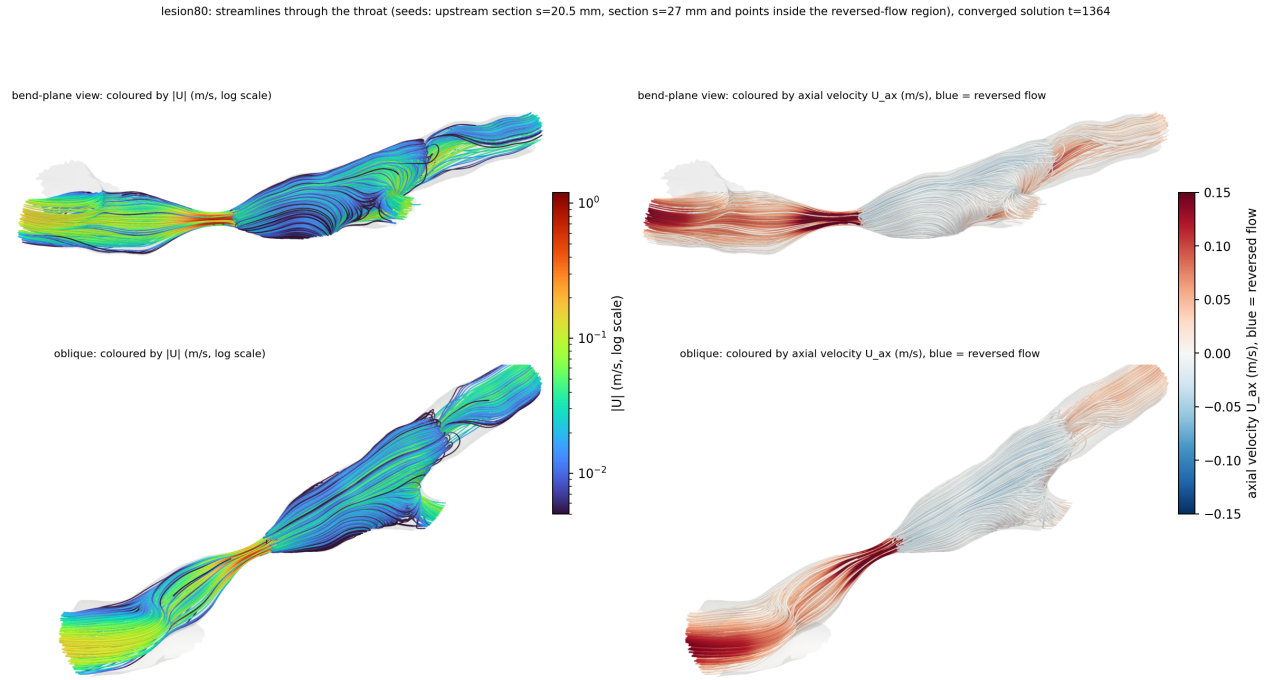


Figure 21: Streamlines through the stenosis (lesion80; seeds on the upstream section at arc 20.5 mm, on the section at 27 mm and inside the reversed-flow region): left coloured by $|U|$ (logarithmic scale; the jet is the red/orange core, the recirculating flow moves at 0.005–0.05 m/s), right coloured by axial velocity on a ± 0.15 m/s scale (blue = reversed). Two views.

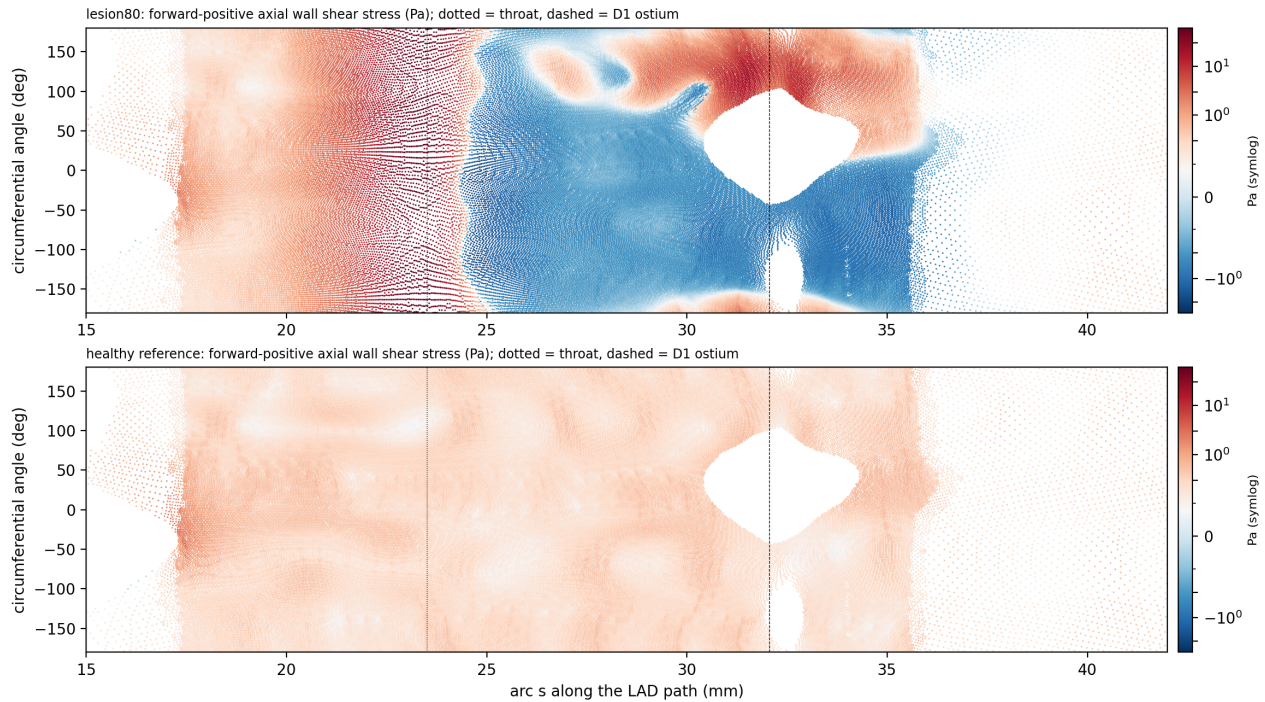


Figure 22: Wall shear stress unrolled onto arc length and circumferential angle (LAD-path wall faces only; forward-positive axial component, symmetric-logarithmic colour scale). Lesion80: forward shear rises to ≈ 51 Pa at the throat; from arc ≈ 24.5 mm to ≈ 35.7 mm the wall shear is reversed over almost the whole circumference (blue), except a forward band on the side facing the D1 ostium (white gap: D1 wall faces, not part of the LAD path); weakly reversed patches continue to ≈ 38 –39 mm. Healthy reference: uniform forward shear of 0.2–0.4 Pa.

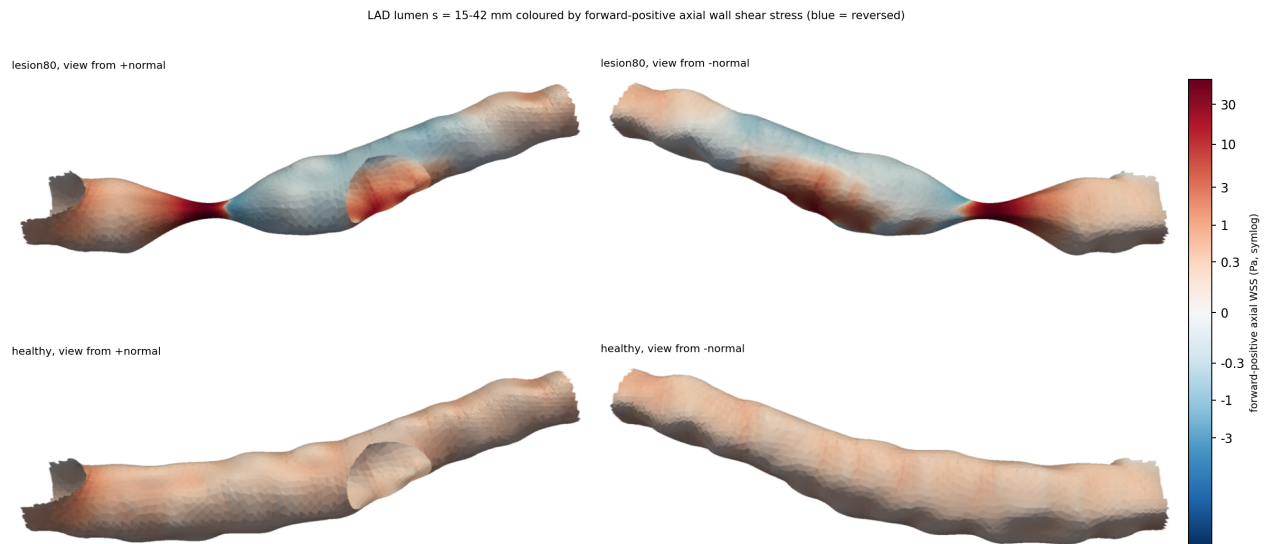


Figure 23: The same wall shear stress on the 3D lumen of the LAD (arc 15–42 mm), viewed from the two sides of the bending plane. Lesion80 top, healthy reference bottom.

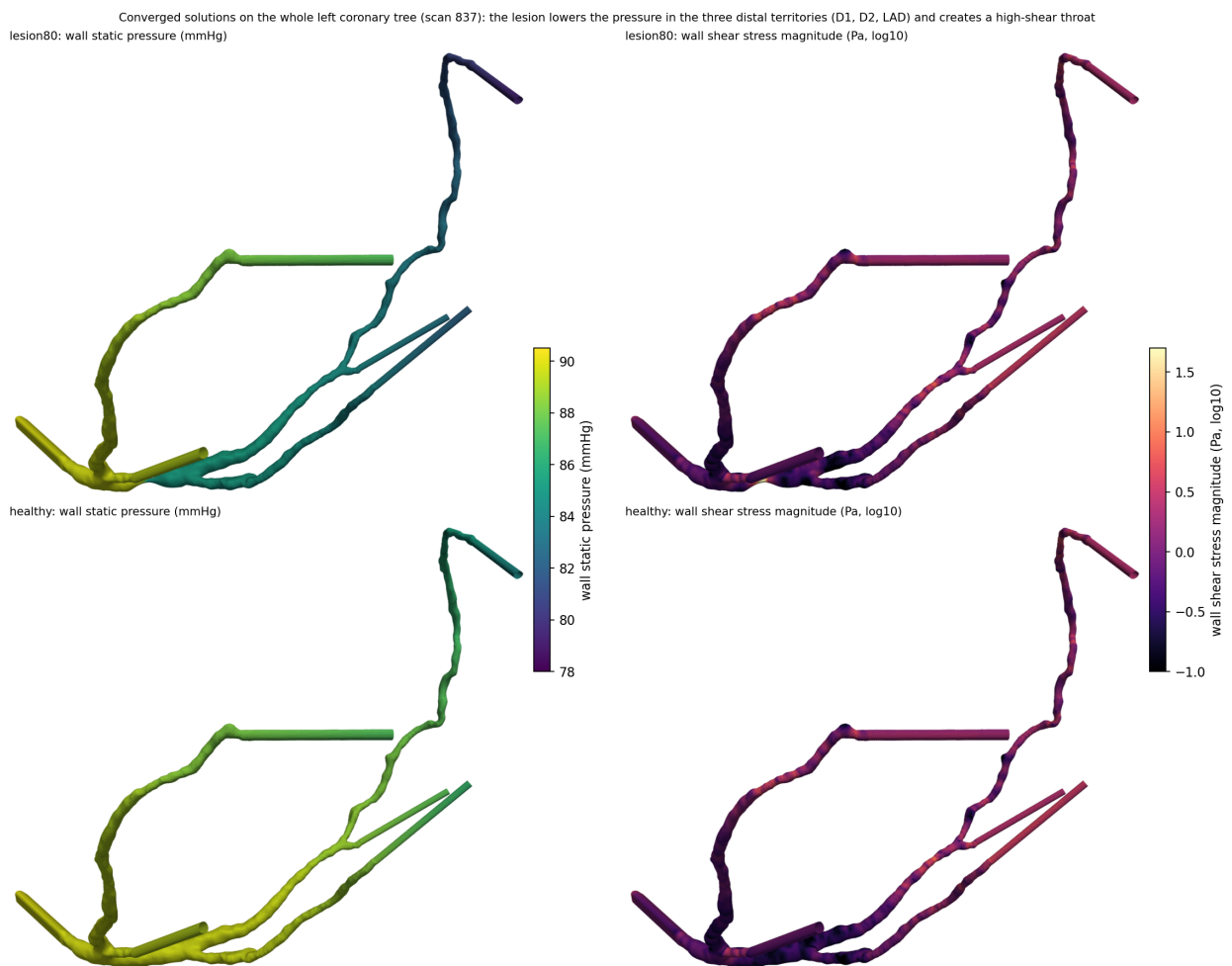


Figure 24: Wall static pressure (left) and wall shear stress magnitude (right) on the whole left coronary tree, lesion80 (top) and healthy reference (bottom), identical colour scales. The lesion lowers the pressure in the three territories distal to it (LAD, D1, D2; outlet pressure ratios 0.88–0.92 against 0.93–0.97) while LCX and IM are unchanged.

Mesh-sensitivity study of the lesion80 throat: design, pre-registered rules and audits (results follow below when the runs have finished). The lesion80 result above is a single-mesh result. Three further meshes of the *same* surface, boundary conditions, numerics and 0D twin were built and run under the same audited launch procedure: **W100** (100 μm cells in the whole refined tube, 1.30 M cells), **T25a** (25 μm for arc 20.5–26.5 mm inside the 50 μm tube, 3.53 M cells) and **T25b** (25 μm for arc 19–28 mm, 5.03 M cells), against the audited **M50** (50 μm , 2.68 M cells; not re-run). Cell sizes are quantised by cfMesh to $0.2\text{ mm}/2^n$, so the levels differ by exactly a factor 2; the sizes were measured, not assumed (p95 of the cell size in the throat zone 103.9, 51.5, 25.6 and 25.6 μm); the throat area of the mesh against the STL is -2.01 , -0.33 , -0.08 and -0.08% respectively, and the boundary-layer stack (three layers, ratio 1.2) scales with the cell size (throat-station stack 75, 38, 20.5 and 20.5 μm). Fig. 25 shows the four levels. **Design changes made before any of these solves, and why.** A planned coarse throat-only level (T100) was rejected by its own gate (flagged cells inside the lesion tube at the size-jump interface); a first second fine mesh (zone 20.5–28 mm) shared its upstream interface with T25a and could not test interface contamination, so T25b moves *both* interfaces (19 and 28 mm against 20.5 and 26.5 mm). A 25 μm zone over the whole analysis window would need $\approx 9\text{ M}+$ cells and exceed the 27 GB of RAM, so the fine side is a throat-zone refinement only; consequently the level set is *not* a systematic refinement family and **no GCI, order of accuracy or uncertainty bound is computed** (Item 2 had already shown a three-level GCI can be untrustworthy). The recirculation tail beyond the refined tube (which ends at arc 36 mm; the recovery of the reversed flow at ≈ 38.5 –39 mm lies in 100–200 μm cells in every level) is resolution-limited by construction whatever its label. **Pre-registered rules (fixed before the solves; engineering judgement, not from the study specification; not changed after seeing any result).** Ten quantities: distal and total flow reduction against the healthy reference (tolerance 0.5 percentage points), the added static pressure loss between arc 18.5 and 29.5 mm (3%; the loss is taken at 29.5 mm because there is no section at 30 mm — an earlier table of this report used the 29.5 mm values under the label 30.0), the pressure minimum (0.3 mmHg), peak velocity (5%), peak throat wall shear (5%), the onset arc (0.25 mm) and recovery arc (1 mm) of reversed wall shear, and the maximum reversed wall-shear fraction and reversed section-area fraction (0.05 each). A quantity is INSENSITIVE only if both 25 μm levels stay within the tolerance of M50 and agree with each other (so the interface positions cannot drive the result), with precedence UNASSESSABLE > SENSITIVE > INTERFACE-CONTAMINATED > NON-MONOTONE > INSENSITIVE; missing, invalid, non-finite, unconverged or time-inconsistent data give UNASSESSABLE and windows are never shrunk; outcome categories (separated or not, recovered or not) are compared before the numbers. INSENSITIVE is the only label that counts as a pass. **Audits.** The three cases and the launch script were confirmed by Fable 5.1 and Gemini 3.8 Flash and, for the arithmetic, checked by all three auditors over eleven rounds: every round found genuine input-handling defects in the analysis script (a silent 30.0-versus-29.5 mm station, missing-data windows, booleans read as numbers, duplicated or quoted headers, duplicate JSON keys, NUL bytes, pandas accepting malformed quoting, an omitted output argument that silently analysed the default directories, and more); each was fixed and each fix tested (162 checks). Since the user’s bug protocol of 2026-09-24 the fixes were made by Opus 5.5 (attempt 1 each) and only audited by the other models; before that date I fixed them myself, and Fable was called 13 times as an auditor (the user’s rule is that the cap of two applies to bug fixes only; from 2026-09-24 the pre-run and post-hoc audits use Sonnet and Opus 5.5, with Gemini and Astra as further audit-only reviewers). The analysis code was frozen with Gemini and Astra confirming the final version (round 11); after the runs, two further independent audits by Opus 5.5 and Sonnet (round 13) each re-derived all ten labels and every level value from the raw analysis, section and wall-shear files with their own code and found no difference (maximum relative difference $< 10^{-6}$; the closest approach to a tolerance is the added static loss at 0.49 of its tolerance for the fine levels) and returned no blockers.

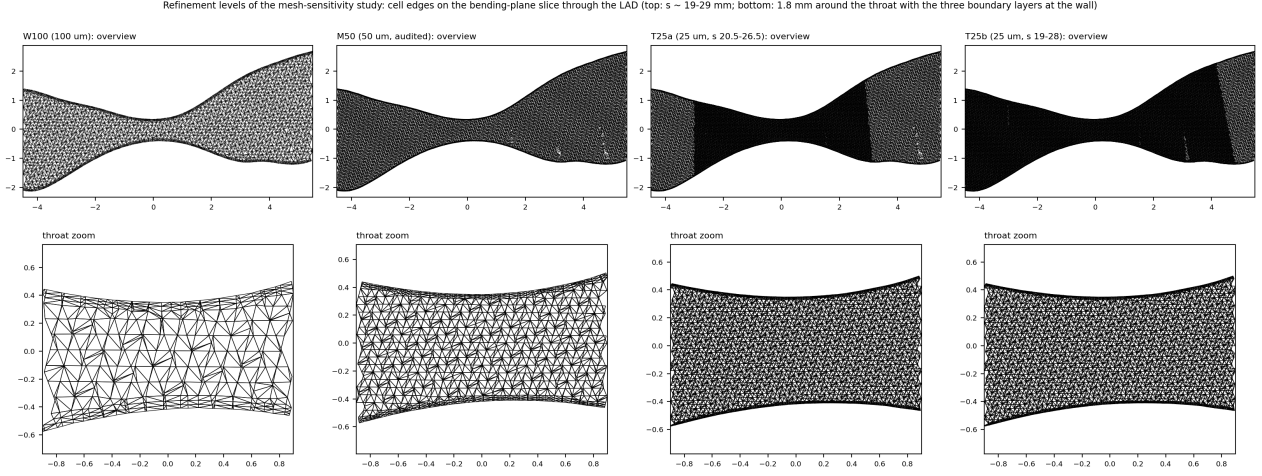


Figure 25: The four refinement levels on the bending-plane slice through the LAD (slice triangulation of the polyhedral cells): W100 (100 μm), M50 (the audited 50 μm mesh), T25a (25 μm for arc 20.5–26.5 mm) and T25b (25 μm for arc 19–28 mm); bottom row: 1.8 mm around the throat with the wall-adjacent boundary layers. The refinement interfaces of the two T25 meshes (density changes in the top row) are at different arcs on both sides.

Mesh-sensitivity results: every pre-registered quantity is INSENSITIVE — with the limits stated below. All three new runs converged under the predefined criteria (W100 in 688 iterations, T25a in 1315, T25b in 1470; final pressure residual $\leq 2 \times 10^{-7}$, velocity residuals $\leq 9 \times 10^{-6}$, every outlet flux stable to $\leq 0.05\%$ over the last 200 iterations (W100 $\leq 0.04\%$, T25a and T25b $\leq 0.0012\%$), mass imbalance $\leq 2 \times 10^{-6}\%$, BC-accuracy error $\leq 5 \times 10^{-4}\%$; **convergence and BC accuracy: PASS** for all three), and were stopped gracefully by the watcher. The frozen analysis script (checksums recorded before any result was computed) then produced, from analysis, section and wall-shear files that share the reconstructed final iteration of every level:

Table 13: Mesh-sensitivity of the lesion80 throat under the pre-registered rules. Flow reductions are relative to the healthy reference; the added static loss is $(P_{18.5} - P_{29.5})_{\text{lesion}} - (P_{18.5} - P_{29.5})_{\text{reference}}$.
[†]Recovery lies outside the refined tube and is resolution-limited by construction (see text).

Quantity	W100	M50	T25a	T25b	Tol.	Label
distal flow reduction (pp of reference)	5.703	5.479	5.398	5.398	0.5 pp	INSENSITIVE
total flow reduction (pp)	3.063	2.943	2.900	2.900	0.5 pp	INSENSITIVE
added static loss 18.5→29.5 mm (mmHg)	4.931	4.741	4.671	4.671	3%	INSENSITIVE
minimum static P/P_{aorta}	0.9335	0.9361	0.9368	0.9368	0.0033	INSENSITIVE
peak axial velocity, arc 22–26 (m/s)	1.200	1.160	1.148	1.148	5%	INSENSITIVE
peak mean wall shear, arc 22–25 (Pa)	57.87	54.74	53.60	53.60	5%	INSENSITIVE
onset of reversed wall shear (arc, mm)	24.25	24.25	24.25	24.25	0.25 mm	INSENSITIVE
recovery of reversed wall shear [†] (arc, mm)	39.00	39.00	39.00	39.00	1 mm	INSENSITIVE
max reversed-wall fraction, arc 24–39	0.942	1.000	1.000	1.000	0.05	INSENSITIVE
max reversed section-area fraction, arc 24–38.5	0.667	0.670	0.671	0.671	0.05	INSENSITIVE

What the numbers say. Refining the throat zone from 50 to 25 μm changes the distal flow reduction by -0.08 percentage points ($5.48 \rightarrow 5.40$), the added static loss by -1.5% (-0.07 mmHg), the pressure minimum by $+0.0007 P_{\text{aorta}}$ (0.06 mmHg), the peak velocity by -1.0% , the peak throat wall shear by -2.1% (-1.1 Pa) and the maximum reversed-wall and reversed-area fractions by ≤ 0.002 ; the onset and recovery arcs of reversed wall shear do not move. The two 25 μm meshes, whose refinement interfaces are at different arcs on both sides, agree with each other to 2×10^{-4} (flows and loss), $< 10^{-3}$ (fractions) and $< 10^{-3}$ relative (velocity and wall shear), so the interface

position has no visible effect on any registered quantity. Coarsening to $100\ \mu\text{m}$ moves the same quantities in the same direction by 2.3–3.7 times as much (loss +4.0%, peak velocity +3.5%, peak wall shear +5.7%, distal flow reduction +0.22 pp): the series W100, M50, T25a is monotone for every quantity and the change per halving shrinks, but the $100\ \mu\text{m}$ mesh differs from the audited one by more than the registered tolerance in three quantities (added loss +0.19 mmHg against 0.14, peak wall shear +3.1 Pa against 2.7, maximum reversed-wall fraction -0.058 against 0.05), so a mesh that coarse would not have been adequate for the throat quantities. Fig. 26 shows the level-to-level differences directly: the axial-velocity profiles of M50, T25a and T25b lie on top of each other (W100 visibly blunts the throat plug and lowers the jet at 26.5 mm), and the zero-velocity and 0.3 m/s contours of all four levels almost coincide.

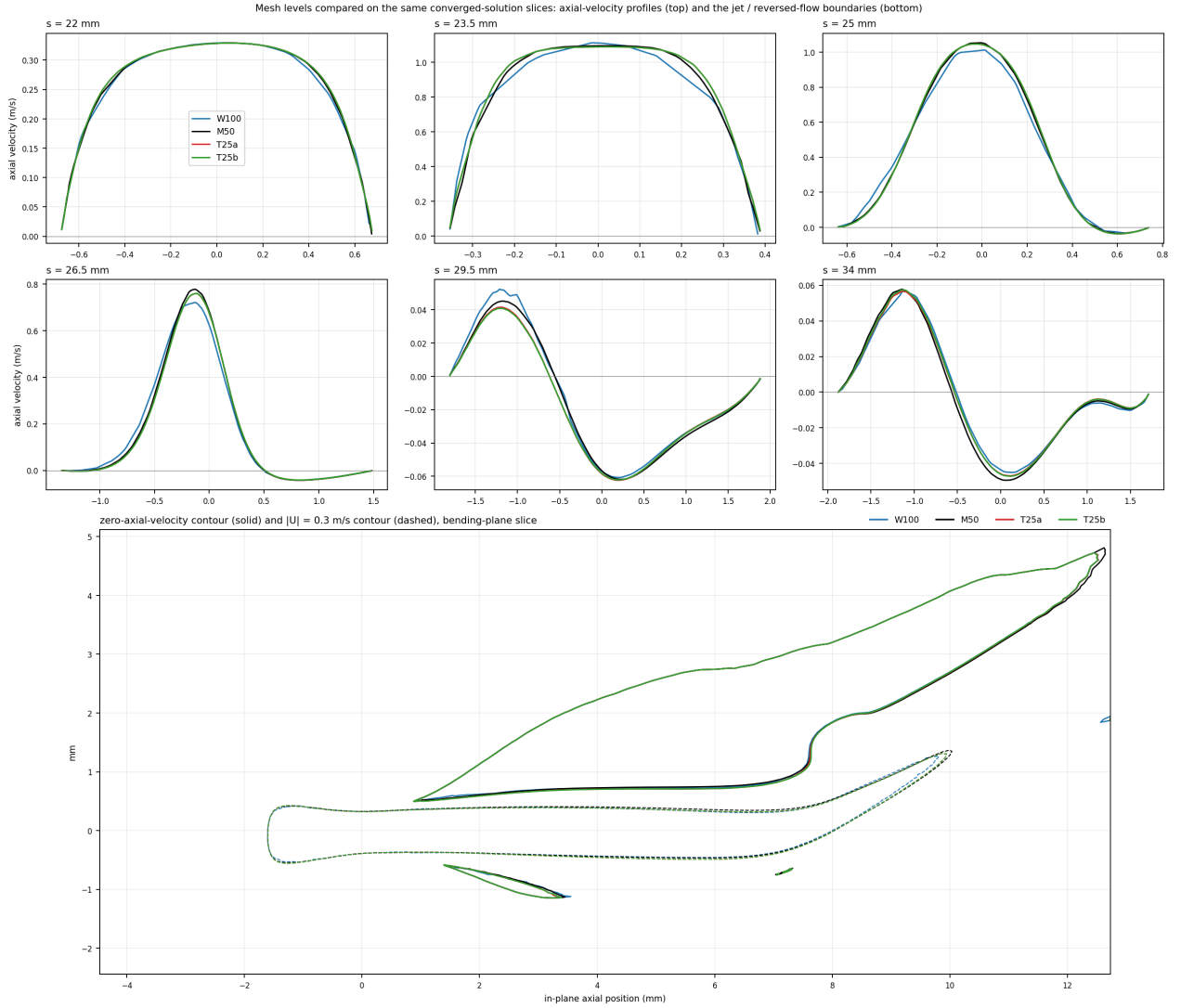


Figure 26: The four mesh levels compared on identical slices of the converged solutions. Top: axial-velocity profiles across the lumen at six stations (W100 blue, M50 black, T25a red, T25b green; T25a is hidden under T25b). Bottom: zero-axial-velocity contour (solid) and $|U| = 0.3\ \text{m/s}$ contour (dashed) on the bending-plane slice, arc $\approx 21\text{--}36\ \text{mm}$.

Limits of this result (all stated in the design before the runs). (i) INSENSITIVE means “within the pre-registered tolerance for a factor-2 refinement”, not “converged”: the tolerances are engineering judgement, no order of accuracy, GCI or uncertainty bound is computed (the level set is not a systematic refinement family), and a further halving is not tested. The lesion80 numbers of the preceding sections may therefore be read as mesh-dependent by roughly the amounts above (order 1–2% for the throat pressure loss and peak velocity, 2% for the peak wall shear between

50 and 25 μm), which is a description of what was observed, not a bound. (ii) The onset arc (24.25 mm) and recovery arc (39.0 mm) are identical in all four levels, but the recovery lies in cells of 100–200 μm outside the refined tube and is quantised to the 0.25 mm station grid; agreement there shows the outcome is not sensitive to the throat-zone resolution, not that the recirculation extent is converged. The reversed-flow region itself was refined only up to 50 μm in its core. (iii) The fine side is a throat-zone refinement (25 μm over 6–9 mm of the ≈ 15 mm long recirculating region); the effect of a finer expansion-zone mesh is not tested. (iv) The healthy reference was not re-run per level (its split moved $\leq 0.01\%$ between 1.1 M and 3.25 M cells). (v) The STL facet size (0.017–0.24 mm edges in the window) and the steady laminar assumption are untouched by this study. (vi) One geometry, one flow, one patient. **Pass/fail:** all ten registered quantities are INSENSITIVE; that is the study’s pass criterion, met, with the caveats above. **Effect on the 0D–3D comparison:** the area-matched 0D twin’s added loss (4.89 mmHg) is +3.1% above M50 and +4.7% above T25a, and the EDT-radius 0D twin’s (7.34 mmHg) is +57% above T25a; the conclusion that the radius definition, not the loss law, dominates the difference is unchanged.

Still open for Items 3 and 4. A finer expansion-zone mesh and a further throat halving (the sensitivity study brackets the 50 μm result but proves no convergence); the jet-reattachment length of an *isolated* lesion (this geometry’s lesion sits 8.5 mm upstream of a bifurcation, which the recirculation crosses); manual-repair minutes, which were not timed to the precision the design asks for; and a single end-to-end Item 4 runner. If the remeshing step is reinstated for future cases, its target edge length must be constrained to $\lesssim r/3$ of the locally smallest vessel radius (here ~ 0.2 mm on 0.6 mm branches) and gated by a pre-/post-remesh cross-section deviation check, not left at a single global value.

11.6 Items 3 and 4, continued: hyperaemic flow, extended refined tube, isolated lesion, transient check, end-to-end runner

This subsection reports the follow-up work commissioned after the mesh-sensitivity study (2026-09-24). Every new setup went through a read-only pre-run audit by GPT-6 Astra and Gemini 3.8 Flash before its solves started, every defect found (in the assistant’s own scripts or by an auditor) was handed to Opus 5.5 (at most three attempts per defect set) and then Fable 5.1 (at most two), and finished runs received a post-hoc audit; the record is in §11.6.

Hyperaemic flow: definition. The study’s 0D demand is $Q = k r_{\text{in}}^3$ with $k = 562 \text{ s}^{-1}$ (the 0D code labels this demand “hyperaemic”) applied to the 0D inlet radius, which is the mask-EDT radius (1.057 mm) and is biased low: the 3D inlet patch of the two meshes has an area of 7.553 mm² (radius 1.5505 mm). The lower-flow pair of this report (called “resting” in tables and figures; the word labels the protocol demand of 0.6636 mL/s, not a measured resting state) used exactly that demand. The hyperaemic pair applies the same law to the 3D area-equivalent inlet radius: $Q_{\text{hyp}} = 562 (1.5505 \text{ mm})^3 = 2.0950 \text{ mL/s}$, $3.157\times$ the resting demand. No parameter was tuned, and the size of the ratio carries no physiological claim: it is a synthetic flow increase, not a measured flow reserve and not maximal hyperaemia. The bed constant of the 0D twin was recalibrated for Q_{hyp} (C from 246.94 to 73.17, all outlet resistances $\times 0.2963$, relaxation factors 0.113/0.050/0.094/0.062/0.214 for LCX/IM/D1/D2/LAD). Meshes (the audited 2.68 M-cell lesion mesh and 3.25 M-cell healthy reference mesh), numerics, boundary-condition formulation and post-processing are the resting pair’s; a file-level diff shows that only the outlet values, resistances and relaxation factors and a two-line comment differ. The outlet patches sit 20 mm beyond the 0D leaves and their extension resistance is not compensated (about 6% of R_{out} at this flow, against 1.7% at rest).

Hyperaemic flow: validity of the pair. Both solves converged under the pre-registered strict rules (residuals $< 10^{-5}$, mass imbalance $< 10^{-6}\%$, boundary-condition error $\leq 0.0011\%$, flux and pressure bands $< 0.1\%$ over 200 iterations, all monitors of equal length, logs finished): the healthy

reference at iteration 1010 and the lesion case at 2106; the measurement-plane pressure ratio at four stations changed by at most 9×10^{-6} (lesion) and 9×10^{-5} (reference) between the final write and a write 356 and 260 iterations earlier (requirement $< 10^{-3}$).

Table 14: Hyperaemic pair against the resting pair (lesion80, same meshes). Flow changes are lesion minus healthy reference in the same flow state; “0D EDT” and “0D area” are the two 0D lesion variants.

	resting	hyperaemic	0D EDT (hyp.)	0D area (hyp.)
total inflow, healthy reference (mL/s)	0.6608	2.0248	2.0478	2.0478
total flow change, 3D (%)	-2.94	-11.93	-17.03	-13.04
D1 / D2 / LAD outlet change (%)	-5.5/ -5.4/ -5.5	-23.6/ -23.2/ -22.8	-33.1/ -32.9/ -32.9	-25.4/ -25.2/ -25.2
throat flow (mL/s) / Re (area-equiv.)	0.338 / 151	0.807 / 362		
peak axial velocity, arc 22–26 mm (m/s)	1.16	2.46		
peak circumferential-mean wall shear, arc 22–25 mm (Pa)	54.7	180.0		
static loss added by the lesion, arc 18.5→29.5 mm (mmHg)	4.74	20.64		
p/P_{ao} at arc 40 mm, lesion / healthy reference	0.944 / 0.996	0.762 / 0.983	0.676 / 0.985	0.749 / 0.985

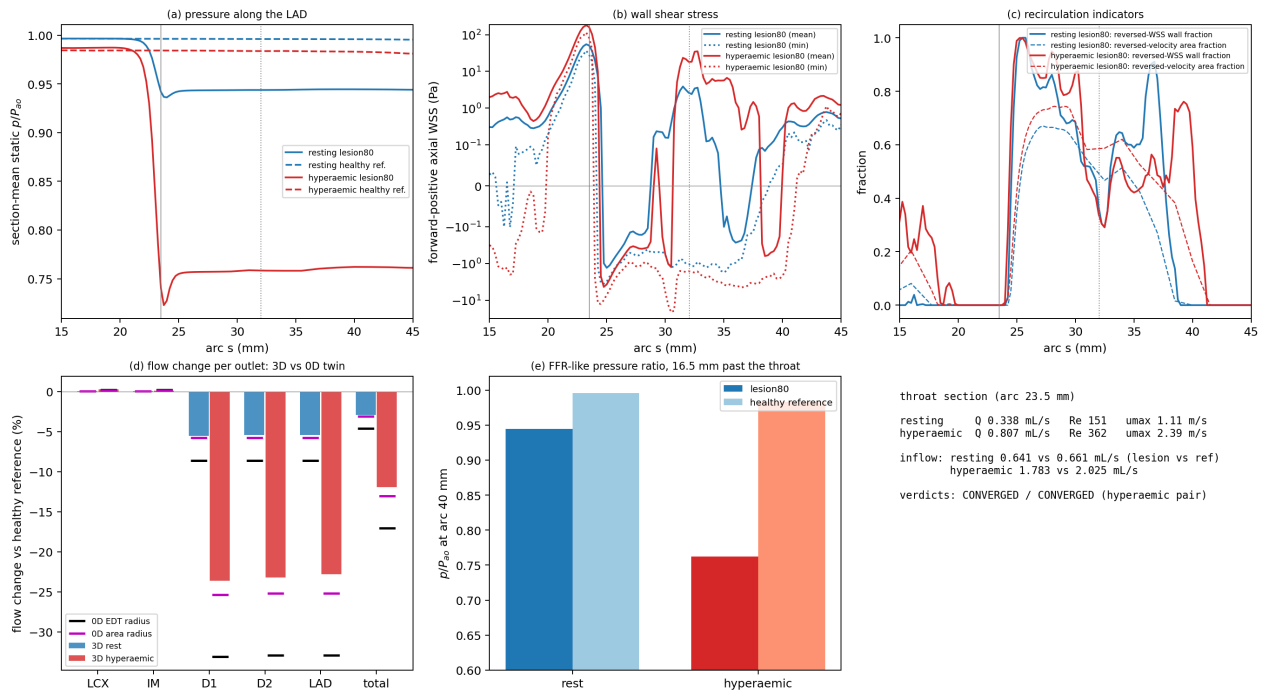


Figure 27: Resting (blue) and hyperaemic (red) lesion80 against their own healthy references. (a) Section-mean static pressure ratio along the LAD; (b) forward-positive axial wall shear (symmetric log axis); (c) reversed-wall-shear wall fraction and reversed-velocity section-area fraction; (d) outlet flow change against the two 0D variants; (e) pressure ratio 16.5 mm past the throat. The reversed-flow extent in (c) beyond arc 34 mm lies outside the $50 \mu m$ refined tube (which ends at 36 mm) and is diagnostic only.

Hyperaemic flow: results. (i) *Flow.* At $3.16\times$ the flow the lesion removes 11.9% of the total inflow and about 23% of the flow of the three distal outlets, against 2.9% and 5.5% at rest. The 3D change is smaller in magnitude than both 0D variants at every distal outlet and in total (pre-registered category (i), “smaller than the area-matched 0D variant”), as it was at rest; the LCX outlet, whose change is only +0.25%, falls in category (iii). This is a statement about ranking and sign, not agreement: the 3D distal reduction lies 1.7–2.4 percentage points below the area-matched variant and 9.5–10.2 below the EDT variant. (ii) *Pressure.* The 3D static loss between arc 18.5 and 29.5 mm rises from 4.74 to 20.64 mmHg (a factor 4.35 for a throat-flow factor 2.39). The 0D variants bracket the loss up to the throat (22.2 mmHg in 3D against 25.0 for the EDT variant and 19.1 for the area variant) but disagree qualitatively downstream of it: the 3D solution

recovers 1.6 mmHg of pressure between the throat and arc 29.5 mm while both 0D variants keep losing (2.3–3.0 mmHg). The pressure ratio at arc 40 mm, a static, area-averaged FFR-like quantity relative to the 90 mmHg inlet total pressure, is 0.762 in 3D against 0.749 (area variant) and 0.676 (EDT variant); the healthy reference gives 0.983. (iii) *Loss law*. Solving $\Delta P = aQ + bQ^2$ exactly from the two flow levels (throat flows 0.338 and 0.807 mL/s) gives $a = 7.7 \times 10^8 \text{ Pa s m}^{-3}$ and $b = 3.3 \times 10^{15} \text{ Pa s}^2 \text{ m}^{-6}$, so the quadratic term carries 78 % of the loss at the hyperaemic flow; $\Delta P/Q^2$ falls from 5.5×10^{15} to 4.2×10^{15} . The 0D expansion coefficient of the lesion window is $5.7 \times 10^{15} \text{ Pa s}^2 \text{ m}^{-6}$ (all four 0D loss nodes together, including three native narrowings, 6.4×10^{15}). Two points solve the two coefficients exactly and therefore do not test the quadratic form; the comparison is descriptive. (iv) *Jet and wall shear*. Peak axial velocity doubles (1.16 to 2.46 m/s) and the peak circumferential-mean wall shear rises from 54.7 to 180 Pa; separation still starts at arc 24.25 mm; the reversed-flow region reaches 41 mm in wall shear (38.75 mm at rest) and 40 mm in reversed-velocity area (38.5 at rest), and the reattachment outcome is again censored by the D1 ostium at arc 32.05 mm.

Hyperaemic flow: what is and is not a finding. The recirculation extent beyond arc 34 mm is **not mesh-supported**: the 50 μm tube ends at 36 mm, so the tail lies in 100–200 μm cells; the extents above (41 and 40 mm) are diagnostic values and are not presented as results; the pre-registered flag (last reversed station more than 2 mm inside the tube end) reads “NOT MESH-SUPPORTED”. The flow, pressure, peak-velocity and peak-shear results above are not affected by this flag, but no hyperaemic mesh-sensitivity study was made (boundary layers thin with $Re^{-1/2}$, so the resting sensitivity study does not transfer). Further caveats: the healthy hyperaemic reference delivers 2.025 mL/s, 1.1 % below the 0D twin and 3.4 % below the demand, with per-outlet deviations from the 0D split between –6.1 % (D1) and +5.6 % (LAD) against at most +3.1 % at rest (the cause was not isolated); the demand is synthetic (no autoregulation, uniform resistance scaling); the solves are steady and laminar at a throat Re of 362, which could hide unsteadiness (the transient check of this subsection covers only the resting case); the lesion is synthetic and the geometry single.

Extended refined tube (resting flow): the 36 mm tube end does not move the ten registered quantities. The refined 50 μm tube of the resting pair ends at arc 36 mm. To test whether this truncation affects the results, both resting cases were rebuilt with the refined tube extended (requested end 60 mm; realised: the last 50 μm cylinder ends at 58.0 mm, the frame sequence at 58.83 mm), meshed with the same recipe (lesion case 3.87 M cells, healthy reference 4.43 M), and solved with the same boundary conditions, numerics and 0D-derived resistances as the audited pair (16 ranks each, run concurrently). Both converged under the strict rules (lesion at iteration 1450, reference at 728; residuals $< 10^{-5}$, boundary-condition error $< 10^{-3}$ %, all bands < 0.1 %); the measurement-plane pressure ratio at four stations changed by at most 1.2×10^{-5} (lesion, over 450 iterations) and 5.8×10^{-4} (reference, over 478 iterations; the earlier write of the reference case is at iteration 250 because the case converged early, so this change is the least stringent of the four cases but below the 10^{-3} requirement). Table 15 gives the ten quantities of the frozen mesh-sensitivity analysis for the original (M50) and extended (E60) tubes, each pair against its own healthy reference.

Table 15: Extended refined tube (E60) against the audited resting pair (M50). The tolerance is the frozen sensitivity tolerance of the ten quantities; no quantity moved.

quantity	M50	E60	difference	tolerance
distal flow reduction (%)	5.4788	5.4794	+0.0006	0.5
total flow reduction (%)	2.9430	2.9434	+0.0004	0.5
added static loss (mmHg)	4.7410	4.7416	+0.0006	0.142
minimum p/P_{ao}	0.93610	0.93609	-0.00001	0.0033
peak axial velocity (m/s)	1.16027	1.16035	+0.00009	0.058
peak throat wall shear (Pa)	54.740	54.746	+0.006	2.74
separation onset, arc (mm)	24.25	24.25	0	0.25
wall-shear recovery, arc (mm)	39.0	39.0	0	1.0
max reversed wall fraction	1.000	1.000	0	0.05
max reversed area fraction	0.6699	0.6703	+0.0004	0.05

Every difference is below 0.8 % of its tolerance (largest: the maximum reversed area fraction, 0.72 % of its tolerance and 0.054 % of its value). The last reversed station lies at arc 38.5 mm in both tubes (sections and wall), inside both analysed windows (49 mm sections, 45 mm wall); the pressure ratio at arc 40 mm is 0.9444 (lesion) and 0.9958 (reference) for both tubes. The result therefore supports that the resting sensitivity study and the resting lesion80 results were not affected by the truncation at 36 mm; it says nothing about the hyperaemic pair, whose reversed-flow tail (40–41 mm) extends beyond the tube end and which was not repeated on the extended tube (the follow-up work order closed new work on this geometry). The hyperaemic extent quantities therefore remain diagnostic and NOT MESH-SUPPORTED, as stated above. Limits: one geometry and one flow; wall-clock figures of this pair are contended (32 ranks on 16 physical cores until iteration 728 of the lesion case; see the timing table of the returned data).

Audit and defect-handling record. *Roles* (set by the study lead on 2026-09-24): the main assistant does the work; GPT-6 Astra and Gemini 3.8 Flash audit only, before a setup runs and after it has finished, read-only; a defect of any kind (also in the assistant’s own scripts) goes to Opus 5.5 (at most three attempts), then Fable 5.1 (at most two), then stops and is reported; the fixes are verified by the assistant and audited again. *Follow-up tooling of this subsection* (hyperaemic pair, extended tube, isolated-lesion and transient setups, end-to-end runner): six audit rounds by both auditors; reporting and launch tooling needed Opus attempts 1–3 and Fable attempts 1–2 (Fable’s first attempt ended before delivering a report; its second passed 59 of 59 checks); the comparison scripts needed Opus attempts 1–2, the runner Opus attempts 1–2, the transient tooling Opus attempt 1. One auditor finding survived the attempt cap (a helper that could report “end of the reversed region established” when every downstream station was missing) and is disputed between the auditors; it was mitigated by reading the coverage counts of every analysis by hand. The isolated-lesion, transient and hyperaemic-on-extended-tube setups were built and audited but not run. *Task-1 setup* (U_{3D}): two audit rounds by Gemini alone (both READY WITH CONDITIONS) because GPT-6 Astra reached its usage limit until 2026-09-26 18:22 and the study lead decided on 2026-09-24 to proceed on one auditor; every finding was fixed (an outlet-flux sign, a binary-write setting, missing-file handling, two decision rules that contradicted the work order, a premature PASS wording without the shifted-zone family; Opus attempts 1 for each group), the launcher was generalised (coded-BC libraries derived from the case, rank cap 16, coded-free option, per-case iteration budgets: 221 asserting tests; Opus attempt 1 ended before finishing its test run, attempt 2 showed that the five failing checks were errors of the tests, not of the launcher). Astra’s audit of these setups is pending. *Task-3 setups* (prescribed-flow round trip, scan-14 geometry, meshes and cases): one Gemini round each, READY WITH CONDITIONS, no code defect. *Post-hoc audit of the returned data* (Gemini): sound with caveats; every sampled number was reproduced from the raw monitors. *The assistant’s own errors, found by data re-audits and corrected in place in this report*: the Item 6 timing was called isolated but was not; the sten70 level-4 solution was called

converged although its monitor kept drifting and its residuals were high; the A4 round-trip error and the A3 sten50 run were described from an intermediate plateau and as stopped early; the gate of the scan-828/837 selection was misnamed; the count “8 of 8” could not be reproduced (the running summary has seven pass/fail rows). *Mechanical incidents, not defects:* the CLI binaries were briefly unavailable during an automatic update, the Gemini service failed several times with transient DNS or eligibility errors (retried), and a root-disk shortfall (below the launcher’s 10 GB floor) was resolved by moving finished cases to another drive.

11.7 Task 1 of the 2026-09-24 work order: the 3D discretisation uncertainty U_{3D} of sten70

Question. §11.3 left the 3D FFR discretisation uncertainty of sten70 undefined: the four-level ladder was not a systematic refinement family (settings held while the realised boundary-layer stack and the core cell size varied), and its finest level was not converged. The study’s statistics plan has a pre-registered rule waiting on this number. **Design.** The classification, the thresholds (0.005, the noise factor 0.25) and the choice of the Celik–Eça–Hoekstra index were written in the design document (versions 2.1 and 2.2) and audited before any solve; the SETTLED criterion (version 2.4) was fixed after the first wedge run had shown a drifting finest level but before the first 3D result (the code carries a file time 3 min after the S50 launch and 1 h 43 min before its result). The iteration budgets of S25A, S25B, S12A and S12B (versions 2.5–2.7) and the reduction of the S12B budget were set *with earlier results in view*, on settling behaviour only, never on FFR values. Straight tapered vessel of Stage A, 70% diameter stenosis (throat diameter 0.891 mm), one outlet with the coded resistance $R = 6.974826 \times 10^9 \text{ Pa s m}^{-3}$, relax 0.20, inlet total pressure 90 mmHg, steady laminar **simpleFoam**, the audited numerics of Stage A, a fixed iteration budget fixed per level in advance (no residual-based stopping). A finer surface (192 circumferential facets, $20 \mu\text{m}$ axial rings in the lesion) replaced the Stage A STL. Three 3D levels with throat-zone cell sizes of 50, 25 and $12.5 \mu\text{m}$ (cfMesh sizes are $0.2 \text{ mm}/2^n$, so the ratio is exactly 2), four boundary layers of ratio 1.2 whose stack scales with the local cell size, the same lesion zone (11.5–51.5 mm) at $50 \mu\text{m}$ for every level, the throat zone A (29–34 mm) at 25 or $12.5 \mu\text{m}$, and, for the interface-contamination test the work order requires when only a throat zone is refined, a second throat zone B shifted by 1 mm at both ends. A 2D axisymmetric wedge family (blockMesh, 2° , four levels of ratio 2 in every direction, the same boundary conditions with the wedge-scaled resistance) serves as the physics-versus-discretisation cross-check. The pre-registered rules: the FFR of a level is the last-100-iteration mean of the area-averaged pressure at $x = 56.5 \text{ mm}$ over the aortic pressure; a level is analysed only if it is *settled* (drift over the final 500 iterations $\leq 2 \times 10^{-4}$ and last-100 band $\leq 10^{-4}$); rule 1 (the differences shrink in every family, no family is noise-limited and every finest pair is < 0.005) gives U_{3D} as the larger Celik–Eça–Hoekstra $\text{GCI}_{21} = 1.25 |e_{21}| / (2^p - 1)$ of the families; rules 2 and 3 handle flat families with the wedge and the transient fallback; without both families the outcome is only provisional.

Table 16: U_{3D} ladder of the 3D family (measured meshes; every solve on 16 ranks, one at a time; the memory column is the sum over the simpleFoam processes on the host).

level	cells	cells across throat (median/min)	p95 cell (μm)	BL first/stack (μm)	iterations	FFR	wall (h)	RAM (GB)
S50	2,687,640	26.5 / 20	51.2	7.1 / 39.0	3000	0.793362	1.8	4.7
S25A	3,194,196	47.5 / 33	25.5	3.9 / 21.6	4000	0.791103	2.6	5.3
S12A	6,739,168	88.0 / 58	12.7	1.5 / 8.6	6000	0.790308	8.2	9.8
S25B	3,299,948	47.5 / 33	25.6	3.9 / 21.6	4000	0.791197	2.9	5.4
S12B	7,649,800	88.0 / 58	12.7	1.5 / 8.6	3000	0.790418	5.7	10.9

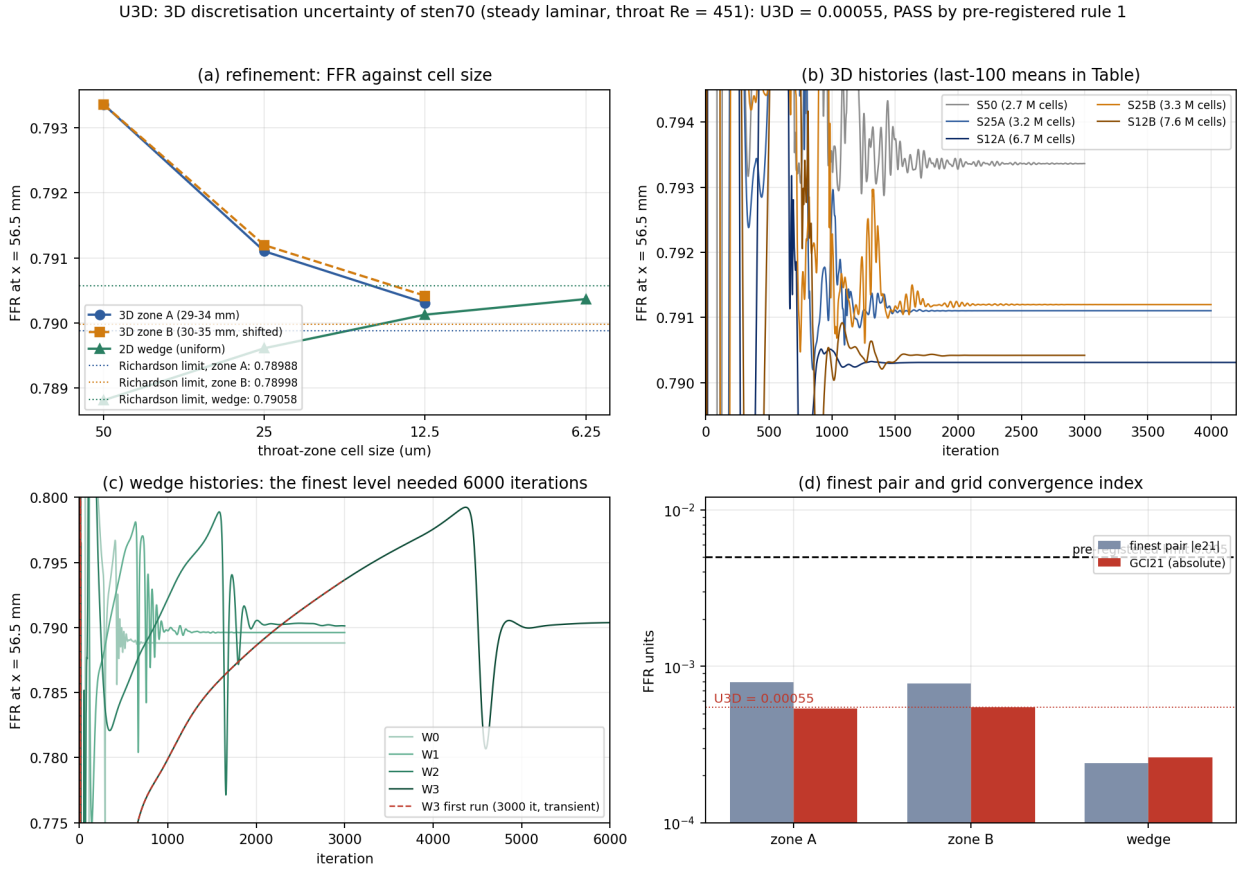


Figure 28: U_{3D} study. (a) FFR at $x = 56.5$ mm against the throat-zone cell size for the 3D zones A and B and the 2D wedge, with the Richardson limits of the three families (the 3D families approach from above, the wedge from below). (b) Iteration histories of the five 3D levels (the last-100-iteration means are in the table). (c) Histories of the four wedge levels; the first 3000-iteration run of the finest level (red) was still drifting and was re-run with 6000 iterations. (d) Finest pair and grid convergence index of the three families against the pre-registered limit 0.005.

Meshes and solves. All five meshes passed the gates that were measured on the mesh, not the dictionary: standard `checkMesh` OK with 0 failed strict checks, maximum non-orthogonality 35–42° and skewness 1.2–1.4, throat-plane area within 0.16% of the surface, and the concave-face sets confined to the ends of the refinement zones and to the zone interfaces. The measured ratio of cells across the throat between levels is 1.8–1.9 (the medians depend on how many diameters are sampled, staircase crossings; the axis-aligned count of 33 and 58 cells equals the diameter over the cell size plus the layers and is the minimum reported), the cell size ratio 2.0, and the realised boundary-layer stack ratio 1.8 and 2.5 (cfMesh sizes it itself; an independent 16-line measurement gives 1.7 and cell heights within about 25% of the reported ones, so the scaling with the cell size holds within that scatter). All levels were settled well before their budgets (S50 at about iteration 1750, S25 at about 2000, S12A at about 1500); the last-100 FFR band at the end is 10^{-8} for S25A and S12A, 10^{-7} for S25B and S12B and 10^{-5} for S50. The strict verdict is CONVERGED for S25A, S12A, S25B and S12B; S50 reads UNCONVERGED only because its velocity residuals (7×10^{-5}) exceed 10^{-5} while its FFR is constant to 10^{-5} . The throat Reynolds number is 451–453 in every solve, with the flux through the throat equal to the outlet flux to within 0.04%.

Result. Zone A gives FFR 0.793362, 0.791103, 0.790308: $e_{32} = 0.002259$, $e_{21} = 0.000794$, a ratio of 2.84 (monotone), observed order $p = 1.51$, $GCI_{21} = 0.00068$ relative or **0.00054** absolute. The shifted zone B gives 0.793362, 0.791197, 0.790418: $e_{32} = 0.002165$, $e_{21} = 0.000779$, $p = 1.47$, $GCI_{21} = 0.00069$ relative or 0.00055 absolute. The interface effect (zone B minus zone A at equal

cell size) is $+0.00009$ at $25\mu\text{m}$ and $+0.00011$ at $12.5\mu\text{m}$, about seven times below e_{21} ; it does not shrink with refinement, so it is the visible floor of refining only a throat zone (zone B refines 1 mm more of the jet region than zone A) and 20% of U_{3D} . Both families shrink, neither is noise-limited, and both finest pairs are far below 0.005, so the pre-registered rule 1 applies. $U_{3D} = 0.00055$, in absolute FFR units at $x = 56.5\text{ mm}$ (0.07% of the FFR of 0.79): the larger, over the two throat-zone families, of the GCI_{21} of the steady laminar FFR of sten70 under a 50/25/12.5 μm throat-zone refinement with boundary-layer stack scaling with the cell size. It is well below the pre-registered 0.005; the study's recipe (throat zone refined with the boundary-layer stack scaling with the cell size) is therefore accepted. **Read with:** U_{3D} is the uncertainty of the finest solution; the 50 μm production base itself is 0.0031 (S50–S12A) or 0.0029 (S50–S12B) above it, and the 25 μm level 0.0008 above the 12.5 μm level.

Wedge cross-check. The four wedge levels (11,232, 44,928, 179,712 and 718,848 cells; cell sizes 50, 25, 12.5 and 6.25 μm at the throat) give FFR 0.788811, 0.789611, 0.790127 and 0.790367: successive differences $+0.00080$, $+0.00052$, $+0.00024$ (shrinking), apparent order 1.11 and $\text{GCI}_{21} = 0.00026$ on the three finest levels, so the axisymmetric family converges cleanly. The finest wedge level (0.790367) lies within 0.0001 of the two finest 3D levels (0.790308 and 0.790418), but the families approach from opposite sides (3D from above, wedge from below), so the agreement of the finest levels is partly a crossing; the Richardson limits are 0.78988 and 0.78998 (3D zones A, B) against 0.79058 (wedge), a gap of 0.0006–0.0007 that is comparable to U_{3D} and lies inside the GCI bands. The wedge is therefore consistent with the 3D result within its uncertainty and supports that the 3D differences measure discretisation and not a meshing artefact, without confirming the limit; its own GCI is weakly determined (two levels carry a drift of about 10^{-4} and the observed order of 1.1 or 0.63 for a formally second-order uniform family means it is not yet in its asymptotic range). The finest wedge level was still drifting after its first 3000 iterations (FFR 0.7934 with a band of 4×10^{-4} , a transient, kept in the returned table as W3_3000it); the re-run with a budget of 6000 rose to a hump of 0.799 near iteration 4300 and then settled to the value above (drift over the last 500 iterations 6×10^{-5} , last-100 band 10^{-5}). The wedge is not used by rule 1. All wedge levels read UNCONVERGED under the strict verdict through the azimuthal velocity component, whose normalised residual in a one-cell wedge stays of order 10^{-2} whatever the convergence, and W2 and W3 additionally through U_y residuals of 3×10^{-5} and 1.1×10^{-5} ; the FFR itself is settled to 10^{-5} at every level.

Deviations and limits, disclosed. (i) Budgets and thresholds: the SETTLED drift threshold of 2×10^{-4} was fixed after W2 (drift 1.2×10^{-4} , passes with 40% margin) and the drifting first W3 run had been seen, and the version-2.4 note said the W3 re-run would use 9000 iterations while it used 6000; the first wedge budget (3000) was too short for the finest level, so the SETTLED criterion and per-level budgets were pre-registered before the 3D runs; S25A and S25B were given 4000 iterations, S12A 6000 and, after S12A had settled at about 1500 iterations, S12B was reduced from 6000 to 3000 before it started (the settled criterion was applied at its end). (ii) U_{3D} refers to the refined throat zone only: the lesion zone at 50 μm and the regions beyond do not change between levels, and the geometry, the flow, and the steady laminar assumption are those of Stage A; a converged steady solution to 10^{-8} does not exclude an unsteadiness the steady solver cannot see. (iii) The boundary-layer stack ratio between levels is not exactly 2 (1.8 and 2.5). (iv) The setups were audited by one auditor (Gemini 3.8 Flash, two rounds), Astra being unavailable (§11.6).

11.8 Task 3 of the 2026-09-24 work order: Gate M1 on the shipped scan-14 packages

Status and verdict (2026-09-26). The three shipped packages of scan 14 (left LAD, proximal, 20 mm lesion of 80%DS: the unmodified lumen, the lesion, and the lesion plus a deleted side branch) were built end to end by a script without manual geometry repair, meshed, and solved in resistance mode, and the prescribed-flow mode with a round trip was run on the lesion case and on the missed-branch case (the clean case is still running at the time of writing; the tables of this subsection are

regenerated from the data files). **Gate M1 is not a clean pass:** all solves converge and both boundary-condition modes are stable with round trips closing to $\leq 0.005\%$, but three geometry gates fail and are reported as results (self-intersection of all three surfaces; the literal absolute throat check of the two lesion surfaces) and the strict `checkMesh` reports three failed checks per mesh (localised). The clock of the work order (five working days without a scriptable path) is not met: about 5.5 calendar days were spent in total including the scan-837 pilot, interleaved with other items, and a scriptable no-repair path has existed since 2026-09-23. Scan 14 is in the frozen cohort; its membership in the frozen CFD subset could not be asserted because the subset file is absent from the machine, and the study lead approved the logged waiver on 2026-09-26. No 0D twin was run on scan 14 or any cohort scan (blinding).

Build. The frame (the NIfTI affine is RAS: x and y negated) puts 121/121 deletion points of the truncation and 163/163 of the missed-branch package in the lumen (0 without the flip). The shipped 6-connected deletion with protect radius $1.10r$, run once on the union of the sets, was executed verbatim and removes 1310 voxels (clean and baseline) and 2089 voxels (T1), the shipped numbers, leaving two connected components (Figure 29); the shipped “erosion 25/40” has no definition and could not be reproduced, and the retained LAD radius is unchanged except at the truncation end and, for T1, by 0.075, 0.045 and 0.023 mm at three nodes next to the deleted branch. The surface was made by marching cubes, Taubin smoothing (`vtksmoothingsmoothing`, 30 iterations; no `vtksurfaceclipping`), planar outlet clips and straight extensions (inlet 13.01 mm, outlets 3.29–4.73 mm, taken as $5D$ and $3D$ with $D = 2r_{\text{mm}}$ of the package; with the area-equivalent diameter of the clipped loops they would be 0.9–1.6 times as long). The lesion window was subdivided to edges ≤ 0.0288 mm ($= r_{\text{throat}}/8$; about 1.1 M triangles) before the raised-cosine deformation, which follows the smoothing-spline path frames of the scan-837 pipeline (the package’s literal nearest-node rule steps by up to 0.4 mm across Voronoi cells).

Scan 14 (ImageCAS-X label volume, 0.32 x 0.32 x 0.5 mm voxels): the mask edit of the Gate-M1 packages (6-connected deletion, protect radius $1.10r$, run once on the union)

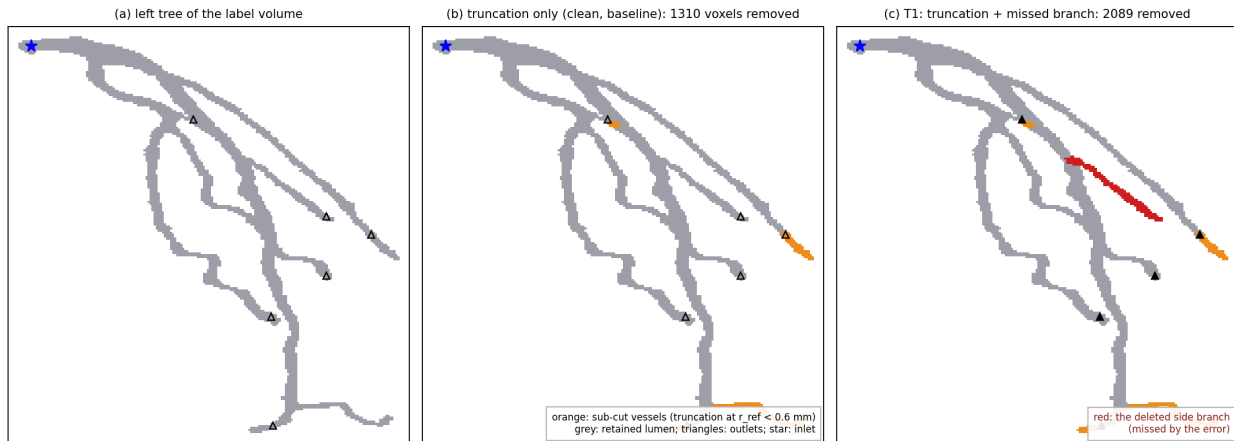


Figure 29: Scan 14, label volume of the left tree (coronal maximum-intensity projection) and the mask edit of the packages. (a) The left tree with the inlet and the outlets of the baseline package. (b) The truncation at $r_{\text{ref}} < 0.6$ mm removes the sub-cut vessels (orange). (c) The T1 package removes, in the same run, a side branch (red) that the error “missed”. The dataset provides the segmentation, not the CT intensities.

Table 17: Gate M1, scan 14: geometry gates (P = pass, F = fail; clean / baseline / T1).

gate	result	verdict
frame flip, deletion points in the lumen	121/121, 121/121, 163/163 (0 without flip)	P
voxels removed, components after the edit	1310, 1310, 2089 (= shipped), 2	P
clip loops with outward normals	7, 7, 6 (expected 7, 7, 6)	P
final surface closed, one part, positive volume	77 k / 1.17 M / 1.17 M triangles	P
lesion purity, fold-over, edge $\leq r_{\text{throat}}/8$	min normal dot 0.37, 100% of window edges	P
throat, <i>relative</i> (as built / undeformed against <code>radial_scale</code> 0.2081)	+0.002 / +0.022 / +0.024% (three radius definitions)	P (decisive, deviation)
throat, <i>literal absolute</i> against $r_{\text{target}} = 0.2306$ mm	+14.0 / +19.5 / -10.1% (inscribed circle / area-equivalent / sphere at the axis)	F
<code>surfaceCheck</code>	14 / 16 / 8 locations	F
<code>-checkSelfIntersection</code> (positive control flagged)		

Failed gates, reported as results. (i) *Self-intersection*: present in the raw marching-cubes surface, hence caused by the mask, not by smoothing or clipping: a 1-voxel-wide whisker of mask at LAD node 285 (all cases) and a thin D1 of 1.7 voxels at node 133 (clean and baseline; gone in T1 where the branch is deleted). Nothing was repaired; cfMesh meshed the surfaces and no mesh entity flagged by the strict check lies within 2 mm of either site. (ii) *Literal throat check*: the package radius (vmtk maximum inscribed sphere on the dataset surface) and this marching-cubes surface differ by -0.3% in the median and by -10 to $+8\%$ (10th to 90th percentile) per node, so the as-built throat cannot reproduce r_{target} within 1% whatever the definition; the gate that was made decisive is therefore the *relative* one (as built over undeformed, against the package’s own `radial_scale`), which passes to 0.03%. This is a deviation from the work order’s rule (“reject if $> 1\%$ ”); the analysis side must confirm or overrule it.

Meshes. cfMesh with four boundary layers of ratio 1.2 (the scan-837 meshes used three), a $25\ \mu\text{m}$ zone of ± 4 mm along the LAD around the throat, $50\ \mu\text{m}$ zones at the inlet and outlets, and $100\ \mu\text{m}$ and coarser elsewhere: 3.89 M (clean), 3.66 M (baseline) and 3.49 M cells (T1) in 1:41–1:57 min on the host (peak 2.8 GB); all inlet and outlet patches present; standard `checkMesh` OK; the strict check fails three checks per mesh (about 1000 faces of poor face-decomposition tets, 41–52 concave cells and 19–30 small-volume-ratio faces), spread over the tree and at the zone ends, none within 2.6 mm of the throat; maximum non-orthogonality 55.6° , skewness 1.85, aspect ratio 11. The lesion meshes hold 36 (minimum) and 39.5 (median) cells across the 0.55 mm throat chord (requirement ≥ 12), a p95 cell size of $25.7\ \mu\text{m}$ and a four-layer stack of $16.5\ \mu\text{m}$ at the throat; the as-meshed throat has an area-equivalent radius of 0.2756 mm (Figure 32b). cfMesh writes the inlet and outlet patches with type `wall`; only that keyword was changed to `patch` in each `boundary` file.

Solves. Both boundary-condition modes use the audited setups of this report (coded resistance boundary condition with $G = R/R_{\text{own}}$ relaxation, R_{own} from the package centreline radii; prescribed mode with `flowRateOutletVelocity` on every outlet at once and zero-gradient pressure; inlet total pressure of the package; 16 ranks, one solve at a time, fixed budget of 3000 iterations, laminar). The audited relaxation rule gives 0.0036–0.040 for the outlets (time constants up to about 280 iterations); it did not prevent convergence in 3000 iterations. Table 18 lists the solves, Table 19 the resistance-mode flows against the package flows, Table 20 the round trips; Figure 30 shows the wall pressure and Figure 31 the pressure along the LAD. The throat Reynolds number is computed from the section flux at the throat probe (an earlier value from an infinite sampling plane that also cut other branches was wrong and was corrected after the post-hoc audit): 123 (baseline, resistance),

106 (baseline, prescribed), 60 (T1) and 29 at the would-be throat station of the clean tree, so the steady laminar assumption is comfortably within range for scan 14.

Table 18: Scan-14 Gate-M1 solves (16 ranks each, one at a time, fixed budget 3000 iterations, strict verdict as stated; the throat Reynolds number is the section flux at the throat probe, at the would-be throat station for the clean tree).

case	BC mode	cells	verdict	inflow (mL/s)	throat Re	wall (h)	RAM (GB)	round trip max error (%)
clean	resistance	3,889,165	converged	0.7119	29	2.34	6.1	–
baseline (80%DS)	resistance	3,657,147	converged	0.6817	123	2.31	5.9	–
baseline (80%DS)	prescribed	3,657,147	converged	0.6490	106	2.17	5.8	0.0013
T1 (lesion; branch deleted)	resistance	3,485,528	converged	0.5796	60	2.09	5.7	–
T1 (lesion; branch deleted)	prescribed	3,485,528	converged	0.6490	106	2.09	5.6	0.0051

Table 19: Resistance mode with the package resistances (**bc_A**): 3D outlet flow against the package target flow (**bc_C**); a positive deviation means that the 3D tree delivers more than the target.

case	outlet	Q_{target} (mL/s)	Q_{3D} (mL/s)	deviation (%)
clean	out_160	0.0953	0.1272	+33.5
clean	out_425	0.1044	0.1054	+1.0
clean	out_600	0.0782	0.1039	+32.8
clean	out_639	0.1126	0.1122	-0.3
clean	out_742	0.1234	0.1258	+1.9
clean	out_868	0.1351	0.1374	+1.7
baseline	out_160	0.0953	0.1105	+16.0
baseline	out_425	0.1044	0.1054	+1.0
baseline	out_600	0.0782	0.0903	+15.4
baseline	out_639	0.1126	0.1122	-0.3
baseline	out_742	0.1234	0.1259	+2.0
baseline	out_868	0.1351	0.1374	+1.7
T1	out_383	0.1044	0.1054	+1.0
T1	out_558	0.1735	0.0985	-43.2
T1	out_597	0.1126	0.1123	-0.3
T1	out_700	0.1234	0.1259	+2.0
T1	out_826	0.1351	0.1375	+1.8

Table 20: Prescribed-flow round trip on scan 14: every outlet flow prescribed at once (**bc_C**), $R_i = (\bar{p}_i - P_v)/Q_i$ derived, re-imposed in resistance mode; error of the re-solved flow against the prescribed one (criterion 0.5%), and derived resistance against the package resistance (**bc_A**).

case	outlet	Q (mL/s)	round-trip error (%)	$R_{\text{derived}}/R_{\text{bc_A}} - 1$ (%)
baseline	out_160	0.0953	+0.0000	+19.5
baseline	out_425	0.1044	-0.0004	+1.0
baseline	out_600	0.0782	-0.0003	+19.0
baseline	out_639	0.1126	-0.0013	-0.3
baseline	out_742	0.1234	-0.0002	+2.1
baseline	out_868	0.1351	+0.0000	+1.8
T1	out_383	0.1044	-0.0004	+1.0
T1	out_558	0.1735	-0.0051	-47.5
T1	out_597	0.1126	-0.0013	-0.3
T1	out_700	0.1234	-0.0002	+2.1
T1	out_826	0.1351	+0.0000	+1.8

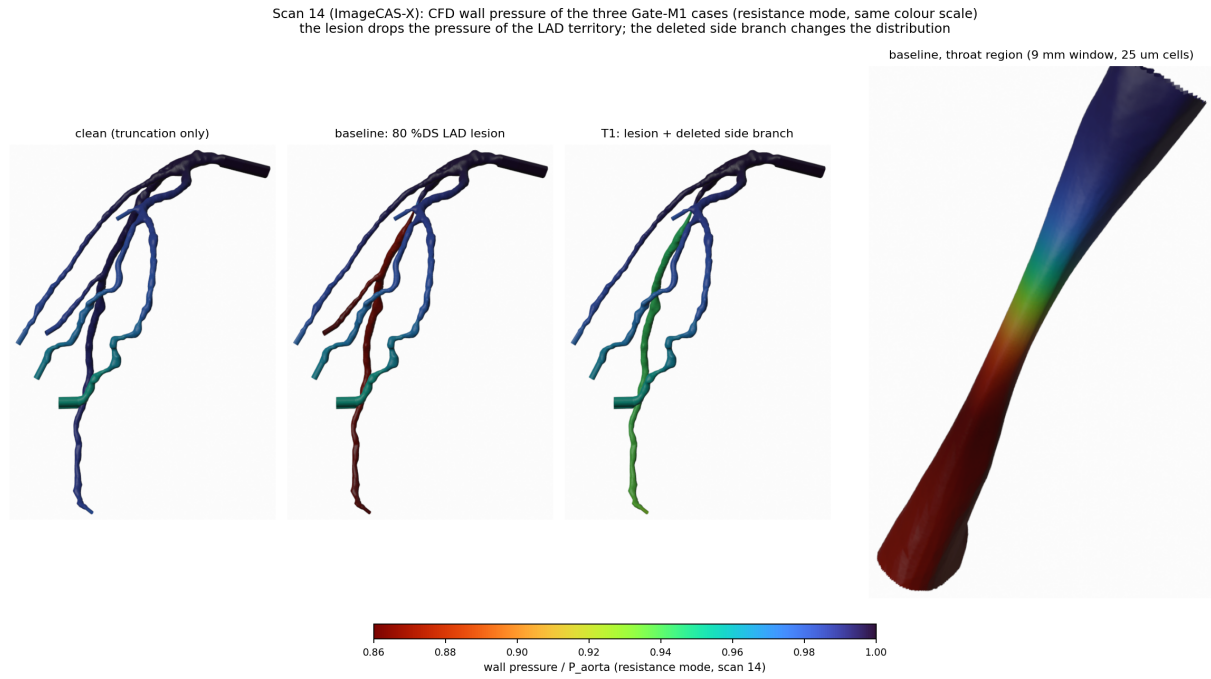


Figure 30: Scan 14, wall pressure over aortic pressure of the three cases in resistance mode (same colour scale; Blender, Workbench engine). The 80%DS lesion drops the pressure of the LAD territory to 0.86–0.87 of the aortic pressure; in T1 the deleted side branch changes the distribution (out_558, whose full-territory target cannot be delivered, is depressed). Right: the throat region of the baseline mesh.

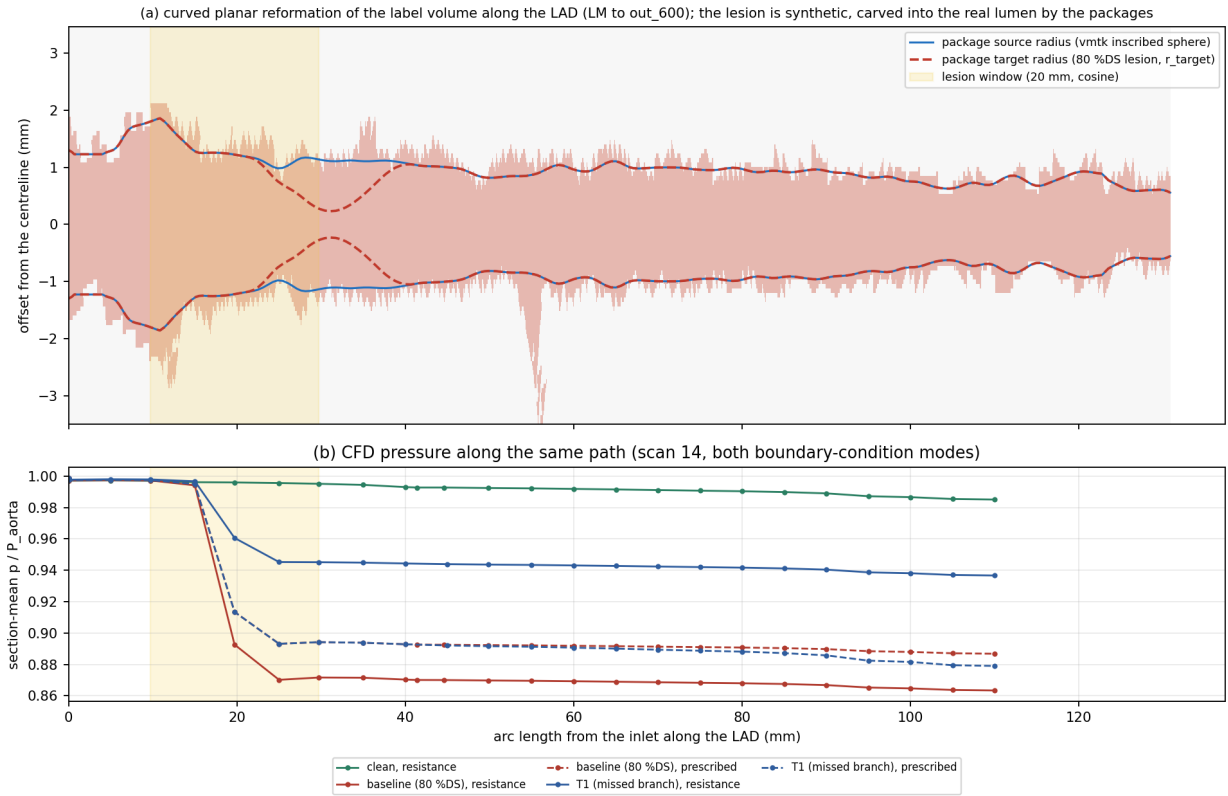


Figure 31: (a) Curved planar reformation of the scan-14 label volume along the LAD (LM to out_600) with the radii of the package: the source radius (blue) and the target radius of the 80%DS lesion (red dashed), which is synthetic and carved into the real lumen. (b) Section-mean pressure of the CFD solves along the same path in both boundary-condition modes.

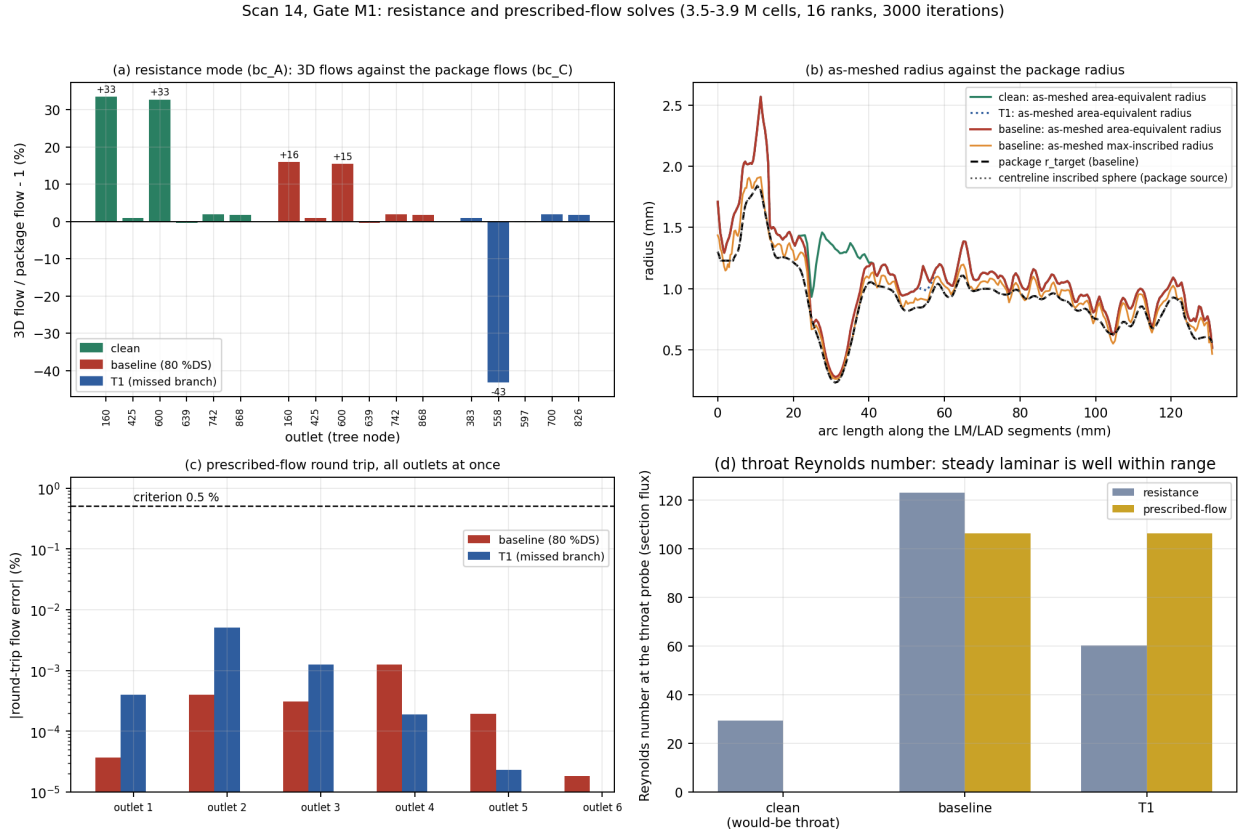


Figure 32: Scan 14, Gate M1: (a) resistance mode, 3D outlet flows against the package flows (bc_C); (b) as-meshed area-equivalent and maximum-inscribed radius along the LM/LAD segments against the package radius; (c) round-trip flow errors of the prescribed-flow mode (all outlets at once); (d) throat Reynolds number from the section flux.

Findings. (i) *Flow split.* With the package resistances the 3D trees deliver +33% and +33% (clean), +16% and +15% (baseline) at the two outlets of territory 0 (out_160 and out_600) against the package flows, and $\leq 2\%$ elsewhere; set-up errors are excluded ($R_{\text{derived}}/R_{\text{bc_A}} = 1.00000$ on all outlets in resistance mode; the two deviating outlets are exactly territory 0 of bc_C). The package's own numbers imply a path resistance to territory 0 about 17 times that of the 3D lumen (the package places territory 0 at 0.749 of the aortic pressure, the 3D clean lumen at 0.98): a 0D-versus-3D question for the analysis side. (ii) *T1.* The T1 package's bc_C holds the full-territory flows, so $P_v + RQ$ at out_558 is 19.1 kPa, above the aortic pressure of 12.0 kPa; the resistance solve delivers 0.098 of the 0.173 mL/s target (-43%), and in prescribed mode the flow is delivered with that outlet at 0.863 of the aortic pressure (labelled pilot: its targets are provisional). The package README's statement that bc_A $\times$ bc_C reconstructs the clean outlet pressures holds only for clean and baseline. (iii) *Round trips.* Prescribing all outlets at once was stable from rest without a ramp; the re-imposed resistances reproduce the prescribed flows to $\leq 0.0013\%$ (baseline) and $\leq 0.0051\%$ (T1), against the 0.5% criterion. This is a fixed-point and stability test of the relaxed coded condition (the resistances are derived from the same converged solution), not a validation of the package flows. (iv) *Cost.* About 2.1–2.3 h and 5.7–6.1 GB per 3.5–3.9 M-cell solve at 16 ranks; the geometry build takes 3–9 min per case and the meshing 2–5 min including the gates, so the whole path from mask to converged solution is scripted and takes about 2.5 h per case and mode. **Limits:** one scan; the geometry gates failed as reported; the strict mesh check is localised, not eliminated; the setups were audited by one auditor (Gemini) and, after the fact, by Fable 5.1 as a stand-in for Astra; the E0 assertion stands on a waiver.

11.9 Item 5: RCR outlet BC verification — **PASS**

The Crank-Nicolson coded RCR BC (§11.2’s design revision, independently verified standalone against scipy to 0.011% at a 10 ms timestep before ever compiling in OpenFOAM) was run on Stage A’s pipe geometry (61,440 cells, PIMPLE, adjustable timestep targeting the stated Courant limit) for 2 full pulsatile cycles (2 s), reaching $t = 2.0000$ s cleanly with no errors, 17,316 timesteps, mean Courant 0.044 (max 0.52, at the 0.5 target). **Independently re-run by the coordinator, not just taken from the fork’s report:** comparing the solver’s own outlet-pressure monitor against a from-scratch scipy RK45 integration of the same RCR ODE (same R_1, R_2, C, P_v , same prescribed $Q(t)$, same $p_C(0) = 0$ initial condition) over the second cycle only gives a **maximum relative error of 0.2827%** (mean 0.042%) — comfortably under the 0.5% pass criterion, using 56% of the tolerance budget (the error is normalised by the maximum $|p_{\text{ref}}|$ over cycle 2, as the pre-registered metric defines it; the point-wise relative error peaks at 0.54% at $t = 1.911$ s, found by the post-hoc audit). No NaN/blow-up/growing oscillation occurs during the backflow window (1,195 of 8,656 cycle-2 samples have $Q < 0$; pressure stays smooth and bounded, 250–4420 Pa throughout, and the worst-case error itself occurs during backflow without escalating). A mass-conservation sanity check (outlet flux vs. the prescribed inlet $Q(t)$ through the rigid pipe, independent of the outlet BC) agrees to 0.25%.

Operational note: a stale 7-line file from an earlier serial compile-smoke-test (run first, deliberately, to force the coded BC to compile before the parallel launch — the same MPI recompilation-race precaution Item 4’s design calls for) sat at the same `postProcessing` path the real run also wrote to; OpenFOAM avoided clobbering it by suffixing the real run’s file (`_0`), and the first comparison attempt’s file glob picked up the stale stub instead, producing an empty cycle-2 array. Caught immediately (an empty array is a loud failure, not a silent wrong answer) and fixed by globbing for the largest matching file rather than an exact name.

11.10 Operational incident: cross-fork CPU oversubscription, diagnosed and fixed

While Items 1, 2, 3+4, 5 and 6 ran in parallel (three background subagents plus this document’s own coordinator process), the machine became severely oversubscribed: load average reached 78 with 48 concurrent OpenFOAM solver processes. Per this document’s standing practice, a three-reviewer panel (Fable 5.1, GPT-6-astra, Gemini 3.8 Flash) was convened on the live problem itself, not just on CFD setups. Two of three reviewers independently confirmed, by reading each `mpirun`’s working directory and `/proc` state directly rather than trusting the coordinator’s own attribution, two facts the coordinator had gotten wrong: (i) a group of three solves the coordinator had attributed to the Item 3+4 fork actually belonged to Item 2 (which had launched `sten60`, `sten65`, and `sten70`’s 4th mesh level concurrently rather than sequentially); (ii) the host has 16 physical cores, not the 32 logical threads `nproc` reports — CFD’s memory-bandwidth-bound solvers gain little from SMT, so the true available budget was roughly 8 free physical cores, not the ~ 24 assumed when each fork was given its own core guidance in isolation. All three reviewers independently identified Open MPI’s default non-yielding busy-wait behaviour (each `mpirun` perceives itself as undersubscribed against the 32-thread host and spins rather than yields) as amplifying the effective slowdown to roughly $5\times$, well beyond the naive $1.5\times$ the raw process-count ratio would suggest — one reviewer measured this directly (wall-time-per-step degrading from ~ 8 s uncontended to ~ 45 s contended on the same case).

The panel’s fix was applied in full: Item 6’s two pulsatile runs were stopped (their deliverable, a wall-clock cost estimate, had already been captured from a smoke test — later found to have run alongside other solves, hence contended, see §11.4 — so no further Item 6 result was lost); Item 2’s `sten60/sten65` solves were stopped (already fully converged and reported, so continuing to run them to their nominal end iteration was pure waste); the Item 3+4 fork was instructed to run its remaining solves strictly sequentially rather than concurrently before launching any of them. All

four stops used OpenFOAM’s own graceful `stopAt writeNow` mechanism, not a hard kill. Load fell from 78 to 33 within a minute of the fix, and the remaining solves (Item 1, Item 5, Item 2’s sten70 4th level) measurably accelerated. No result’s correctness is affected — convergence is independent of wall-clock pacing — but this incident is exactly why Item 6’s own wall-clock numbers are reported from the smoke test (itself contended, see §11.4) rather than the fully contended runs, and it is disclosed here in full rather than folded silently into the numbers, consistent with this document’s practice for every other incident in this study.

12 Conclusions

Stage A. Of the seven pass/fail rows of the running summary (§2), six pass and one is not established. A1, A3 on sten00 and sten50, and A4 are clean, unanimous passes; A3 on sten80 is a flagged pass (numeric tolerance met, the stricter residual criterion not, cause left open between a numerical SIMPLE stall and a slow physical instability; a time-accurate diagnostic was scoped, reviewed and deliberately not run once its cost was weighed against a free check showing the practical effect on FFR to be inside the study’s accepted tolerance, §8); A5 on sten80 passes comfortably, with the new observation that the fluctuation band grows with refinement (mild evidence against a pure numerical-stall origin). **A5 on sten70 is NOT ESTABLISHED:** it met its numeric tolerance at 94.6% of the budget with a non-monotone sequence, and the fourth level of Item 2 did not confirm a convergent trend; the Celik–Eça–Hoekstra analysis is not trustworthy there because the boundary-layer stack was held constant across levels (§11.3). A2 is informational by design and shows the 0D-vs-3D gap growing and changing sign with severity (-3.01% at 0%DS, -1.48% at 50%DS, $+8.68\%$ at 80%DS). The coded resistance-outlet BC, the central risk of the audit-and-fix effort, is confirmed on four straight-vessel geometries (sten80 flagged), an idealised 3-outlet tree and a real 5-outlet coronary tree (scan 837); the prescribed-flow mode, run first on a single outlet (A4), now also closes its round trip on the 3-outlet Item 1 tree at both severities (all outlets prescribed together, largest flow error 0.0007%), but not yet on a real lumen.

Beyond Stage A. Item 1: the idealised branched tree with the automatic relaxation rule gives outlet flows within 1.04% of the 0D twin (both severities) and BC errors below $2 \times 10^{-4}\%$. Item 2: no trustworthy 3D discretisation uncertainty for sten70 exists yet; the severity sweep brackets the sign change of the 0D-vs-3D flow gap between 50 and 60%DS. Items 3 and 4: the meshing blocker of the real-lumen pipeline was root-caused (the surface remesh pinched narrow lumens) and removed; the rebuilt pipeline yields converged 5-outlet solves; for the lesion80 case of scan 837 the ten registered quantities are insensitive to the throat mesh (W100, M50, T25a, T25b), to an extension of the refined tube (Table 15), and the hyperaemic pair (§11.6) shows a 3D flow reduction smaller than both 0D variants at $3.16\times$ the flow, with the recirculation extent beyond arc 34 mm not mesh-supported. Task 1: the 3D FFR discretisation uncertainty of sten70 from a systematic refinement family is $U_{3D} = 0.00055$ (§11.7), a pre-registered PASS. Task 3: the scan-14 pipeline is scripted end to end without manual repair, all solves converge, both boundary-condition modes are stable and the round trips close to $\leq 0.005\%$, but the surfaces fail the self-intersection gate and the literal throat check (§11.8). Item 5: the RCR boundary condition passes. Item 6: the pulsatile runs were stopped; only a cost estimate exists. **Scan 837 is a pipeline pilot outside the frozen cohort; none of this is a Gate M1 result.**

Open items (not run or not finished at the time of writing). (i) Nothing of Task 1 remains open; its wedge cross-check is complete (§11.7). (ii) Gate M1 on the shipped scan-14 packages (§11.8): the geometry, meshes, the three resistance-mode solves and the prescribed-flow solves with round trips of the baseline and missed-branch cases are done; the prescribed-flow solve and round trip of the clean case were still running at the time of writing. The surfaces fail the self-intersection gate and the literal absolute throat check (reported as results, the verdict on them rests with the analysis side), and the E0 membership assertion could not be executed because the frozen-subset file is absent from the machine (the study lead approved the logged waiver on 2026-09-26). (iii) A clean

uncontended pulsatile timing (Item 6, Task 4). (iv) The mesh sensitivity of the hyperaemic case, the isolated-lesion and transient-steadiness checks, which were built and audited but not run because the work order closed new work on scan 837. (v) The sten80 numerical-versus-physical question. (vi) Audit coverage: while this report was being finalised the GPT-6 Astra auditor reached its usage limit (until 2026-09-26 18:22); by the study's rule no new CFD setup runs before it has been audited by both auditors, so nothing new was started in that window.

The sten80 episode is also a demonstration of the review discipline this document has followed throughout: reviewers were asked before hardware time was spent, not after; a specific, checkable claim (Fable's sten70 comparison) was independently verified against this document's own logs rather than trusted on authority; two reviewers caught the same fatal bugs by independently reading the same source code, which is why the fix could be trusted; and when the reviewers' own recommendation turned out to cost far more than estimated, they were brought back and re-consulted rather than the plan being quietly abandoned or quietly pushed through. Gemini 3.8 Flash's unavailability for the last two rounds (7 failed attempts, a Google-side OAuth/DNS issue) is disclosed rather than papered over; the 2-of-3 panel that remained still reached a defensible, well-evidenced joint recommendation.